

Q.1)

Ans) c

Exp) Option c is the correct answer.

The inscriptions of Emperor Ashoka are among the most important historical sources for understanding the Mauryan Empire. These inscriptions were issued by Ashoka to communicate his policy of **Dhamma**, administrative instructions, and moral teachings to the people.

Ashokan inscriptions are broadly classified into four categories:

1. Major Rock Edicts
2. Minor Rock Edicts
3. Major Pillar Edicts
4. Minor Pillar Edicts

There are **14 Major Rock Edicts**. These were a set of 14 edicts engraved on natural rock surfaces across different regions of the Mauryan Empire. They were usually located near trade routes and populated settlements so that the message could reach the public easily.

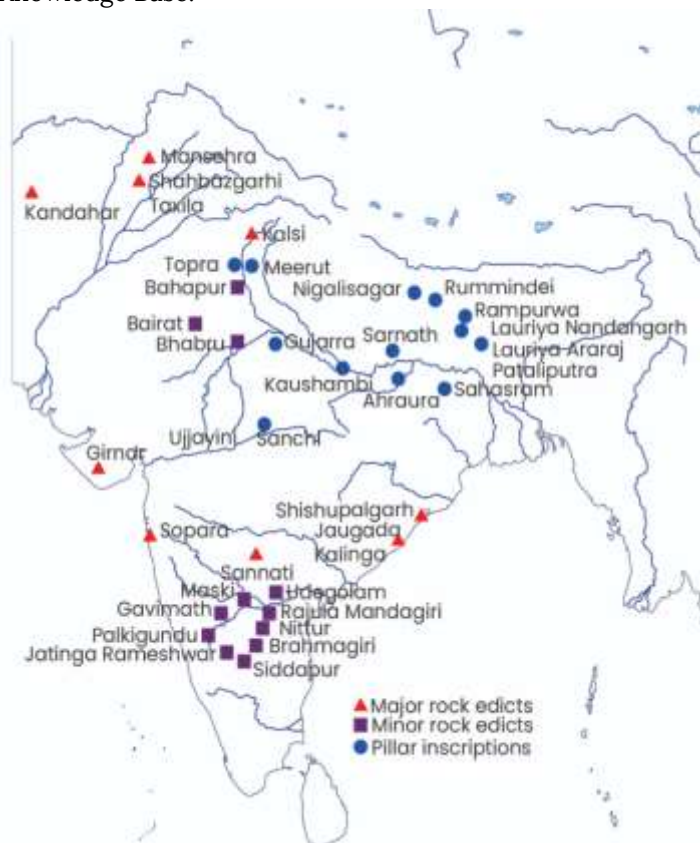
**Option a is correct.** **Dhauli**, located in present-day **Odisha**, is an important site of Ashokan inscriptions. It contains **Major Rock Edicts** along with the famous **Separate Kalinga Edicts**. These inscriptions are associated with the aftermath of the **Kalinga War**, which transformed Ashoka's outlook towards governance and violence.

**Option b is correct.** **Girnar**, located in **Gujarat**, is one of the most important sites where **Major Rock Edicts of Ashoka** are inscribed. The Girnar rock later also carried inscriptions of rulers like **Rudradaman**, making it an important epigraphic site in Indian history.

**Option c is incorrect.** **Sanchi**, located near **Bhopal in Madhya Pradesh**, is famous for the **Sanchi Stupa** and other Buddhist monuments. However, It does **not contain Major Rock Edicts**. Instead, it is known for the **Ashokan Pillar with a lion capital** and other Buddhist structures built during and after the Mauryan period.

**Option d is correct.** **Shahbazgarhi**, located in present-day **Pakistan**, is one of the most significant sites of Ashokan inscriptions. It preserves the **Major Rock Edicts written in the Kharosthi script**, which was used in the north-western regions of the Mauryan Empire.

Knowledge Base:



Source:) <https://egyankosh.ac.in/bitstream/123456789/84170/3/Unit-5.pdf> - Unit 5 - Inscriptions And Prashastis - Ashokan Edicts - Page 80

Subject:) History

Topic:) Ancient History

Subtopic:)

Q.2)

Ans) a

Exp) Option a is the correct answer.

Economic life in ancient and early medieval India was strongly organised around **guilds**, known as **Shreni**. These were corporate bodies of merchants, artisans, and traders. They regulated trade, protected members' interests, and often acted as autonomous economic institutions.

Texts and inscriptions refer to merchant organizations by several terms such as **Naigama, Shreni, Samgha, Sartha and Samuha**. These institutions functioned like professional associations and played a key role in the organisation of production, trade, and long-distance commerce.

**Statement I is correct.** A **Shreni** refers to a corporate association of individuals engaged in the **same occupation or profession**. According to commentator **Medhatithi**, a Shreni could include Traders, Moneylenders and Artisans etc. Thus, it functioned as a professional organisation that regulated economic activity and protected the interests of its members.

**Statement II is correct.** Some merchant guilds maintained their own military force known as **Shrenibala**. These armed groups served multiple purposes: protection of merchant caravans during long-distance trade, security of trade routes and goods, protection from bandits and political instability.

**Statement III is incorrect.** Guilds in ancient India enjoyed **financial autonomy**, including the use and circulation of money in economic activities. Evidence from inscriptions and literary sources suggests that guilds accepted deposits, functioned like banking institutions, and issued **guild seals or coins** used in economic transactions. There is no historical evidence which suggests that the guilds were prohibited by the state from issuing their own currency for economic transactions.

**Knowledge base:** The **Mandsaur inscription provides an important example of guild activity in ancient India. It refers to a guild of silk weavers who had migrated from Lata (modern Gujarat) and later settled at Mandsaur in present-day Madhya Pradesh.** The inscription highlights two significant aspects of guilds: first, the mobility of skilled craftsmen across different regions, and second, the important role of guilds in organising production as well as influencing patterns of settlement.

Source:) <https://www.egyankosh.ac.in/bitstream/123456789/73904/1/Unit-13.pdf> - Unit 13 - Merchant Guilds and Urbanization in North India and South India - Guilds of North India and The Trading Organizations (Guilds) of South India - Page 234; <https://www.egyankosh.ac.in/bitstream/123456789/103488/1/Unit-8.pdf> - UNIT 8 - Urban Classes: Traders And Artisans, Extension Of Agricultural Settlements - Functions Of The Guilds - Page 95; [https://censusindia.gov.in/nada/index.php/catalog/27762/download/30931/DH\\_23\\_2001\\_MAN.pdf](https://censusindia.gov.in/nada/index.php/catalog/27762/download/30931/DH_23_2001_MAN.pdf) Census of India 2001 Mandsaur District - Brief History of the District - Page 6; <https://tehattagovtcollege.ac.in/Pdf/Resources/History/Online-learning/PDF-Book/INDIAN%20HISTORY%20ANCIENT%20TIME/A%20History%20of%20ancient%20an%20early%20medieval%20India%20by%20Upinder%20sing.pdf> - A History of Ancient and Early Medieval India by Upinder Singh - Chapter 8 - Interaction and Innovation, c. 200 BCE -300 CE - Crafts and Guilds - Page 655

Subject:) History

Topic:) Ancient History

Subtopic:)

Q.3)

Ans) d

Exp) Option d is the correct answer.

The **Harappan (Indus Valley) Civilization** covered a very vast geographical area. It extended across present-day **Pakistan, north-western India and parts of Afghanistan**. The total cultural zone is estimated to be between **680,000 and 800,000 square kilometres**, making it one of the largest ancient civilizations.

Archaeological sites of this civilization have been found in several regions such as:

- Punjab and Sindh in Pakistan
- Baluchistan and the North-West Frontier region
- Jammu, Punjab, Haryana, Rajasthan, Gujarat and western Uttar Pradesh in India

Archaeological discoveries help identify the extreme limits of this civilization:

- **Northern-most site:** Manda (Jammu and Kashmir)
- **Southern-most site:** Malvan (Gujarat)
- **Western-most site:** Sutkagen-dor (Makran coast, Pakistan)
- **Eastern-most site:** Alamgirpur (Saharanpur district, Uttar Pradesh)

These sites indicate the enormous spread of Harappan culture across the north-western part of the Indian subcontinent.

Knowledge Base:

- The largest Harappan cities were **Mohenjo-daro** and **Harappa**,
- Major port cities of the Harappan Civilization include **Lothal** etc

Source: <https://tehattagovtcollege.ac.in/Pdf/Resources/History/Online-learning/PDF-Book/INDIAN%20HISTORY%20ANCIENT%20TIME/A%20History%20of%20ancient%20an%20early%20medieval%20India%20by%20Upinder%20sing.pdf> - A History of Ancient and Early Medieval India by Upinder Singh - Chapter 4 - The Harappan Civilization, c . 2600-1900 BCE - Harappan, Indus, or Sindhu-Sarasvati Civilization? - Page 244

Subject: History

Topic: Ancient History

Subtopic:

Q.4)

Ans) a

Exp) Option a is the correct answer.

Sangam literature represents the **earliest body of Tamil literary works**. These texts were composed in **Old Tamil** between approximately **3rd century BCE and 3rd century CE**. They provide rich information about the **political, social, and cultural life of early South India**, especially about the **Chera, Chola, and Pandya kingdoms**.

The Sangam corpus mainly includes three major components:

- 1. Ettutokai (The Eight Collections)** - It consists of **eight separate collections of poems** composed by different poets. These poems deal with themes such as Love and human emotions, Warfare and heroism, Kingship and patronage, Social customs and everyday life
- 2. Pattuppattu (The Ten Songs)** - It contains **ten long poems**, each focusing on particular themes such as Praise of kings and chieftains, Descriptions of landscapes and trade routes, Warfare and political power, These poems also describe famous rulers such as **Karikala Chola**, whose victories over rival kings are celebrated in Sangam literature.
- 3. Tolkappiyam**, an early Tamil grammatical work which also discusses literary conventions.

Source: <https://tehattagovtcollege.ac.in/Pdf/Resources/History/Online-learning/PDF-Book/INDIAN%20HISTORY%20ANCIENT%20TIME/A%20History%20of%20ancient%20an%20early%20medieval%20India%20by%20Upinder%20sing.pdf> - A History of Ancient and Early Medieval India by Upinder Singh

Subject: History

Topic: Ancient History

Subtopic:

Q.5)

Ans) b

Exp) Option b is the correct answer.

Muhammad Ghori launched several military campaigns into the Indian subcontinent during the **second half of the 12th century CE**. At that time, North India was politically fragmented and ruled by several regional kingdoms such as the **Chauhans, Gahadavalas and Chaulukyas (Solankis) etc.**

The absence of political unity among these kingdoms played a major role in enabling the success of Ghori's campaigns.

**Option I is correct.** Prithviraj III (Prithviraj Chauhan) was the ruler of the **Chauhan dynasty** with his capital at **Ajmer and Delhi**. He fought Muhammad Ghori in the famous:

- **First Battle of Tarain (1191)** – Prithviraj Chauhan defeated Muhammad Ghori.
- **Second Battle of Tarain (1192)** – Muhammad Ghori defeated Prithviraj Chauhan, leading to the establishment of Turkish rule in North India.

**Option II is correct.** Jayachandra belonged to the **Gahadavala dynasty** and ruled from **Kannauj and Varanasi**. He was another powerful ruler in North India during the same period. Muhammad Ghori later defeated him in the **Battle of Chandawar (1194)**.

**Option III is correct.** Bhimdev II was a ruler of the **Chaulukya (Solanki) dynasty of Gujarat**. During Muhammad Ghori's campaign into Gujarat in **1178 CE**, Ghori was **defeated near Mount Abu (Battle of Kayadara)** by the forces of the Chaulukya kingdom.

**Option IV is incorrect.** Govindachandra was a ruler of the **Gahadavala dynasty**, but he ruled **earlier in the 12th century (c. 1114–1155 CE)**. He died before the major invasions of Muhammad Ghori began in North India.

**Option V is incorrect.** Dharmapala was a ruler of the **Pala dynasty of Bengal**, but he lived much earlier, during the **late 8th and early 9th century CE**. Thus, he did not rule during the period of Muhammad Ghori's invasions.

Source:) Tamil Nadu Board - Class 11 History - 2024 Edition - Chapter 8 - Harsha and Rise of Regional Kingdoms - Pala Rulers - Page 111, Chapter 10 - Advent of Arabs

and Turks - Muhammad Ghori - Page 138 and 139; NCERT History of Medieval India by Satish Chandra - Chapter 5 - The Age of Conflict (Circa 1000–1200) - The Rajput States - PAGE 78, The Turkish Conquest of North India - Page 83

Subject:) History

Topic:) Medieval History

Subtopic:)

Q.6)

Ans) c

Exp) Option c is the correct answer.

The period between **600 CE and 1200 CE** witnessed significant literary activity in Sanskrit, Prakrit and regional languages. Many important texts were written not only by scholars but also by **ruling kings themselves**, reflecting royal patronage of learning.

**Pair I is correct.** The work **Samarangana Sutradhara** is traditionally attributed to **Raja Bhoja**, the celebrated ruler of the **Paramara dynasty of Malwa in the 11th century CE**. Raja Bhoja is remembered in Indian historical tradition as a **scholar-king who displayed deep interest in several branches of knowledge such as architecture, grammar, medicine, astronomy and poetics**.

The text **Samarangana Sutradhara** deals extensively with **architecture, temple construction, town planning and sculpture**. It even contains fascinating discussions on **mechanical devices and automata**, reflecting the sophisticated technical imagination of the period.

**Pair II is correct.** **Kavirajamarga** is regarded as **one of the earliest extant works of literary criticism and poetics in the Kannada language**. It is generally attributed to **Amoghavarsha I**, the illustrious ruler of the **Rashtrakuta dynasty in the 9th century CE**.

Amoghavarsha I was known not only as a political ruler but also as a **patron of literature and learning**. The work **Kavirajamarga** discusses **poetic conventions, literary styles and the proper use of the Kannada language in literary composition**. It is therefore extremely important for understanding the **early development of classical Kannada literature**.

**Pair III is correct.** *Harshacharita* was composed by **Banabhatta**, who served as the **court poet of King Harsha in the 7th century CE**. The work is written in ornate Sanskrit prose and is widely regarded as **one of the earliest historical biographies in Sanskrit literature**.

The text narrates the **life and achievements of King Harsha**, while also providing valuable descriptions of **court life, social customs, religious practices and political conditions in early medieval North India**. Although it contains elements of literary embellishment typical of courtly writing, it remains an **important historical source for the period of Harsha's rule**.

**Pair IV is incorrect.** *Rajatarangini* is a celebrated Sanskrit chronicle that presents a **systematic historical account of the rulers of Kashmir**. However, this important work was composed by **Kalhana**, a Kashmiri Brahmana scholar, in the **12th century CE**. Kalhana's *Rajatarangini* is particularly significant because it attempts to **present history in a chronological and critical manner**, drawing upon earlier chronicles, inscriptions and local traditions.

On the other hand, **Bilhana** was another famous Sanskrit poet of the same broad period, but he is best known for composing **Vikramankadeva Charita**, a courtly biography praising the **Chalukya ruler Vikramaditya VI**.

Source:) [https://ignca.gov.in/Asi\\_data/64800.pdf](https://ignca.gov.in/Asi_data/64800.pdf); Tamil Nadu Board - Class 11 History - 2024 Edition - Chapter 8 - Harsha and Rise of Regional Kingdoms - Krishna I and His Successors - Page 113, Harsha as a Patron of Art and Literature - Page 114,

NCERT - History of Medieval India by Satish Chandra - Chapter 10 - Struggle for Empire in North India-I (Circa 1400-1525) - Northwest And North India-The Sharqis, The Lodi Sultans And Kashmir - Kashmir - Page 196;

[https://tehattagovtcollege.ac.in/Pdf/Resources/History/Online-learning/PDF-](https://tehattagovtcollege.ac.in/Pdf/Resources/History/Online-learning/PDF-Book/INDIAN%20HISTORY%20ANCIENT%20TIME/A%20History%20of%20ancient%20an%20early%20medieval%20India%20by%20Upinder%20sing.pdf)

[Book/INDIAN%20HISTORY%20ANCIENT%20TIME/A%20History%20of%20ancient%20an%20early%20medieval%20India%20by%20Upinder%20sing.pdf](https://tehattagovtcollege.ac.in/Pdf/Resources/History/Online-learning/PDF-Book/INDIAN%20HISTORY%20ANCIENT%20TIME/A%20History%20of%20ancient%20an%20early%20medieval%20India%20by%20Upinder%20sing.pdf) - A History of Ancient and Early Medieval India by Upinder Singh - Chapter 10 - Emerging Regional Configurations, c. 600-1200 CE - Sources, Literary and Archaeological - Page 879

Subject:) Art and Culture

Topic:) Literature

Subtopic:)

**Q.7)**

**Ans) d**

**Exp) Option d is the correct answer.**

Buddhist philosophy places great emphasis on understanding the **nature, origin and cessation of suffering (Dukkha)**. One of the most important doctrines that explains **why suffering arises in the world** is the principle known as **Pratityasamutpada**. This concept forms the philosophical foundation of the **Second Noble Truth**, which explains the **cause of suffering**.

The term **Pratityasamutpada** literally means **“dependent origination” or “arising in dependence upon causes.”** It explains that **nothing in the world exists independently; every phenomenon arises due to certain causes and conditions**.

In Buddhist teaching, suffering is explained through a **chain of twelve interconnected causes**, often referred to as the **Twelve Nidanas**. These links describe how ignorance leads to desire, attachment, rebirth and ultimately suffering.

The doctrine emphasises that **since suffering arises from causes, it can also cease when these causes are removed**. Therefore, by eliminating **ignorance and desire (tanha)** through ethical living, meditation and wisdom, a person can attain **Nirvana**, the state free from suffering.

Knowledge Base:

- The concept of **Anatta** means **“non-self” or “absence of a permanent soul.”** According to this doctrine, what we call a person is actually a **combination of constantly changing physical and mental elements**.
- **Sasvatavada** literally means **“the doctrine of eternalism.”** It is the belief that **the soul or self is permanent and unchanging** and continues eternally in the same form.



- The term **Ucchedavada** means “**the doctrine of annihilationism.**” It refers to the belief that **the individual self completely ceases to exist after death** and that there is no continuity of existence or moral consequences beyond this life.

Source:) <https://www.egyankosh.ac.in/bitstream/123456789/35230/1/Unit-4.pdf> - Unit 4 - The Philosophy of Buddhism - The Four Noble Truths - Page 1

Subject:) History

Topic:) Ancient History

Subtopic:)

Q.8)

Ans) b

Exp) Option b is the correct answer.

**Mahmud Gawan served as the Prime Minister (Wazir) and a leading military commander of the Bahmani Kingdom in the 15th century.** Originally a Persian merchant who came to the Deccan, he gradually rose in prominence due to his **administrative ability, diplomatic skill and military competence.**

Under his guidance, the Bahmani kingdom expanded its authority and successfully conducted several campaigns. **Mahmud Gawan led military expeditions that brought regions such as Hubli, Belgaum and Bagalkot under Bahmani control,** thereby strengthening the empire's position in the Deccan.

He also introduced important **administrative and revenue reforms**, reorganising the kingdom into **smaller and more manageable provinces** to improve governance. Unfortunately, due to **court intrigues and rivalry between the Deccani and Afaqi factions of nobles**, Mahmud Gawan was falsely accused of treason and executed in **1481 CE**. His death greatly weakened the Bahmani state and eventually contributed to its decline.

Knowledge Base:

- The **Vijayanagara Kingdom** was one of the most powerful Hindu empires in South India during the medieval period. It was founded in **1336 by Harihara and Bukka** and frequently came into conflict with the Bahmani rulers over control of strategic territories such as the **Tungabhadra Doab and the Raichur Doab.**
- The **Bijapur Sultanate** was one of the **five Deccan Sultanates that emerged after the disintegration of the Bahmani kingdom in the early 16th century.** It was ruled by the **Adil Shahi dynasty.**
- The **Golconda Sultanate**, ruled by the **Qutb Shahi dynasty**, was another successor state that emerged after the **break-up of the Bahmani kingdom.** It became famous later for its **diamond trade and the prosperity of Hyderabad-Golconda region.**

Source:) <https://egyankosh.ac.in/bitstream/123456789/73302/1/Unit-5.pdf> - Unit 5 - Deccan Kingdoms - Conquest and Consolidation of The Bahmani Power - Page 122

Subject:) History

Topic:) Medieval History

Subtopic:)

Q.9)

Ans) a

Exp) Option a is the correct answer.

The history of medieval Kashmir witnessed the rule of several important dynasties. Among them, **Zain-ul-Abidin stands out as one of the most liberal and enlightened rulers**, remembered for his policies of religious tolerance, cultural patronage and economic development.

**Zain-ul-Abidin (c. 1420-1470 CE)** was a ruler of the **Shah Mir dynasty of Kashmir** and is widely remembered by the title “**Bud Shah**” meaning “**The Great King.**” Due to his tolerant and inclusive policies, later historians often referred to him as the “**Akbar of Kashmir.**”

He reversed the harsh policies of his predecessor **Sikandar Shah**, who had pursued intolerant measures against non-Muslims. **Zain-ul-Abidin invited back those groups who had fled Kashmir and allowed people who had been forcibly converted to return to their original faith.** He also **abolished the jaziya tax and prohibited cow slaughter**, which helped restore social harmony in the region.

A remarkable aspect of his reign was his **patronage of learning and scholarship**. Zain-ul-Abidin himself was well-versed in **Persian, Sanskrit, Kashmiri and Tibetan**. Under his patronage, several important Sanskrit works were translated into Persian. Notably, he **ordered the translation of the Mahabharata and Kalhana's Rajatarangini (the celebrated historical chronicle of Kashmir) into Persian**, thereby making these works accessible to a wider intellectual audience.

Knowledge Base:

- **Sikandar Shah**, who ruled Kashmir before Zain-ul-Abidin, is often remembered in historical sources for his **intolerant religious policies**. Because of his actions against temples and non-Muslim communities, he earned the title **"Sikandar Butshikan" (destroyer of idols)**.
- **Lalitaditya Muktapida** was a famous ruler of the **Karkota dynasty of Kashmir** who ruled in the **8th century CE**, much earlier than the medieval period discussed here. He is known for **military conquests and the construction of the grand Martand Sun Temple**, one of the finest examples of ancient Kashmiri architecture.
- **Yusuf Shah Chak** was a later ruler of Kashmir belonging to the **Chak dynasty in the 16th century**. His reign came just before the **annexation of Kashmir by the Mughal emperor Akbar in 1586**.

Source:) <https://egyankosh.ac.in/bitstream/123456789/73296/1/Unit-1.pdf> - UNIT 1 - Survey of Sources - Sanskrit Literature - Sanskrit Translations in Persian - Page 38;  
[https://digital.nios.ac.in/content/315en/315\\_History\\_Eng\\_Lesson9.pdf](https://digital.nios.ac.in/content/315en/315_History_Eng_Lesson9.pdf) - Chapter 9 - Establishment and Expansion of the Delhi sultanate - Kashmir - Page 149; Ancient and Medieval India by Poonam Dahiya - Chapter 12 - Provincial Kingdoms Of Medieval India - North India: Kashmir - Zain-ul-Abidin (c. 1420-1470 CE) - Page 625

Subject:) History

Topic:) Medieval History

Subtopic:)

Q.10)

Ans) d

Exp) Option d is the correct answer.

The constitutional history of British India witnessed several legislative reforms introduced by the British Parliament. These Acts gradually expanded administrative participation but often retained ultimate authority in British hands.

The **Government of India Act, 1935** was the **most comprehensive constitutional reform introduced by the British in India** before independence. One of its most important provisions was the introduction of **provincial autonomy**.

Under this system, **provinces were granted the authority to function with elected ministries responsible to the provincial legislatures**. The earlier system of **dyarchy in provinces was abolished**, and the provincial governments received control over subjects such as **education, agriculture, local government and public health**.

Although the British Governors retained certain **special powers and safeguards**, the Act allowed Indian leaders to form ministries in the provinces after the **1937 elections**. This development significantly expanded **Indian participation in governance at the provincial level**.

Knowledge Base:

- The **Government of India Act, 1858** was passed after the **Revolt of 1857** and marked the **transfer of power from the East India Company to the British Crown**. It established the **office of the Secretary of State for India** and brought India directly under British government control.
- The **Indian Councils Act, 1909**, also known as the **Morley-Minto Reforms**, mainly focused on **expanding legislative councils and introducing limited electoral representation**. A notable feature of this Act was the **introduction of separate electorates for Muslims**, which had long-term political consequences for Indian politics.

- The **Government of India Act, 1919**, commonly known as the **Montagu–Chelmsford Reforms**, introduced the system of **dyarchy in the provinces**. Under dyarchy, provincial subjects were divided into “**reserved**” subjects controlled by the Governor and “**transferred**” subjects administered by Indian ministers responsible to the legislature. Since real authority still remained with the Governor for key areas such as finance and law and order, **full provincial autonomy was not granted**.

Source:) <https://egyankosh.ac.in/bitstream/123456789/19086/1/Unit-4.pdf> - Unit 4 - Administrative System under 1935 Act - Provincial Autonomy - Page 52

Subject:) History

Topic:) Modern History

Subtopic:)

Q.11)

Ans) a

Exp) Option a is the correct answer.

The **84th Constitutional Amendment Act (2001)** amended **Articles 82 and 170** of the Constitution and provided that the **total number of seats** in the **Lok Sabha** and the **State Legislative Assemblies** shall **remain frozen** until the **first Census conducted after the year 2026**.

- The objective of this provision was to ensure that states which had successfully implemented **population control measures** were not disadvantaged in terms of parliamentary representation compared to states with higher population growth.
- However, the amendment allowed readjustment of territorial constituencies within states on the basis of the **1991 Census**, without altering the total number of seats.
- Later, the **87th Constitutional Amendment Act (2003)** further modified this provision by allowing delimitation of constituencies based on the **2001 Census**, while **still retaining the freeze** on the total number of Lok Sabha seats until the Census after 2026.

Thus, the amendment responsible for continuing the freeze on Lok Sabha seats until the Census after 2026 is the 84th Constitutional Amendment Act (2001).

Source:) <https://www.thehindu.com/news/national/census-may-be-held-next-year-to-match-delimitation-schedule-says-minister-kishan-reddy/article69522508.ece>

<https://www.thehindu.com/news/national/centre-announces-population-census-caste-to-be-counted/article69656894.ece>

<https://www.thehindu.com/news/national/andhra-pradesh/delimitation-ensure-justice-to-all-states-by-making-constitutional-amendment-jagan-tells-pm/article69361195.ece>

Subject:) Polity

Topic:) Constitutional Amendment Acts

Subtopic:)

Q.12)

Ans) b

Exp) Option b is the correct answer.

**Statement I is correct:** The **Preamble** to the Constitution of India is **neither a source of power for the legislature nor a limitation or prohibition upon its powers**. It primarily serves as an introductory statement reflecting the philosophy, objectives, and guiding principles of the Constitution. This interpretation was affirmed by the Supreme Court in the **Kesavananda Bharati v. State of Kerala**, which clarified that the Preamble is part of the Constitution but not an independent source of power.

**Statement II is correct:** The **Preamble is non-justiciable**, meaning it **cannot be enforced in a court of law by itself**. No legal remedy can be claimed solely on the basis of the Preamble. However, courts often use it as an interpretative aid to understand the spirit and objectives of the Constitution when interpreting constitutional provisions.



**Why Statement II does not explain Statement I:**

- The fact that the Preamble is non-justiciable only means that it cannot be directly enforced in courts. This does not explain why the Preamble is not a source of legislative power.
- These are two **independent and distinct characteristics of the Preamble**.
- This position was clarified by the Supreme Court in **Kesavananda Bharati v. State of Kerala**, which held that the Preamble is a part of the Constitution and can guide interpretation, but it does not grant powers by itself.

Therefore, **Both Statement I and Statement II are correct but Statement II does not explain Statement I.**

Source:) Mastering Indian Polity by Forum IAS, Chapter: 5 - PREAMBLE OF THE INDIAN CONSTITUTION, Page: 37.

[https://digital.nios.ac.in/content/338en/338\\_Introduction\\_To\\_Law\\_Eng\\_L18.pdf](https://digital.nios.ac.in/content/338en/338_Introduction_To_Law_Eng_L18.pdf)

Subject:) Polity

Topic:) Preambles of the Constitution

Subtopic:)

**Q.13)**

**Ans) c**

**Exp) Option c is the correct answer.**

The **Sixth Schedule** of the Constitution contains provisions relating to the **administration of Tribal Areas in the states of Assam, Meghalaya, Tripura and Mizoram**. These provisions are made under **Article 244(2)** and **Article 275(1)** of the Constitution.

The Schedule provides for the establishment of **Autonomous District Councils** and **Regional Councils** to ensure a significant degree of self-governance for tribal communities.

**Option a is incorrect: Autonomous District Councils:**

The **Sixth Schedule (Articles 244(2) and 275(1))** provides for the **establishment of Autonomous District Councils (ADCs)** for the administration of Tribal Areas in Assam, Meghalaya, Tripura and Mizoram.

- These councils are vested with legislative, administrative and limited judicial powers over specified subjects such as land, forests (other than reserved forests), shifting cultivation, village administration and social customs of tribes.

**Option b is incorrect: Regional Councils:**

The **Sixth Schedule** also provides for the **creation of Regional Councils within an Autonomous District** where different tribal communities inhabit distinct regions. These councils are meant to ensure separate representation and self-governance for specific tribal groups within the district.

**Option c is correct: Tribes Advisory Council:**

The **Tribes Advisory Council (TAC)** is **not part of the Sixth Schedule**. It is provided under the **Fifth Schedule** of the Constitution, which deals with the administration of Scheduled Areas in states other than those covered by the Sixth Schedule. The TAC advises the Governor on matters relating to the welfare and advancement of Scheduled Tribes in such areas.

**Option d is incorrect: Tribal Areas**

The **title of the Sixth Schedule** itself is "**Provisions as to the Administration of Tribal Areas** in the States of Assam, Meghalaya, Tripura and Mizoram."

Thus, the **term "Tribal Areas"** is **explicitly used in the Sixth Schedule**. In contrast, the Fifth Schedule uses the term "Scheduled Areas", not "Tribal Areas."

Source:) <https://www.mea.gov.in/images/pdf1/s5.pdf>

<https://www.mea.gov.in/Images/pdf1/S6.pdf>

Subject:) Polity

Topic:) Schedules of the Constitution

Subtopic:)

Q.14)

Ans) d

Exp) Option d is the correct answer.

**Article 20** and **Article 22** of the Constitution of India both **provide safeguards in the criminal justice process**, but they deal with different subject matters.

- **Article 20** provides **protection in respect of conviction for offences**, i.e., safeguards available during trial and punishment for a crime.
- **Article 22**, on the other hand, **provides protection against arbitrary arrest and detention**, i.e., safeguards available immediately after a person is arrested or detained.

**Option a is incorrect:** This provision relates to **protection against ex post facto laws under Article 20(1) of the Constitution**. It means that a person cannot be convicted for an act that was not an offence under the law at the time when it was committed, nor can a heavier penalty be imposed than what was prescribed at that time.

**Option b is incorrect:** This protection relates to the **principle of double jeopardy under Article 20(2) of the Constitution**. It ensures that no person shall be prosecuted and punished for the same offence more than once, thereby protecting individuals from repeated trials and punishments for the same offence.

**Option c is incorrect:** This protection is **provided under Article 20(3) of the Constitution** and relates to **self-incrimination**. It states that no person accused of an offence shall be compelled to be a witness against himself, meaning the accused cannot be forced to give evidence or testimony that could incriminate him.

**Option d is correct:** **This right is specifically guaranteed under Article 22(1) of the Constitution**. It provides that a **person who is arrested must be informed**, as soon as possible, **of the grounds for such arrest**, and also has the **right to consult and be defended by a legal practitioner of his/her choice**. This safeguard ensures protection against arbitrary arrest and detention.

Source:) Mastering Indian Polity by Forum IAS, Chapter: 10 - FUNDAMENTAL RIGHTS, Page: 126 & 137.

<https://www.thehindu.com/opinion/lead/the-danger-of-an-unchecked-pre-crime-framework/article70140459.ece>

<https://www.thehindu.com/news/national/communication-of-grounds-of-arrest-in-writing-in-accuseds-language-is-a-must-supreme-court/article70249585.ece>

Constitution of India

Subject:) Polity

Topic:) Fundamental Rights

Subtopic:)

Q.15)

Ans) b

Exp) Option b is the correct answer.

**Fundamental Duties** are the **moral and civic obligations of citizens of India**, intended to promote a sense of responsibility and patriotism among them. They are **contained in Part IV-A** of the Constitution under **Article 51A**.

- These duties were not originally included in the Constitution. They were **incorporated by the 42nd Constitutional Amendment Act** on the recommendation of the **Swaran Singh Committee**. Later, one more duty was added by the **86th Constitutional Amendment Act**.
- At present, the Constitution contains **11 Fundamental Duties**, which apply to citizens of India.
- Unlike Fundamental Rights, **Fundamental Duties are non-justiciable**, meaning they are not directly enforceable by courts, but they serve as important guiding principles for responsible citizenship and for the interpretation of laws by the judiciary.

**Option I is incorrect:** **To raise the level of nutrition and the standard of living and to improve public health.**

**This is a Directive Principle of State Policy under Article 47 of the Constitution of India in Part IV of the Constitution of India, not a Fundamental Duty.**

**Option II is correct:** **To safeguard public property and to abjure violence.**

**This is a Fundamental Duty under Article 51A(i) of the Constitution of India.**

**Option III is correct:** To protect and improve the natural environment including forests, lakes, rivers and wildlife, **and to have compassion for living creatures.**

This is also a **Fundamental duty** provided under **Article 51A(g)** of the Indian Constitution.

**Option IV is incorrect:** To provide early childhood care and education for all children until they complete the age of six years.

This is a **Directive Principle under Article 45** of the Constitution of India (after the 86th Constitutional Amendment Act, 2002), not a Fundamental Duty.

The **Fundamental Duty added by the 86th Constitutional Amendment Act is different**; it requires parents or guardians to provide opportunities **for education** to his child or, as the case may be, ward **between the age of six and fourteen years.**

**Option V is correct:** To strive towards excellence in all spheres of individual and collective activity.

This is a **Fundamental Duty under Article 51A(j).**

It states “to strive towards excellence in all spheres of individual and collective activity so that the nation constantly rises to higher levels of endeavour and achievement”.

Source:) Mastering Indian Polity by Forum IAS, Chapter: 12 - Fundamental Duties, Page: 186.

<https://dsel.education.gov.in/sites/default/files/Fundamental-Duties-English.pdf>

<https://www.mea.gov.in/images/pdf1/part4.pdf>

Subject:) Polity

Topic:) Fundamental Duties

Subtopic:)

**Q.16)**

**Ans) d**

**Exp) Option d is the correct answer.**

**Parliamentary privileges** are **special rights, immunities and exemptions enjoyed by each House of Parliament**, its members and committees, which enable them to perform their legislative functions independently and without interference. These privileges are primarily provided under **Article 105** of the Constitution (**for Parliament**) and **Article 194** (for **State Legislatures**).

- **Article 105(1):** Freedom of speech in Parliament.
- **Article 105(2):** Immunity from court proceedings for anything said or any vote given in Parliament or its committees.
- **Article 105(3):** Other privileges of the House and its members.

**Statement I is correct:** Members of Parliament enjoy freedom from arrest in civil cases during the session of Parliament and for forty days before and forty days after the session.

This privilege ensures that members can attend parliamentary proceedings without obstruction. However, **this immunity does not apply to criminal arrest or preventive detention.**

**Statement II is correct:** Parliament has not enacted any comprehensive law to codify parliamentary privileges in India.

According to **Article 105(3)**, until such privileges are defined by law, the powers, privileges and immunities of each House of Parliament and its members are those that existed in the House of Commons of the United Kingdom at the commencement of the Constitution.

**Statement III is correct:** A question of privilege can be raised by a member only after giving prior notice to the Speaker (Lok Sabha) or the Chairman (Rajya Sabha).

The presiding officer decides whether the motion should be admitted and may refer the matter to the Committee of Privileges for examination and report. The House ultimately decides the matter.

Source:) Mastering Indian Polity by Forum IAS, Chapter: 16 - Legislative Branch, Page: 364.

<https://www.thehindu.com/opinion/op-ed/protecting-the-freedom-of-speech-of-mps/article70663727.ece>

<https://www.thehindu.com/news/national/jmm-bribery-case-sc-overrules-majority-verdict-says-lawmakers-who-indulge-in-graft-have-no-immunity/article67912361.ece>

<https://www.thehindu.com/news/national/no-privilege-to-mps-from-criminal-proceedings-during-session-naidu/article65733637.ece>

<https://www.thehindu.com/news/national/parliamentary-privilege-breach-jagdeep-dhankhar-rajya-sabha-explained/article66548815.ece>

Subject:) Polity

Topic:) Parliamentary Privileges

Subtopic:)

Q.17)

Ans) b

Exp) Option b is the correct answer.

**Statement I is correct:** Under **Article 341(1)** of the Constitution, the **President specifies** the castes, races or tribes that shall be deemed **to be Scheduled Castes** for a particular State or Union Territory. **In the case of a State**, this notification is issued after **consultation with the Governor** of the concerned State.

The initial notification is issued by the President, and any **subsequent inclusion/exclusion requires a law by Parliament** under **Article 341(2)**.

**Statement II is incorrect:** The **Constitution does not provide** a sociological **definition of Scheduled Castes** based on the **historical practice of untouchability**. Instead, **Article 366(24)** defines Scheduled Castes simply as those castes, races or tribes that are deemed to be Scheduled Castes under Article 341.

**Statement III is correct:** The **Constitution provides reservation of seats for Scheduled Castes in legislatures:**

- **Article 330** provides for reservation of seats for Scheduled Castes in the **House of the People (Lok Sabha)**.
- **Article 332** provides for reservation of seats for Scheduled Castes in the **Legislative Assemblies of the States**.

**Knowledge base:** In the landmark **2024 judgment (State of Punjab v. Davinder Singh)**, a seven-judge Constitution Bench of the **Supreme Court** headed by the Chief Justice of India **upheld the power of States to create sub-classifications within the SC category**. This **overruled the earlier E.V. Chinnaiah (2004) judgment**, which had disallowed such sub-classification.

- The court ruled that **this does not violate Article 341** or the principle of equality.
- The Court held that States can create sub-quotas within SCs to ensure equitable distribution of reservation benefits, especially for more backward groups within SCs.

Source:) <https://www.thehindu.com/news/national/cji-gavai-backs-exclusion-of-creamy-layer-in-reservation-to-scheduled-castes/article70286763.ece>

<https://www.thehindu.com/news/national/sc-notice-to-centre-on-plea-to-introduce-a-system-similar-to-creamy-layer-concept-for-scst-reservations/article69927216.ece>

<https://egyankosh.ac.in/bitstream/123456789/10142/1/Unit%201.pdf> Page: 6

<https://frontline.thehindu.com/social-issues/social-justice/subcategorisation-of-scst-community-creamy-layer-new-reservation-policy/article68525021.ece>

<https://www.constitutionofindia.net/articles/article-342-scheduled-tribes/>

<https://www.thehindu.com/news/national/why-has-supreme-court-upheld-states-right-to-sub-categorise-scheduled-castes-for-quota-benefits/article68472172.ece>

Subject:) Polity

Topic:) Constitution of India

Subtopic:)

Q.18)

Ans) c

Exp) Option c is the correct answer.

The **Election Commission of India (ECI)** is an **independent constitutional authority** established under **Article 324** of the Constitution to **ensure the free and fair conduct of elections in the country**.

- It is responsible for the **superintendence, direction and control of elections** to the Lok Sabha, Rajya Sabha, State Legislative Assemblies and Councils, and the offices of the President and Vice-President.
- The Commission currently functions as a **three-member body** comprising the **Chief Election Commissioner (CEC)** and **two Election Commissioners**, who are **appointed by the President of India**.

**Statement I is correct:** The ECI has the **authority to recognize political parties and allot election symbols to them under the Election Symbols (Reservation and Allotment) Order, 1968**, which was issued by the Commission in exercise of its powers under Article 324 and the Representation of the People Act, 1951.

- This order empowers the Commission to classify parties as national or state parties and allocate reserved election symbols to them.
- The Commission also **decides symbol disputes** between rival factions within a political party.

**Statement II is incorrect:** The **Chief Election Commissioner does not possess overriding powers over the other Election Commissioners**. In a multi-member Commission, all members have **equal decision-making power**, and **decisions are taken by consensus** or, if necessary, **by majority**. The CEC acts primarily as **first among equals**, not as a superior authority.

**Statement III is correct:** The **Election Commission advises the President or the Governor on the disqualification of legislators**.

- Under **Article 103** of the Constitution of India, the President decides questions of disqualification of Members of Parliament after obtaining the opinion of the ECI.
- Under **Article 192** of the Constitution of India, the Governor decides disqualification of Members of State Legislatures, again based on the ECI's opinion.

Source:) Mastering Indian Governance by Forum IAS, Chapter: 7 – Constitutional Bodies, Page: 105.

<https://www.thehindu.com/news/national/how-are-symbols-allotted-to-political-parties-explained/article68029670.ece>

<https://www.thehindu.com/news/national/on-the-independence-of-the-election-commission/article70666857.ece>

<https://indianexpress.com/article/what-is/what-is-article-324-in-the-indian-constitution-5729800/>

Subject:) Polity

Topic:) Election Commission of India

Subtopic:)

Q.19)

Ans) a

Exp) Option a is the correct answer.

The **Sustainable Harnessing and Advancement of Nuclear Energy for Transforming India (SHANTI) Act, 2025** is a landmark legislation enacted to **modernise India's civil nuclear energy governance framework**.

**Statement I is correct:** The **SHANTI Act, 2025** consolidates India's civil nuclear legal framework by **repealing the Atomic Energy Act, 1962** and the **Civil Liability for Nuclear Damage Act, 2010**. By replacing these earlier laws, it establishes a single comprehensive legal regime governing nuclear energy development, regulation, and liability in India's civil nuclear sector.

**Statement II is correct:** For the first time, the **Act permits private sector participation in the construction and operation of nuclear power plants**, ending the Union government's exclusive control over the sector. However, the **strategic and sensitive activities** such as enrichment of nuclear materials, production of heavy water, and management of spent nuclear fuel **remain under the control of the Central Government**.

The **Act modifies India's nuclear liability framework**. It **removes or significantly restricts supplier liability** that previously existed under the Civil Liability for Nuclear Damage Act, 2010. At the same time, it **introduces a capped operator liability**, with the liability limit linked to the capacity of the nuclear reactor.

- It **omits the "right of recourse"** that **allows operators to sue suppliers for accidents** caused by defective equipment.
- The operator's liability is capped between ₹100 crore for small plants and ₹3,000 crore for the largest plants.



**Statement III is incorrect:** The Act does not transfer authority to State Governments. Instead, it **strengthens central oversight** by **granting full statutory status to the Atomic Energy Regulatory Board (AERB)**. Previously, the AERB functioned under executive orders; under the SHANTI Act, it is an independent statutory body directly accountable to the Parliament.

Source:) <https://www.thehindu.com/news/national/a-case-against-the-shanti-act-explained/article70623943.ece>

<https://www.thehindu.com/news/national/what-does-the-shanti-bill-change-explained/article70473931.ece>

<https://www.thehindu.com/opinion/op-ed/when-shanti-means-nuclear-energy/article70584823.ece>

<https://www.thehindu.com/news/national/need-to-find-balance-between-visible-national-interest-and-hypothetical-loss-in-case-of-nuclear-accident-says-sc-on-shanti-act/article70685367.ece>

[https://prsindia.org/files/bills\\_acts/bills\\_parliament/2025/Sustainable\\_Harnessing\\_and\\_Advancement%20of\\_Nuclear\\_Energy\\_Bill,2025.pdf](https://prsindia.org/files/bills_acts/bills_parliament/2025/Sustainable_Harnessing_and_Advancement%20of_Nuclear_Energy_Bill,2025.pdf)

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2206598&reg=3&lang=1>

Subject:) Polity

Topic:) Acts/Bills

Subtopic:)

Q.20)

Ans) b

Exp) Option b is the correct answer.

**Statement I is correct:** In 2024, a nine-judge Constitution Bench of the **Supreme Court in Property Owners Association v. State of Maharashtra clarified the scope of Article 39(b)** and held that **not every privately owned resource can be treated as a “material resource of the community”** to be **used by the government to serve the ‘common good.’**

The Court observed that whether a resource qualifies as a material resource must be determined case-by-case, considering factors such as the nature of the resource, its scarcity, and its importance to the community. Hence, the State cannot automatically treat all private property as community resources for acquisition or redistribution.

**Statement II is correct: Article 39(b)** (a Directive Principle of State Policy) provides that the State shall direct its policy towards securing that the **ownership and control of the material resources of the community are distributed** so as to **best subserve the common good.**

**Why Statement II does not explain Statement I:**

Statement II merely states the constitutional provision under Article 39b), whereas Statement I refers to the Supreme Court’s interpretation that limits the scope of this provision, clarifying that all private property cannot automatically be treated as a material resource of the community. Therefore, **Statement II does not explain Statement I.**

Hence, **Both Statement I and Statement II are correct but Statement II does not explain Statement I**

Source:) <https://www.thehindu.com/news/national/private-properties-cannot-be-taken-over-by-government-supreme-court/article68831508.ece>

<https://www.thehindu.com/news/national/can-all-private-properties-be-acquired-by-the-state-explained/article68840840.ece>

<https://indiankanoon.org/doc/555882/>

Subject:) Polity

Topic:) Directive Principles of State Policy (DPSPs)

Subtopic:)

Q.21)

Ans) c

Exp) Option c is the correct answer.

**Statement I is correct: Sodium-ion batteries** are considered a **promising alternative to lithium-ion batteries, particularly for large-scale energy storage applications** such as **grid storage** and **renewable energy systems.**

This is mainly because sodium is **widely available, inexpensive, geographically abundant**, unlike lithium which is relatively scarce and concentrated in limited regions. It also offers **improved safety, and good performance at low temperatures**. These factors make sodium-ion technology attractive for cost-effective stationary energy storage.

**Statement II is incorrect: Sodium-ion batteries do not have higher energy density than lithium-ion batteries.** In fact, their energy density is generally lower.

This is because sodium ions are larger and heavier than lithium ions, which limits how much energy can be stored per unit mass. As a result, **lithium-ion batteries remain more suitable for applications where high energy density is required**, such as **electric vehicles and portable electronics**.

Hence, **Statement I is correct but Statement II is incorrect**.

Source:)

<https://indianexpress.com/article/upsc-current-affairs/upsc-essentials/sodium-ion-batteries-represent-more-than-just-a-technological-shift-for-india-10547782/>

<https://www.thehindu.com/sci-tech/science/rethinking-battery-strategy-in-india-the-case-for-sodium-ion-technology/article70595967.ece#:~:text=India's%20dependence%20on%20lithium%2Dion,potential%20to%20strengthen%20energy%20security>

[thehindubusinessline.com/news/science/tcg-crest-scientists-develop-sodium-ion-battery-with-5-minute-charge-capability/article70169769.ece](https://thehindubusinessline.com/news/science/tcg-crest-scientists-develop-sodium-ion-battery-with-5-minute-charge-capability/article70169769.ece)

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2129649&reg=3&lang=2>

Subject: Environment

Topic: Energy

Subtopic:)

Q.22)

Ans) c

Exp) Option c is the correct answer.

**Statement I is correct:** The International Union for Conservation of Nature (IUCN) conducted the first Green Status of Species assessment of the Gharial (*Gavialis gangeticus*) and classified it as “Critically Depleted.”

This category indicates that although the species survives in the wild and conservation actions have helped prevent its extinction, it is still very far from full ecological recovery across its historical range.

**Statement II is incorrect:** The IUCN Green Status of Species does not primarily evaluate the risk of extinction. That is the **primary role of the IUCN Red List of Threatened Species** (which classifies the gharial as **Critically Endangered** based on extinction risk criteria like population decline and threats).

The **Green Status complements the Red List** by **measuring a species' progress toward full ecological recovery** and the impact of conservation actions rather than extinction risk.

**Statement III is correct:** Although **conservation initiatives** like "Project Crocodile" and the establishment of the National Chambal Sanctuary have **successfully prevented the Gharial from going extinct**.

However, the **species remains far from full ecological recovery across most of its historical range**. It remains **absent from nearly 98% of its historical river systems** and is currently restricted to small, fragmented pockets, meaning it is far from full ecological recovery.

**This condition is precisely what leads to its classification as “Critically Depleted” under the IUCN Green Status framework.**

Therefore, **only Statement III is correct and it explains Statement I, while Statement II is incorrect**.

Hence, the **correct answer is c): Only one of the Statements II and III is correct and that explains Statement I**

Source:)

<https://timesofindia.indiatimes.com/city/meerut/depleted-river-dependent-iucns-first-gharial-green-status-assessment-calls-for-river-protection-to-save-species/articleshow/124421785.cms>

<https://www.iucnredlist.org/about/green-status-species>

Subject: Environment

Topic: Conservation Measures

Subtopic:)

Q.23)

Ans) b

Exp) Option b is the correct answer.

'Global Geoparks' are part of the UNESCO Global Geoparks Programme, an initiative of the United Nations Educational, Scientific and Cultural Organization (UNESCO).

- These are **single, unified geographical areas** where sites and landscapes of international geological significance are managed with a holistic concept of protection, education, and sustainable development.
- The programme promotes conservation of geological heritage, geoscience education, and sustainable local development, particularly through geotourism.
- The formal label "UNESCO Global Geopark" was **endorsed by UNESCO Member States in 2015**.

Knowledge Base: **Global Status of UNESCO Global Geoparks**

- As of **2025–2026**, there are **229 UNESCO Global Geoparks across 50 countries**.

**Examples of well-known geoparks worldwide include:**

- Zhangjiajie Global Geopark (China)
- Khorat Global Geopark (Thailand)
- Arouca Global Geopark (Portugal)
- Jeju Island Global Geopark (South Korea)

**Status of Global Geoparks in India**

- **India currently has no UNESCO Global Geopark.**
- However, India has many geo-heritage sites and is working toward establishing geoparks due to its rich geological diversity.

Source:) <https://www.unesco.org/en/igpp/geoparks/about>

Subject:) Environment

Topic:) Conservation Measures

Subtopic:)

Q.24)

Ans) b

Exp) Option b is the correct answer.

A **Keystone species** is a species that has a **disproportionately large influence on the structure and functioning of an ecosystem** relative to its abundance. The presence of such species helps **maintain ecological balance, species diversity, and ecosystem stability**.

- If a keystone species is removed, it can trigger significant changes in the ecosystem, often leading to trophic cascades, reduction in biodiversity, or even ecosystem collapse.
- **Keystone species may function as top predators, ecosystem engineers, or key mutualists** that regulate populations and shape habitats.
- The **concept was first proposed by Robert T. Paine** in 1969 through his studies on *Pisaster* in intertidal ecosystems.

**Option I is correct: Sea Otter**

**Sea otters** are a **classic example of a keystone species in kelp forest ecosystems**. It controls populations of sea urchins, which would otherwise overgraze kelp forests. Without sea otters, urchin populations explode and create "urchin barrens," destroying the entire habitat for hundreds of other species. By regulating sea urchins, sea otters help maintain the kelp ecosystem.

**Option II is correct: Beaver**

The **Beaver is considered a keystone species** because it significantly modifies habitats by building dams. These dams create wetlands that support many plants, fish, amphibians, birds, and mammals. Hence, it is **also termed an ecosystem engineer**.

**Option III is incorrect: Water Hyacinth**

While it has a massive impact on ecosystems (especially in India, where it is known as the "Terror of Bengal"), it is an **Invasive Species**. A **keystone species maintains the balance of an ecosystem**; an invasive species like **Water Hyacinth destroys it** by clogging water bodies, depleting oxygen, and killing native flora and fauna.

**Option IV is incorrect: Indian Star Tortoise**

The Indian star tortoise is a herbivorous reptile **found in dry grasslands and scrub forests of the Indian subcontinent**. While ecologically relevant, it does not play a disproportionately large ecological role, and therefore **it is not classified as a keystone species**.

**Option V is correct: Starfish**

**Certain species of Starfish** (especially the Pisaster sea stars) are **classic keystone predators** in rocky intertidal ecosystems. By preying on mussels, they prevent mussels from dominating the habitat, thereby maintaining species diversity.

Therefore, **three species (Sea otter, Beaver, and Starfish) are keystone species**.

Source:) <https://www.thehindu.com/children/guardians-of-our-lakes-and-rivers/article69429691.ece>

<https://www.thehindu.com/sci-tech/science/seals-birds-under-threat-in-new-red-list-of-endangered-species/article70157826.ece>

<https://www.thehindu.com/news/international/worrying-biodiversity-loss-in-finnish-coastal-waters-report/article67980640.ece>

<https://timesofindia.indiatimes.com/life-style/spotlight/web-stories/10-keystone-species-you-should-know-about/photostory/117636114.cms>

<https://www.thehindu.com/sci-tech/energy-and-environment/study-brings-indian-star-tortoise-to-evidence-based-conservation/article68961578.ece>

<https://egyankosh.ac.in/bitstream/123456789/85706/1/Unit-7.pdf> Page: 141.

Subject:) Environment

Topic:) Species Diversity

Subtopic:)

**Q.25)**

**Ans) a**

**Exp) Option a is the correct answer.**

The **Ramsar Convention on Wetlands** is an **international treaty adopted in 1971 at Ramsar, Iran**, for the **conservation and sustainable use of wetlands**.

- Wetlands recognized under this convention are called **Ramsar Sites or Wetlands of International Importance**. These sites are identified for their ecological significance, biodiversity value, and role as habitats for migratory waterbirds.
- **India became a party to the Ramsar Convention in 1982**. The **first Ramsar Sites in India** were **Chilika Lake (Odisha)** and **Keoladeo National Park (Rajasthan)**, designated in 1981.
- **As of 2026, India has a total of 98 Ramsar Sites**, covering more than 13.8 lakh hectares, making India the **country with the largest number of Ramsar Sites in Asia**.

**Option a is correct: Pulicat Lake**

Pulicat Lake is a **large brackish water lagoon** located on the **border of Andhra Pradesh and Tamil Nadu** and is the **second-largest brackish lagoon in India after Chilika Lake**. It is an important habitat for migratory birds such as flamingos and supports rich estuarine biodiversity. However, **it has not been designated as a Ramsar Site in India**, even though it has high ecological importance.

**Option b is incorrect: Nanda Lake**

**Nanda Lake was designated as a Ramsar Site in June 2022**. It is **located in Curchorem, Goa**, and is the **state's first and only Ramsar wetland**. It is a **freshwater wetland** known for supporting rich biodiversity including aquatic plants, birds, and amphibians.

**Option c is incorrect: Wular Lake**

**Wular Lake**, located in **Jammu and Kashmir**, is **one of the largest freshwater lakes in Asia**. It plays an important role in flood control, fisheries, and maintaining the hydrological regime of the Jhelum basin. Due to its ecological significance, it was **designated as a Ramsar Site in 1990**.

**Option d is incorrect: Chilika Lake**

A **brackish water lagoon in Odisha** and the **largest coastal lagoon in India**. It was **designated a Ramsar Site in 1981** and later became the **first site in Asia to be removed from the Montreux Record** in 2002 after successful restoration. It is globally known for large congregations of migratory birds and the presence of the **Irrawaddy dolphin**.

Source:) <https://www.ramsar.org/country-profile/india>

<https://timesofindia.indiatimes.com/city/goa/nanda-lake-in-curchorem-is-states-first-ramsar-site/articleshow/93332412.cms>

<https://frontline.thehindu.com/environment/conservation/photo-essay-the-importance-of-protecting-pulicat-lake/article38011059.ece>

Subject:) Environment

Topic:) Wetlands

Subtopic:)

Q.26)

Ans) b

Exp) Option b is the correct answer.

The **Global Methane Status Report 2025** provides a **comprehensive assessment of global methane emissions**, their sectoral sources, and mitigation opportunities. It examines **methane emission trends across major sectors** such as **agriculture, energy, and waste**, and evaluates global efforts to reduce methane as part of climate change mitigation.

**Statement I is correct:** The **Global Methane Status Report 2025** is **published by the United Nations Environment Programme (UNEP)** in collaboration with the **Climate and Clean Air Coalition (CCAC)**. It was launched on the sidelines of **COP30 in Belem, Brazil**. It assesses global methane emissions and mitigation opportunities.

**Statement II is correct:** A **primary purpose of the report** is to **track progress toward the Global Methane Pledge (GMP)**. The GMP is a voluntary commitment (launched at COP26 in 2021) to **reduce global methane emissions by at least 30% from 2020 levels by 2030**.

**Statement III is incorrect:** Globally, the **Agriculture sector is the largest contributor to anthropogenic (human-caused) methane emissions**, accounting for **approximately 42% of the total**, primarily from livestock digestion (enteric fermentation), rice cultivation, and manure management. The **Energy sector** follows as the **second-largest** at roughly 38%, and the **Waste sector is third** at about 20%.

**Statement IV is incorrect:** **India is the world's third-largest methane emitter in absolute terms (after China and the United States)**. India emitted approximately 31 million tonnes (Mt) in 2020, accounting for nearly 9% of global anthropogenic methane emissions. China led with 58 Mt (16%), followed by the US at 35 Mt (10%).

**In India, agriculture is the dominant source of methane emissions**, particularly from **livestock** (enteric fermentation) and **rice cultivation**. India contributes about 12% of global agricultural methane emissions, making it one of the largest contributors worldwide in this sector.

Source:)

<https://www.thehindu.com/data/are-methane-emissions-in-india-being-missed/article70399086.ece>

<https://www.thehindu.com/news/cities/mumbai/un-report-flags-indias-rising-methane-emissions-highlights-agricultural-gaps/article70298356.ece#:~:text=The%20report%20underscores%20that%2072,36%25%20between%202020%20and%202030>

<https://wedocs.unep.org/items/5711c0ab-d298-4d2d-8283-387d89b2325b>

<https://www.downtoearth.org.in/climate-change/global-methane-status-report-warns-of-rising-emissions-but-charts-path-to-2030-progress>

<https://www.downtoearth.org.in/climate-change/global-methane-status-report-warns-of-rising-emissions-but-charts-path-to-2030-progress>



<https://www.thehindu.com/news/cities/mumbai/un-report-flags-indias-rising-methane-emissions-highlights-agricultural-gaps/article70298356.ece>

Subject:) Environment

Topic:) Climate Change

Subtopic:)

Q.27)

Ans) c

Exp) Option c is the correct answer.

**Statement I is correct:** The Nilgiri Tahr (*Nilgiritragus hylocrius*) is a **mountain ungulate endemic to the Western Ghats**. Its habitat is specifically restricted to the **high-altitude open montane grasslands** (often associated with shola forests), typically at elevations between 1,200 and 2,600 meters.

It is well adapted to **steep rocky cliffs and rugged terrain**, where it grazes mainly on grasses and shrubs.

The Nilgiri Tahr is currently **classified as “Endangered,” not “Critically Endangered,”** on the Red List of the International Union for Conservation of Nature (IUCN).

**The Nilgiri tahr faces a multitude of threats such as:**

- Habitat loss due to rampant deforestation, competition with domestic livestock, hydroelectric projects, and monoculture plantations.
- Occasional hunting for its meat and skin.

**Statement II is incorrect:** The species exhibits **sexual dimorphism**, where **males are larger, darker**, and more robust **than females**. Adult males also develop a distinctive silvery-grey “saddle” patch on the back, giving them the name **“saddlebacks.”**

**Statement III is correct:** Eravikulam National Park in Kerala hosts the **largest population of the Nilgiri Tahr**.

- According to the **2025 synchronized Nilgiri Tahr census** conducted by the Kerala and Tamil Nadu Forest Departments, a total population of **2,668 Nilgiri Tahrs** was recorded.
- Of these, **1,365 individuals** were found in Kerala, and about **841** were **recorded in Eravikulam National Park alone**, making it the largest single population centre for the species in the Western Ghats.

Source:) <https://www.thehindu.com/sci-tech/energy-and-environment/survey-finds-1365-nilgiri-tahrs-in-kerala/article69897466.ece>

<https://www.thehindu.com/news/cities/Coimbatore/nigiris-tahr-interpretation-centre-to-come-up-at-gandhipuram/article70337894.ece>

[https://www.wwfindia.org/about\\_wwf/priority\\_species/threatened\\_species/nigiri\\_tahr/](https://www.wwfindia.org/about_wwf/priority_species/threatened_species/nigiri_tahr/)

<https://tnprojectnilgiritahr.com/wp-content/uploads/2024/02/Project-Tahr-Edu-material-Spreads-R4.pdf>

Subject:) Environment

Topic:) Species

Subtopic:)

Q.28)

Ans) b

Exp) Option b is the correct answer.

**Oxy-combustion is one of the major carbon capture and storage technologies** used to reduce carbon dioxide (CO<sub>2</sub>) emissions from industrial sources.

- **Carbon capture technologies** aim to separate CO<sub>2</sub> from gas streams released during processes such as power generation, cement production, steel manufacturing, and chemical industries. **Broadly, CO<sub>2</sub> capture methods are classified into three categories: post-combustion capture, pre-combustion capture, and oxy-fuel (oxy-combustion) capture.**

**Statement I is correct:** Oxy-combustion is a **carbon capture technology** in which **fuel** (usually coal or natural gas) is **burned using nearly pure oxygen instead of atmospheric air**.

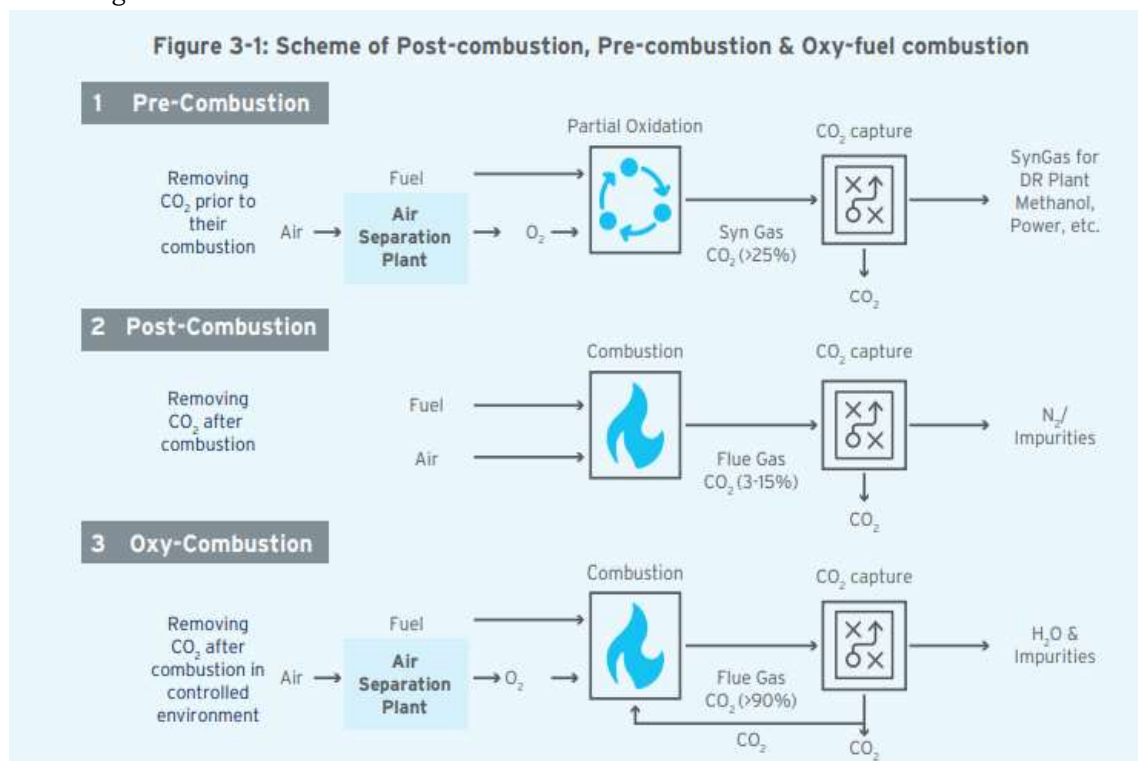
In conventional combustion, air contains about 78% nitrogen, which dilutes the flue gas. By removing nitrogen from the oxidant stream, oxy-combustion produces a flue gas mainly composed of CO<sub>2</sub> and water vapour, which facilitates carbon capture.

**Statement II is correct:** Unlike **post-combustion capture**, which relies on chemical absorbents (such as amines) or membranes to separate carbon dioxide from flue gas, **oxy-combustion** inherently produces a concentrated CO<sub>2</sub> stream. Therefore, it **does not require complex separation technologies** like **membranes or chemical solvents for carbon sequestration**.

**Statement III is correct:** Because nitrogen is largely absent during combustion, the **flue gas primarily contains carbon dioxide and water vapour**. After condensing the water vapour, the **remaining gas is highly concentrated CO<sub>2</sub>** (often **more than 90%**). This high purity of CO<sub>2</sub> makes it suitable for compression, transport, and storage in Carbon Capture and Storage (CCS) systems.

**Statement IV is incorrect:** Currently, the **most widely deployed commercial CCS method is post-combustion capture**, especially in industrial facilities and power plants. **Oxy-combustion** is still largely at the **demonstration or pilot stage** and has not yet become the dominant commercial technology.

Knowledge Base:



Source: <https://www.dghindia.gov.in/assets/downloads/62bbbedfa49947DraftUFCCRoadmap2030.pdf>

<https://www.niti.gov.in/sites/default/files/2022-12/CCUS-Report.pdf>

Subject: Environment

Topic: Mitigation Techniques

Subtopic:

Q.29)

Ans) d

Exp) Option d is the correct answer.

**Bioremediation** is an environmental conservation technique that **uses living organisms** such as **microorganisms** (bacteria, fungi), **plants, or algae to degrade, detoxify, or remove pollutants from contaminated soil, water, or air**. These

organisms utilize pollutants as a source of energy or nutrients and **convert them into less toxic or harmless substances**, such as carbon dioxide, water, and biomass.

- It is considered an **eco-friendly, cost-effective, and sustainable method** for cleaning environments affected by contaminants such as **oil spills, pesticides, plastics, heavy metals, industrial wastes, and organic pollutants**.
- Bioremediation may occur naturally or may be enhanced through human intervention to accelerate the cleanup process.

**Option a is incorrect:** Certain naturally occurring bacteria (such as *Pseudomonas* and *Alcanivorax*) can **metabolize hydrocarbons present in crude oil**. When such microbes are introduced or their activity is stimulated in an oil spill area, they break down complex petroleum compounds into simpler substances like carbon dioxide, water, and biomass.

Hence, the **use of oil-eating bacteria for cleaning marine oil spills is a classic example of bioremediation**.

**Option b is incorrect:** Some plants, known as **hyperaccumulators**, are **capable of absorbing and accumulating heavy metals** such as cadmium, arsenic, nickel, or lead from contaminated soils.

This process is called **phytoremediation, a type of bioremediation** in which plants are used to remove, stabilize, or detoxify contaminants from soil and groundwater.

**Option c is incorrect:** Certain fungi possess enzymes **capable of breaking down complex and persistent organic pollutants**, including pesticides and industrial chemicals. By decomposing these toxic compounds into less harmful forms, fungi help restore contaminated soils.

This biological degradation using fungi is referred to as **mycoremediation, which falls under the broader category of bioremediation**.

**Option d is correct:** Activated carbon filtration is a **physical-chemical treatment method** in which pollutants are removed from water by adsorption onto the surface of porous carbon materials. The process **does not involve any biological organism or metabolic degradation of pollutants**. Therefore, **it is considered a conventional water treatment technique, not a form of bioremediation**.

Knowledge Base: **Types of Bioremediation:**

#### 1. In situ bioremediation:

It refers to the treatment of contaminated soil or water **directly at the site without removing the contaminated material**. **Examples:** Biosparging, Bioventing, Biostimulation/Bioaugmentation and Phytoremediation.

#### 2. Ex situ bioremediation:

It involves **removing the contaminated soil or water from the original site and treating it elsewhere** under controlled conditions. **Examples:** Landfarming, Composting, Biopile/Soil Mounds and Bioreactor.

Source:) <https://www.thehindu.com/sci-tech/science/why-does-india-need-bioremediation-explained/article70346747.ece>

<https://egyankosh.ac.in/bitstream/123456789/95596/1/Unit-10.pdf>

Subject:) Environment

Topic:) Mitigation Techniques

Subtopic:)

Q.30)

Ans) c

Exp) Option c is the correct answer.

**Warm-blooded vertebrates**, also known as **endotherms**, are animals that can **maintain a relatively constant internal body temperature independent of external environmental conditions**. This thermoregulation is achieved through internal metabolic processes such as heat production from food metabolism.

- **Among vertebrates, only two groups are warm-blooded: mammals** (Class Mammalia) and **birds** (Class Aves).
- In contrast, **reptiles, amphibians, and most fishes are cold-blooded (ectothermic)** and depend largely on external environmental conditions to regulate their body temperature.

**Option I is correct: Dolphin**

**Dolphins are marine mammals** belonging to the order *Cetacea*. Like all mammals, they possess features such as lungs for breathing, mammary glands for feeding young, and a **constant internal body temperature maintained through metabolic heat**. Hence, **dolphins are warm-blooded vertebrates**.

**Option II is correct: Pangolin**

**Pangolins are mammals** classified under the order *Pholidota*. **Despite their reptile-like scaly appearance, they are true mammals** and exhibit typical mammalian traits such as endothermy, live birth (in most cases), and nursing of young. Therefore, **they are warm-blooded vertebrates**.

**Option III is correct: Penguin**

**Penguins belong to the class Aves (birds)**. All birds are endothermic, maintaining a stable body temperature through high metabolic activity and insulation provided by feathers and body fat. Thus, **penguins are warm-blooded vertebrates**, even though they inhabit cold marine environments.

**Option IV is correct: Bat**

**Bats are mammals** belonging to the order *Chiroptera*. They are the **only mammals capable of sustained flight** and maintain a constant body temperature through internal metabolic processes, making them **warm-blooded vertebrates**.

**Option V is incorrect: Crocodile**

**Crocodiles belong to the class Reptilia**. Reptiles are ectothermic, meaning they depend on external heat sources (such as sunlight) to regulate their body temperature. Therefore, **crocodiles are cold-blooded, not warm-blooded**.

Source:) <https://www.thehindu.com/books/rohan-chakravarty-bird-business-book-juggernaut-hornbill-black-drongo-bearded-vulture-saurus-crane-monal-world-migratory-bird-day/article69543912.ece>

<https://www.thehindu.com/sci-tech/science/european-perch-larger-body-size-warm-water/article66873491.ece>

<https://ncert.nic.in/textbook/pdf/kebo104.pdf>

Subject:) Environment

Topic:) Species

Subtopic:)

**Q.31)**

**Ans) b**

**Exp) Option b is the correct answer.**

Rivers have been the major reason for cultural growth and settlement since the early beginning of human settlement. Rivers in India originate from diverse physiographic regions such as the Himalayas, the Western Ghats, and the central highlands.

**Statement I is correct.** The **Godavari**, often referred to as the “**Dakshin Ganga**” or the **Ganga of the South**, rises from the **Trimbak (Tryambak) Hills of the Western Ghats near Nashik in Maharashtra**. From its source, the river flows eastward across the **Deccan Plateau** before draining into the **Bay of Bengal**.

**Statement II is correct.** The **Narmada River** originates from the **Amarkantak Plateau in Madhya Pradesh**, which lies at the **meeting point of the Vindhya and Satpura ranges**. Unlike most peninsular rivers that flow eastwards, the Narmada flows **westward through a rift valley** and finally drains into the **Arabian Sea**.

**Statement III is incorrect.** The **Kaveri River does not originate from the Nallamala Hills**. Instead, its source lies at **Talakaveri in the Brahmagiri Hills of the Western Ghats in present-day Karnataka (Kodagu district)**. From this mountainous source, the river flows southeastward through **Karnataka and Tamil Nadu** before reaching the **Bay of Bengal**. The **Nallamala Hills**, on the other hand, are part of the **Eastern Ghats system**.

Source:) <https://egyankosh.ac.in/bitstream/123456789/106548/1/Unit-2.pdf> - Unit 2 - Origin of Rivers, Journey, Tributaries, upstream and downstream linkages - Origin of Rivers - Important rivers in India and their Source - Page 37;

<https://timesofindia.indiatimes.com/religion/religious-places/trimbakeshwar-temple-the-divine-spiritual-history-of-a-powerful-jyotirlinga/articleshow/126672470.cms>

Subject:) Geography

Topic:) Drainage [India]

Subtopic:)

Q.32)

Ans) a

Exp) Option a is the correct answer.

India's power sector has undergone significant diversification in recent decades with a strong policy push towards **renewable energy expansion and a balanced electricity mix**. Different states have emerged as leaders in specific energy segments due to **geographical advantages, natural resource availability, and historical infrastructure development**.

**Pair I is correct. Rajasthan has the largest cumulative installed solar power capacity in India.** According to recent data from the **Ministry of New and Renewable Energy (MNRE)**, the state has installed **over 38 GW of solar capacity**, the highest in the country followed by Gujarat which has over 27 GW solar capacity. Rajasthan's dominance in solar energy is largely due to **extensive desert terrain, very high solar insolation levels**. Mega projects such as the **Bhadla Solar Park**, have significantly strengthened Rajasthan's position as India's leading solar power producer.

**Pair II is incorrect. Gujarat currently has the largest installed wind power capacity in India.** According to recent government data, the state has an installed wind capacity of **about 15197 MW (15.197 GW)**, reflecting rapid expansion of wind farms along its **long coastline and favourable wind corridors in regions such as Kutch and Saurashtra**. These geographical advantages have enabled Gujarat to emerge as the **top wind power generating state in the country**. Gujarat is followed by Tamil Nadu with over 12 GW of wind power capacity.

**Pair III is incorrect. Tamil Nadu has the highest installed nuclear power capacity among Indian states with 2440 MW capacity.** The state hosts major nuclear facilities such as the **Kudankulam Nuclear Power Plant**, which significantly contributes to India's nuclear electricity generation.

Source:) <https://mnre.gov.in/en/physical-progress/> ;

<https://cdnbbsr.s3waas.gov.in/s3716e1b8c6cd17b771da77391355749f3/uploads/2026/02/202602091660392380.pdf> ;

<https://www.aerb.gov.in/english/regulatory-facilities/nuclear-power-plants>

<https://www.deccanchronicle.com/nation/current-affairs/110716/2nd-unit-of-kudankulam-plant-attains-nuke-criticality.html>

Subject:) Geography

Topic:) Energy Resources [India]

Subtopic:)

Q.33)

Ans) c

Exp) Option c is the correct answer.

India's tropical and subtropical climatic regions support a remarkable diversity of **evergreen plant species**, particularly in areas receiving **high rainfall and warm temperatures throughout the year**. Evergreen trees are characterised by their **ability to retain foliage in all seasons**, which allows forests to remain green year-round. Many such species occur naturally in **tropical evergreen forests**, while some are widely cultivated as **ornamental or landscape plants** in gardens, parks and plantations.

**Option I is correct. Ebony (Diospyros ebenum) is a commercially valuable hardwood species found in tropical evergreen forests**, particularly in regions receiving **very high rainfall**. The tree is known for its **dense, dark-coloured wood**, which is widely used in **fine furniture, carving and musical instruments**. As a typical component of evergreen forests, ebony retains its leaves throughout the year and forms part of the **multi-layered forest structure** characteristic of tropical evergreen vegetation.

**Option II is correct. Mahogany (Swietenia mahagoni) is another important tropical hardwood species** associated with evergreen or semi-evergreen forest ecosystems. It is valued for its **strong, durable and reddish-brown timber**, which is widely used in **furniture making, shipbuilding and decorative woodwork**. In suitable tropical climates, mahogany behaves as an **evergreen tree with persistent foliage**, contributing to dense forest canopies.

**Option III is correct. Rosewood (Dalbergia latifolia) is a highly prized timber tree** commonly found in the **moist evergreen and semi-evergreen forests of India**, particularly along the **Western Ghats**. The species is well known for its



strong, fragrant and beautifully grained wood, making it valuable for luxury furniture and decorative items. In favourable climatic conditions with abundant rainfall, the tree maintains year-round foliage, typical of evergreen vegetation.

Option IV is incorrect. Shisham (*Dalbergia sissoo*), also known as Indian Rosewood, is a prominent timber tree widely distributed across the Indo-Gangetic plains and the foothills of the Himalayas. It thrives particularly well in alluvial soils along riverbanks and floodplains. The species belongs to the tropical deciduous forest type, where trees typically shed their leaves during the dry season to reduce moisture loss. This seasonal leaf fall is a key ecological adaptation to climatic conditions in northern India.

Source:) <https://ncert.nic.in/textbook/pdf/iess105.pdf> - Chapter 5 - Natural Vegetation and Wildlife - Page 40

Subject:) Geography

Topic:) Vegetation [India]

Subtopic:)

Q.34)

Ans) a

Exp) Option a is the correct answer.

India's land-use statistics offer an important window into how agricultural land, forests, fallow lands and non-agricultural uses compete for the country's finite land resources. The official publication Land Use Statistics at a Glance 2023-24 compiled by the Ministry of Agriculture tracks long-term trends since 1950-51 and helps analyse the structural shifts in agricultural land availability, land-use efficiency and regional disparities in cultivation.

Statement I is correct. According to the 2023-24 land-use statistics, the Net Area Sown stood at about 138.99 million hectares. This category represents the actual land used for crop cultivation during the agricultural year. Among all land-use categories, this remains the largest single category of land utilisation in India, indicating that agriculture continues to dominate the country's land-use pattern despite rapid urbanisation and industrial expansion.

Statement II is correct. Over the decades, the share of agricultural land in India's total reported area has gradually declined. In the early years after Independence, agriculture occupied a significantly larger share of the country's land resources.

In 1950-51, agricultural land accounted for about 66.7% of the reported geographical area, whereas by 2023-24 this share had declined to around 58.6%. This long-term decline reflects structural changes in the economy, including urbanisation, industrial growth, infrastructure expansion and diversion of agricultural land to non-agricultural purposes.

However, even with this decline, agriculture still occupies well over half of India's reported land area, highlighting the continued importance of agriculture in sustaining livelihoods and food security.

Statement III is incorrect. In agriculture, cropping intensity refers to the number of crops cultivated on the same field within a single agricultural year. It measures the intensity of land use for cultivation and indicates how often crops are planted, harvested, and replanted. Cropping intensity is expressed as the ratio of the Gross Cropped Area to the Net Sown Area. Cropping Intensity =  $(\text{Gross Cropped Area} / \text{Net Sown Area}) \times 100$  (Measures in percentage)

Cropping Intensity in agriculture has increased from around 111.1 % to 156.8 % (not more than doubled) between 1950-51 and 2023-24.

Source:) <https://desagri.gov.in/wp-content/uploads/2025/07/LUS-2023-24-to-be-printed-Uploaded.pdf> - Land Use Statistics At A Glance 2023-2024 - Page 12, 14 and Charts

Subject:) Geography

Topic:) Land Use Statistics

Subtopic:)

Q.35)

Ans) c

Exp) Option c is the correct answer.

The **Dharwar rock system** belongs to one of the oldest geological formations of India, dating back to the **Precambrian period**. These rocks were originally **sedimentary deposits**, but over millions of years they were subjected to **heat and pressure inside the Earth**, which transformed them into **metamorphic rocks**.

Because of these geological processes, the Dharwar rocks became **rich repositories of metallic minerals**. Many important mineral deposits of India are associated with these rocks, including **Iron ore, Manganese, Gold, Copper, Chromite and others**. This is why regions dominated by Dharwar rocks such as parts of **Karnataka, Jharkhand, Odisha, and Chhattisgarh**, have historically been major **mining and metallurgical centres**.

Knowledge Base:

| Rock System        | Key Feature                               |
|--------------------|-------------------------------------------|
| <b>Dharwar</b>     | Metallic minerals (iron, manganese, gold) |
| <b>Vindhyan</b>    | Sandstone, limestone, shale               |
| <b>Gondwana</b>    | Coal deposits                             |
| <b>Deccan Trap</b> | Basaltic lava flows                       |

Source:) <https://ncert.nic.in/textbook/pdf/legy205.pdf> -Unit 3 - Chapter 5 - Mineral and Energy Resources - Page 55

Subject:) Geography

Topic:) Mineral Resources [India]

Subtopic:)

Q.36)

Ans) c

Exp) Option c is the correct answer.

**Niobium** is a strategically important **critical mineral** used in several advanced industrial applications. It significantly improves the **strength, durability, and heat resistance of steel alloys**, and is also used in **superconducting materials, aerospace components, electronics, nuclear technology and steel manufacturing**. Because of its strategic and technological importance, the global distribution of niobium production is highly concentrated.

**Brazil is the world's largest producer of Niobium**, accounting for **more than three-fourths (around 80-90%) of global production**. The country hosts some of the **largest niobium reserves in the world**, particularly in the **Araxa and Catalao deposits**. Brazil's dominance in niobium production makes it a **critical player in global supply chains**, especially for industries that require **high-strength steel alloys and superconducting materials**.

Source:) <https://www.indiatoday.in/auto/bikes/story/niobium-powers-landspeed-record-candidate-lightning-tachyon-nb-2288155-2022-10-21> ;

<https://www.thehindubusinessline.com/business-tech/recent-successes-in-science-ledatmanirbharta/article70550569.ece>

[https://www.statista.com/statistics/1026173/niobium-mine-production-worldwide-country/?srsltid=AfmBOorJDDPOARwcGwNXGhjbtgdCIvNx\\_JT8-6ehadIttsiFgFj8k5dm](https://www.statista.com/statistics/1026173/niobium-mine-production-worldwide-country/?srsltid=AfmBOorJDDPOARwcGwNXGhjbtgdCIvNx_JT8-6ehadIttsiFgFj8k5dm) ;

<https://www.mbmng.mtech.edu/pdf-publications/fs23.pdf>

Subject:) Geography

Topic:) Mineral Resources [World]

Subtopic:)

Q.37)

Ans) d

Exp) Option d is the correct answer.

India possesses some of the most ecologically rich **coral reef ecosystems** in regions such as **Lakshadweep, Gulf of Mannar, Gulf of Kachchh, and the Andaman & Nicobar Islands**. Coral reefs play a crucial role in **marine biodiversity conservation, coastal protection, fisheries productivity, and climate resilience**.

The Government of India has recently decided to establish the country's **first 'National Coral Reef Research Institute' in the Andaman & Nicobar Islands**. The region contains **some of the most diverse and well-developed coral reef ecosystems in India**, with high species richness and relatively healthier reef systems compared to many other regions. The institute is expected to strengthen **scientific research, reef monitoring, climate impact studies, and conservation strategies**, thereby supporting **sustainable management of India's coral reef resources**.

Source:); <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2195434&reg=3&lang=2> ;  
<https://www.thehindubusinessline.com/news/national/andaman-to-get-indias-first-coral-reef-research-centre/article70338234.ece>  
 Subject:; Geography  
 Topic:; Flora and Fauna [India]  
 Subtopic:;

Q.38)

Ans) b

Exp) Option b is the correct answer.

**Brunei is a small Southeast Asian country located on the northern coast of the island of Borneo, and it directly faces the South China Sea.** Because of this geographical location, Brunei is one of the **claimant states in the South China Sea disputes**, particularly regarding parts of the **Spratly Islands and surrounding maritime zones**.

**Option a is incorrect.** Japan is located to the **northeast of the East China Sea and the Pacific Ocean**. The country is involved in maritime disputes with China over the **Senkaku/Diaoyu Islands in the East China Sea, but Japan does not border the South China Sea**.

**Option c is incorrect.** North Korea **does not have a border with the South China Sea**.

**Option d is incorrect.** Laos is a landlocked country and **does not have a border with the South China Sea**.

Knowledge Base:



Source:) <https://www.thehindu.com/news/international/storms-brewing-in-east-south-china-seas-explained/article68685920.ece>

Subject:) Geography

Topic:) World Physical Geography

Subtopic:)

Q.39)

Ans) d

Exp) Option d is the correct answer.

India's natural vegetation varies according to **rainfall, temperature, and soil conditions**. In areas where rainfall is extremely low, plants evolve special adaptations to survive **water scarcity and high evaporation**. Such ecological conditions give rise to **Tropical Thorn Forests and Scrub vegetation**.

**Statement I is correct.** Tropical Thorn Forests develop in **semi-arid climatic regions where annual rainfall is generally below about 70 cm**. Due to such limited precipitation and high evaporation rates, only **drought-resistant vegetation** can survive in these areas.

**Statement II is correct.** In India, these forests are mainly found in the **north-western and central semi-arid belt**, including **Rajasthan, Gujarat, south-west Punjab, Haryana, and parts of Madhya Pradesh and Uttar Pradesh**. These regions experience **irregular rainfall and prolonged dry seasons**, which favour the growth of thorny scrub vegetation.

**Statement III is correct.** The dominant plant species in Tropical Thorn Forests include **babool (Acacia), ber, khair, Khejri, neem, palas and wild date palm**. These species are well adapted to **dry climatic conditions** and are commonly found scattered across these semi-arid landscapes.

Plants growing in thorn forests show **distinct structural adaptations to conserve water (xerophytic adaptations)**. They usually possess **deep root systems to tap underground moisture, thick or small leaves to minimise evaporation**, and sometimes **succulent stems for water storage**. These features help plants survive **long dry periods and intense heat**.

Source:) <https://ncert.nic.in/textbook/pdf/iess105.pdf> - Chapter 5 - Natural Vegetation and Wildlife - Tropical Thorn Forests and Scrubs - Page 40

<https://sathee.iitk.ac.in/ncert-books/class-11/geography/india-physical-environment/chapter-05-natural-vegetation/>

Subject:) Geography

Topic:) Vegetation [India]

Subtopic:)

Q.40)

Ans) b

Exp) Option b is the correct answer.

**Salinity** refers to the **total concentration of dissolved salts present in water**. It is measured as the **amount of salt (in grams) dissolved in 1,000 grams (1 kg) of water** and is generally expressed in **parts per thousand (‰ or ppt)**. Salinity is a crucial property of seawater as it influences **ocean circulation, marine ecosystems, and water density**. Ocean salinity varies across different seas and oceans depending on factors such as **latitude, evaporation rates, freshwater inflow from rivers, precipitation, and degree of enclosure from open oceans**.

**Statement I is incorrect.** Although the **North Sea lies in higher latitudes**, it does **not record low salinity**. In fact, the **North Sea records relatively higher salinity** because **saline waters from the North Atlantic are brought into the sea by the North Atlantic Drift**. This inflow of saline Atlantic water offsets the effect of freshwater inflow from rivers and maintains relatively higher salinity levels.

**Statement II is correct.** The **Arabian Sea records relatively high salinity** mainly because of **high evaporation rates under tropical climatic conditions**. In addition, the **freshwater inflow from rivers is comparatively limited**, and evaporation removes water but leaves salts behind, thereby **increasing the salt concentration in the sea water**.



**Statement III is incorrect.** The Caspian Sea does not record high salinity. Although it is **landlocked**, it receives **significant freshwater inflow from rivers, especially the Volga**, which greatly reduces the salt concentration. As a result, the **salinity of the Caspian Sea is considerably lower than that of the open ocean**.

Source:) <https://ncert.nic.in/textbook/pdf/kegy212.pdf> - Fundamentals of Physical Geography -

Chapter 12 - Water (Oceans) - Salinity of Ocean Waters - Page 104

<https://www.worldatlas.com/rivers/volga-river.html>

Subject:) Geography

Topic:) Oceanography

Subtopic:)

Q.41)

Ans) d

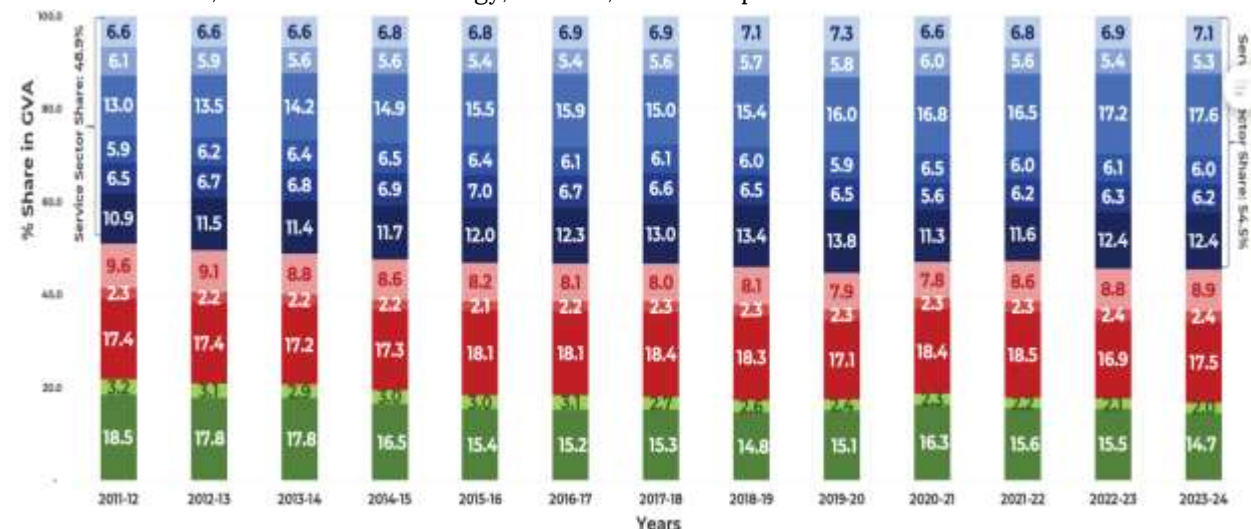
Exp) Option d is the correct answer.

**Statement I is incorrect:** Among the major economies of the world (such as the United States, United Kingdom, Japan, Germany, France, China, etc.), **India does not account for the highest share of the services sector in its GVA**. India's services sector contributes around 53–56% of GDP/GVA in recent years, which is significant for a developing economy but lower than that of most advanced major economies. **For instance:**

- **United States:** ~78–80%
- **United Kingdom:** ~70–73%
- **European Union:** ~65–68%
- **China:** ~53–57%

Advanced economies generally have services shares in the 65–80% range due to structural transformation, whereas India's is still transitioning.

**Statement II is correct:** In India, the **services sector contributes more than 50% of the country's GVA in recent years**, according to official economic statistics. Its share has **increased from about 50.6% in FY2014 to around 55% in FY 2025**. It is the **dominant sector of the Indian economy**, and includes activities such as trade, transport, communication, financial services, information technology, tourism, and other professional services.



Hence, the **correct answer is: c) Statement I is not correct but Statement II is correct.**

Source:) <https://www.indiabudget.gov.in/economicsurvey/doc/eschapter/echap07.pdf> Page: 260 -261.

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2097889>

Subject:) Economy

Topic:) National Income Accounting

Subtopic:)



Q.42)

Ans) a

Exp) Option a is the correct answer.

**Statement I is correct:** The "China Plus One" (C+1) strategy has been increasingly adopted by multinational corporations and several major economies to reduce excessive dependence on China for manufacturing.

- As companies seek alternative production destinations, countries such as India, Vietnam, Thailand, and Mexico are emerging as potential manufacturing hubs.
- India, with its large domestic market, improving infrastructure, and initiatives such as Production Linked Incentive (PLI) schemes and Make in India, can attract investments from firms relocating or expanding their manufacturing bases.

Hence, this strategy can create opportunities for India to expand its manufacturing sector and integrate more deeply into global supply chains.

**Statement II is correct:** The China Plus One strategy refers to a manufacturing and supply-chain diversification strategy in which companies maintain manufacturing operations in China but simultaneously establish additional production facilities in other countries.

The objective is to reduce supply-chain risks arising from excessive reliance on a single country, especially in the context of rising labour costs in China, trade tensions, geopolitical risks, and disruptions such as the COVID-19 pandemic.

**Statement II explains Statement I:**

Since companies are diversifying their manufacturing bases and supply chains beyond China, they are looking for alternative locations for production. This diversification creates opportunities for countries like India to attract manufacturing investments and become part of global supply chains. Therefore, the supply-chain diversification described in Statement II directly explains why India may benefit as mentioned in Statement I.

Thus, Both Statement I and Statement II are correct and Statement II explains Statement I.

Source:) <https://www.thehindu.com/business/china-plus-one-is-an-opportunity-for-indian-firms/article65443033.ece>

<https://www.thehindu.com/news/cities/Hyderabad/taking-all-efforts-to-attract-firms-with-china-1-strategy-telangana-cm/article69815063.ece>

<https://www.thehindubusinessline.com/economy/japanese-firms-eye-india-as-part-of-china-plus-one-strategy/article69225927.ece>

<https://www.thehindu.com/opinion/op-ed/intent-china-plus-one-impact-india-minus-one/article69036321.ece>

Subject:) Economy

Topic:) External Sector

Subtopic:)

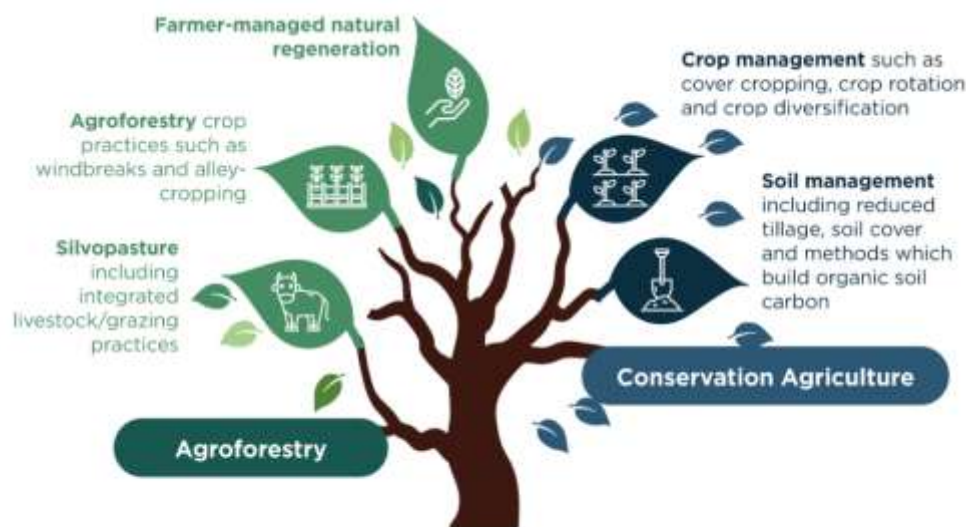
Q.43)

Ans) a

Exp) Option a is the correct answer.

**Regenerative agriculture** refers to a set of farming practices aimed at restoring and enhancing the health of soil, ecosystems, and biodiversity while maintaining sustainable agricultural productivity.

- Unlike conventional agriculture, which often focuses mainly on maximizing yields by relying heavily on chemical inputs, regenerative agriculture emphasizes improving soil fertility, increasing organic matter, and strengthening natural ecological processes.
- It promotes practices such as crop diversification, cover cropping, reduced or no tillage, agroforestry, and integrated livestock management. These practices help improve soil structure, enhance water retention, reduce soil erosion, and increase soil carbon sequestration.
- By restoring degraded soils and improving the resilience of farming systems, regenerative agriculture can contribute to sustainable food production, climate change mitigation through carbon sequestration, and greater resilience of agriculture to climate-related stresses such as droughts and floods.



Regenerative agriculture focuses on improving soil health through methods including crop rotation and reduced ploughing.

Image: IUCN

**Statement I is correct:** Regenerative agriculture is widely recognized as a crucial dual-purpose strategy. It helps mitigate climate change (primarily through carbon sequestration in soils, reducing net GHG emissions from agriculture) and supports adaptation to climate impacts (e.g., better resilience to droughts, floods, erratic rainfall via improved soil structure).

At the same time, it enhances long-term agricultural productivity, yield stability, and nutritional quality of food, making it a key approach for ensuring food security in a warming world with increasing climate stresses.

Reports highlight that these practices build resilient food systems capable of sustaining production amid population growth and environmental challenges.

**Statement II is correct:** Farming practices such as crop diversification (including crop rotations), cover cropping, and reduced tillage (or no-till/conservation tillage) are core regenerative techniques. These practices improve soil structure, soil organic matter, and microbial activity. This leads to better water retention, improved nutrient availability, and reduced soil erosion, which enhance agricultural productivity.

These practices also increase carbon sequestration in soils, thereby reducing the concentration of atmospheric carbon dioxide and contributing to climate change mitigation.

**Statement II explains Statement I:**

Because regenerative practices improve soil health and water-holding capacity, they support sustained crop productivity and yield stability, which strengthens food security. At the same time, the sequestration of carbon and improved ecosystem resilience help address climate change. Therefore, **Statement II provides the underlying mechanism that explains why regenerative agriculture can address both food security and climate change challenges.**

Hence, **Both Statement I and Statement II are correct and Statement II explains Statement I.**

Source: <https://www.thehindubusinessline.com/economy/agri-business/regenerative-agriculture-the-future-of-indias-cultivation-practices/article69774214.ece>

<https://www.thehindubusinessline.com/economy/agri-business/regenerative-agriculture-how-soil-health-can-strengthen-indias-textile-supply-chains/article70226702.ece>

[https://www.business-standard.com/industry/agriculture/up-s-resilient-farming-model-exemplary-for-small-farmers-world-bank-chief-125101700012\\_1.html](https://www.business-standard.com/industry/agriculture/up-s-resilient-farming-model-exemplary-for-small-farmers-world-bank-chief-125101700012_1.html)

<https://www.weforum.org/stories/2022/10/what-is-regenerative-agriculture/>

Subject: Economy

Topic: Sustainable Development

Subtopic:)

Q.44)

Ans) b

Exp) Option b is the correct answer.

**In the context of the Indian economy**, the term “**gilt-edged market**” (or gilt-edged securities market) specifically **refers to the market for government securities** (also known as **G-Secs**). These include **treasury bills, dated securities** (government bonds), and **other debt instruments issued by the Central Government** (and sometimes **state governments**).

The name “gilt-edged” originates from the historical practice in the **UK** (and adopted in India and other Commonwealth countries) where high-quality government bonds had gilded (gold-edged) borders on their certificates, symbolizing their superior quality and safety.

**Key reasons why government securities are called gilt-edged:**

- They are **considered virtually risk-free** (zero default risk) because they are **backed by the sovereign guarantee of the Government** of India.
- They **offer stable, predictable returns** through fixed interest (coupon) payments.
- They are **highly liquid** and form a major part of the debt market regulated by the Reserve Bank of India (RBI).

Source: <https://www.rbi.org.in/commonman/English/Scripts/FAQs.aspx?Id=711>

<https://www.thehindubusinessline.com/opinion/editorial/gilt-edged-issue/article70208068.ece>

<https://m.economictimes.com/wealth/invest/what-are-government-securities-and-how-do-they-function/articleshow/127813469.cms>

Subject:) Economy

Topic:) Capital Market

Subtopic:)

Q.45)

Ans) b

Exp) Option b is the correct answer.

**Human capital formation** refers to the **process of enhancing the knowledge, skills, health, and productive capacities of individuals** through deliberate investments in areas such as **education, healthcare & nutrition and training & skill development**.

It focuses on improving the quality of the labour force, enabling individuals to become more productive and contribute more effectively to economic growth and national development.

**Key sources of human capital formation include:**

1. **Education** (schools, universities, professional courses)
2. **Skill development and vocational training**
3. **Healthcare and nutrition**
4. **On-the-job training and research**

A well-developed stock of human capital leads to **higher productivity, technological innovation, better income levels, and sustainable economic growth**. Hence, investments in human capital are considered crucial for long-term development of a country.

**Option I is correct:** **Expansion of public healthcare facilities** and nutrition programmes improves the health status, physical capacity, and productivity of individuals. A healthier population is able to work more efficiently and for longer periods.

Therefore, **expenditure on healthcare and nutrition is considered an important component of human capital formation**.

**Option II is correct:** **Establishing a vocational training institute** enhances the technical and professional skills of workers. Skill development improves the efficiency, employability, and productivity of the workforce.

Hence, such investments directly contribute to human capital formation.

**Option III is incorrect:** Construction of an expressway improves transportation networks, reduces logistics costs, and facilitates trade and mobility. However, it represents investment in physical infrastructure, which is a **form of physical capital formation, not human capital formation.**

**Option IV is correct:** Spending on higher education and professional training increases an individual's knowledge, expertise, and productive capabilities. Such educational investment improves earning potential and productivity, making it a **key example of human capital formation.**

**Option V is incorrect:** Purchasing automated machinery increases the productive capacity of a firm by adding to its stock of machines and equipment. **This is an example of physical capital formation,** as it does not directly enhance the skills, education, or health of individuals.

Source:) <https://ncert.nic.in/textbook/pdf/keec104.pdf>

<https://indianexpress.com/article/upsc-current-affairs/upsc-essentials/how-human-capital-acquires-a-centre-stage-in-indias-growth-strategy-10549864/>

<https://indianexpress.com/article/upsc-current-affairs/upsc-essentials/why-human-capital-needs-to-be-the-anchor-of-growth-10487177/>

Subject:) Economy

Topic:) Human Capital

Subtopic:)

Q.46)

Ans) c

Exp) Option c is the correct answer.

The term '**Friendshoring**' (also called **allyshoring**) refers to an emerging strategy in the global economy where **countries and multinational companies shift production facilities and supply chains to nations that are trusted political and economic partners or geopolitical allies.** The objective is to reduce geopolitical risks, enhance supply-chain resilience, and avoid over-dependence on rival or unstable countries.

- The concept gained prominence after disruptions caused by events such as the **COVID-19 pandemic** and **rising geopolitical tensions** (such as the **Russia-Ukraine War**), which exposed vulnerabilities in highly concentrated global supply chains and when several countries sought to make supply chains more resilient by relying on trusted allies rather than purely cost-efficient locations.
- Under friendshoring, production is not merely moved for cost advantages or geographic proximity, but to countries that share strategic, political, or economic alignment.

Knowledge Base: Some **other related supply-chain strategies** used in the global economy:

#### 1. Nearshoring:

It refers to the **relocation of production or business processes to nearby or neighbouring countries** to reduce transportation costs, improve supply-chain efficiency, and shorten delivery time. Geographic proximity is the key factor.

#### 2. Offshoring:

It means **shifting production or services to distant foreign countries**, usually to benefit from **lower labour costs, cheaper resources, or favourable regulations.**

#### 3. Reshoring (Onshoring):

It refers to **bringing manufacturing and production activities back to the home country** after they were previously moved abroad. Governments often encourage this to strengthen domestic industries and employment.

#### 4. Diversified Supply-Chain Strategy:

Instead of relying on one country, firms **spread production and sourcing across multiple countries** to reduce risks arising from geopolitical tensions, trade disputes, or disruptions.

Source:) <https://www.thehindu.com/business/budget/economic-survey-flags-growing-challenges-in-global-trade/article69164846.ece>

<https://indianexpress.com/article/explained/explained-economics/what-is-friendshoring-8277847/>

<https://www.weforum.org/stories/2023/02/friendshoring-global-trade-buzzwords/>  
<https://www.economist.com/the-economist-explains/2023/08/30/what-is-friendshoring>  
 Subject:) Economy  
 Topic:) External Sector  
 Subtopic:)

Q.47)

Ans) d

Exp) Option d is the correct answer.

**The Reserve Bank of India has recently proposed a unified cross-border digital payment system linking the Digital Rupee with the CBDCs of BRICS countries.**

**Statement I is correct:** Digital Rupee or e₹, is India's Central Bank Digital Currency (CBDC). It is the digital form of India's physical currency, the Rupee (₹). e₹ is issued by the Reserve Bank of India (RBI) in digital form and e₹ is at par with the physical currency, offering features similar to physical cash like convenience of use, guarantee of RBI, finality of settlement, etc. e₹ is stored in the user's digital wallet and can be used to receive / send money, and / or make payment for transactions, just like any physical ₹ note.

In terms of **Section 26 of the Reserve Bank of India (RBI) Act, 1934, every e₹ bank note shall be legal tender at any place in India** in payment or on account for the amount expressed therein and is guaranteed by the Central Government. e₹, being a digital form of the ₹ bank note, is legal tender and is the liability of the Reserve Bank of India.

The logo and tagline for India's CBDC is as under:



**Statement II is correct: The RBI has introduced two variants of CBDC:**

- **e₹-W (Wholesale CBDC):** Used mainly by **banks and financial institutions** for **interbank transactions** and settlement of government securities.
- **e₹-R (Retail CBDC):** Designed for public use, enabling individuals and businesses to conduct everyday payments.

**Statement III is correct: Given the cash-like features of e₹, there is no interest payable on the e₹ wallet balances.** The Digital Rupee does not earn interest, unlike bank deposits kept in savings or fixed deposit accounts. It functions similarly to physical cash. However, it maintains full convertibility and can be exchanged for bank deposits or physical currency, ensuring parity with existing forms of money.

Source:) : <https://www.rbi.org.in/commonman/English/scripts/FAQs.aspx?Id=3686>

<https://www.thehindu.com/business/Economy/rbi-proposes-linking-brics-digital-currencies/article70525693.ece>

<https://www.thehindu.com/opinion/editorial/building-bridges/article70534042.ece>

Subject:) Economy

Topic:) Money and Banking

Subtopic:)

Q.48)

Ans) b

Exp) Option b is the correct answer.



**Gross Domestic Product (GDP)** is the **total monetary value of all final goods and services produced within the domestic territory of a country during a specific period, usually one year**. It is the most widely used indicator to measure the size and performance of an economy.

GDP includes production carried out within the geographical boundaries of the country, irrespective of whether the producers are citizens or foreigners.

#### Key Features of GDP

- It **measures only final goods and services to avoid double counting**.
- It includes **current production** within the country during the given period.
- It is **measured in monetary terms** to aggregate diverse goods and services.
- It reflects economic activity within the **domestic territory**.

#### Major Components of GDP

GDP is **commonly calculated using the expenditure method**:

$$\text{GDP} = C + I + G + (X - M)$$

Where:

- C – Private Final Consumption Expenditure
- I – Gross Investment (capital formation)
- G – Government Final Consumption Expenditure
- X – M – Net Exports (Exports – Imports)

#### Transactions Not Included in GDP:

Certain transactions are excluded because they do not represent current production, such as:

- **Transfer payments** (pensions, remittances, scholarships)
- **Sale/purchase of Second-hand goods**
- **Pure financial transactions** (purchase of shares/bonds)
- **Illegal economic activities** (in official statistics)

#### Option I is correct: Salaries paid to employees of the Government of India – Included

Government employees provide services such as administration, defence, policing, etc. The wages and salaries paid by the government are payments for current services, therefore **they form part of government final consumption expenditure and are included in GDP**.

#### Option II is incorrect: Remittances sent by Indians working abroad to their relatives in India – Not Included

**These are transfer payments and do not arise from production within India**. Since GDP counts only the value of goods and services produced domestically, **such remittances are excluded**.

#### Option III is incorrect: A citizen selling his five year old used car to his friend – Not Included

The car was produced in an earlier year and its value was already counted in GDP when it was originally produced. Including it again would lead to **double counting**. Hence, **sale of second-hand goods is excluded** (though any brokerage involved would be counted).

#### Option IV is correct: Brokerage earned by a share broker on the sale of equity shares – Included

The shares themselves are financial assets, so their transfer is not counted. However, the **brokerage represents payment for a service** provided during the current period, so **it contributes to GDP**.

Hence, **payment for this service is therefore included in the calculation of GDP**.

#### Option V is correct: Purchase of a newly constructed residential house by an Indian Individual – Included

A newly built house is a final good produced in the current period. It represents new production in the construction sector and is **included in GDP under gross capital formation (investment expenditure)**.

Source:) <https://ncert.nic.in/textbook/pdf/leec102.pdf>

[https://mospi.gov.in/sites/default/files/reports\\_and\\_publication/statistical\\_manual/Chapter%208.pdf](https://mospi.gov.in/sites/default/files/reports_and_publication/statistical_manual/Chapter%208.pdf)

<https://www.egyankosh.ac.in/bitstream/123456789/117462/1/Unit-2.pdf>

Subject:) Economy

Topic:) National Income Accounting

Subtopic:)

Q.49)

Ans) c

Exp) Option c is the correct answer.

Economic activities in an economy are commonly classified into three broad sectors based on the nature of the activity.

**1. The Primary Sector** includes activities that involve the **extraction or direct use of natural resources**, such as agriculture, fishing, forestry, mining, and animal husbandry.

**2. The Secondary Sector** consists of **activities related to manufacturing and construction**, where **raw materials obtained from the primary sector are processed or transformed into finished or semi-finished goods**, such as steel, textiles, cement, and machinery.

**3. The Tertiary Sector**, also known as the **service sector**, includes **activities that provide services** rather than producing tangible goods, such as banking, transport, trade, communication, education, healthcare, and tourism.

**Option I is correct: Processing of raw cotton into finished textile garments in a factory:**

This involves manufacturing, where raw cotton is converted into garments. Hence, it **belongs to the Secondary Sector**.

**Option II is correct: Construction of residential buildings by a real estate company:**

**Construction is considered part of the Secondary Sector** because it involves transforming raw materials (cement, steel, bricks) into infrastructure.

**Option III is correct: Conversion of iron ore into steel in a steel plant:**

**This is a manufacturing activity**, where raw material (iron ore) is processed into steel. Hence, it **belongs to the Secondary Sector**.

**Option IV is incorrect: Banking services provided to customers through commercial banks:**

Banking provides financial services and does not involve production of goods. Therefore, it **belongs to the tertiary (service) sector**.

**Option V is correct: Manufacturing of cement using limestone and other raw materials:**

This involves industrial production, converting raw materials into cement, so it **belongs to the Secondary Sector**.

Knowledge Base:



Source:) <https://ncert.nic.in/textbook/pdf/jess202.pdf> Page: 20.

<https://ncert.nic.in/textbook/pdf/fees114.pdf> Page: 199.

Subject:) Economy

Topic:) Sectors of Economy

Subtopic:)

Q.50)

Ans) b

Exp) Option b is the correct answer.

The **value of the Indian Rupee in the foreign exchange market is determined by the demand and supply of foreign currencies**, particularly the US dollar. When the **demand for foreign currency rises relative to the rupee**, the **rupee tends to depreciate** against major global currencies.

To manage excessive volatility or depreciation, the Reserve Bank of India and the Government of India may **adopt various monetary, fiscal, and external sector policy measures**, such as influencing capital flows, regulating imports, and intervening in the foreign exchange market. These steps aim to stabilize the currency and maintain external sector stability.

**Option I is correct: Increase in the policy repo rate by the Reserve Bank of India**

An increase in the repo rate raises domestic interest rates. Higher interest rates tend to attract foreign portfolio investment into government securities and financial markets, **increasing the demand for the rupee and helping stabilize the currency by addressing its depreciation.**

**Option II is incorrect: Purchase of large quantities of US dollars by the RBI from the foreign exchange market**

When the RBI purchases US dollars, it increases the demand for dollars and injects rupees into the market. This can **further weaken (depreciate) the rupee rather than support (appreciate) it.** To defend the rupee, the RBI typically sells dollars, not buys them.

**Option III is correct: Relaxation of norms for External Commercial Borrowings (ECBs) by the Reserve Bank of India**

Relaxing ECB norms allows Indian firms to borrow more from international markets, bringing foreign currency inflows into the country. These inflows increase the supply of foreign exchange, **helping reduce pressure on the rupee.**

**Option IV is incorrect: Reduction in customs duties on a wide range of imported goods**

Reducing import duties makes imports cheaper and can increase import demand. This raises the demand for foreign currency, which can **worsen the depreciation of the rupee.**

Source:) <https://www.thehindu.com/opinion/lead/the-solution-to-the-falling-rupee-lies-in-diplomacy/article70556822.ece>

<https://www.thehindu.com/business/Economy/rbi-does-not-target-any-band-for-rupee-allows-it-to-find-its-own-level-says-governor-sanjay-malhotra/article70360998.ece>

<https://www.thehindu.com/business/we-do-not-target-any-level-of-rupee-depreciation-due-to-demand-for-us-dollar-rbi-governor-sanjay-malhotra/article70302916.ece>

Subject:) Economy

Topic:) External Sector

Subtopic:)

Q.51)

Ans) a

Exp) Option a is the correct answer.

The **provision for National Emergency** is contained in **Article 352** of the Constitution of India. It **empowers the President of India to proclaim an emergency** when the security of India or any part of its territory is threatened by war, external aggression, or armed rebellion.

**Statement I is correct:** Under Article 352 of the Constitution of India, the **President can proclaim a National Emergency on three grounds:**

- War
- External aggression
- Armed rebellion

Originally, the Constitution used the term “internal disturbance,” but it was replaced by “armed rebellion” by the **44th Constitutional Amendment Act, 1978** to prevent misuse of emergency powers.

The **resolution approving the proclamation of National Emergency must be approved by both Houses of Parliament within one month from the date of its issuance**. The **approval must be passed by a special majority** (meaning a majority of the total membership of the House and not less than two-thirds of the members present and voting) **in either house of Parliament**.

- Originally, the Constitution provided that a proclamation of National Emergency had to be approved by both Houses of Parliament within two months.
- However, the **44th Constitutional Amendment Act, 1978** reduced this approval period to one month.
- Once the proclamation is approved by both Houses of Parliament the National Emergency continues for six months. It can be extended indefinitely, but each extension requires fresh approval by Parliament every six months.

**Statement II is incorrect:** The **suspension of freedoms guaranteed under Article 19** during a National Emergency is governed by **Article 358**. According to this provision, **Article 19 is automatically suspended only when the Emergency is proclaimed on the grounds of war or external aggression**. However, if the **Emergency is declared on the ground of armed rebellion, Article 19 does not get automatically suspended**, but rather continues to operate.

**Statement III is incorrect:** The **President can proclaim a National Emergency only on the written recommendation of the Union Cabinet, not merely the Prime Minister**.

This safeguard was **introduced by the 44th Constitutional Amendment Act, 1978** to ensure that the decision is taken collectively by the Cabinet rather than by a single individual.

Source:) Mastering Indian Polity by Forum IAS, Chapter: 10 – Emergency, Page: 239.

<https://www.mea.gov.in/Images/pdf1/Part18.pdf>

<https://egyankosh.ac.in/bitstream/123456789/57928/1/Unit11.pdf>

Subject:) Polity

Topic:) Emergency Provisions

Subtopic:)

Q.52)

Ans) b

Exp) Option b is the correct answer.

The **Seventh Schedule** of the Constitution of India **defines the distribution of legislative powers between the Union and the States**, forming the basis of India's federal structure. **It contains three lists of subjects** on which different levels of government can make laws.

- **Union List (List I):** Subjects of national importance on which **only Parliament can legislate** (e.g., defence, foreign affairs, currency, insurance).
- **State List (List II):** Subjects primarily of local or state importance on which **State Legislatures normally legislate** (e.g., public order, police, public health, fisheries).
- **Concurrent List (List III):** Subjects on which **both Parliament and State Legislatures can make laws** (e.g., education, forests, marriage and divorce). In case of conflict, the Union law prevails.

Additionally, residuary powers (subjects not mentioned in any of the three lists) are vested in Parliament under Article 248 and Entry 97 of the Union List.

**Pair I is correct: Fisheries**

**Fisheries fall under the State List (Entry 21)**, which includes fisheries within territorial waters and inland fisheries. Therefore, the State Legislatures have the power to make laws regarding fisheries within their territorial waters.

**Pair II is correct: Insurance**



**“Insurance” is included in the Union List (Entry 47).** Hence, only Parliament has the power to legislate on matters relating to insurance.

**Pair III is incorrect: Education**

**Originally, education was a subject in the State List.** However, **it was shifted to the Concurrent List by the 42nd Constitutional Amendment Act, 1976.** Now both Parliament and State Legislatures can make laws on education.

**Pair IV is incorrect: Cyber security**

**“Cyber security” is not explicitly mentioned in any of the three lists of the Seventh Schedule.** Therefore, it **falls under residuary powers**, which **belong to Parliament** under **Article 248** of the Constitution of India and **Entry 97** of the **Union List**. Hence, it is not part of the Concurrent List.

Source:) Mastering Indian Polity by Forum IAS, Chapter: Appendices- Appendix X, Page: 553.

<https://www.mea.gov.in/images/pdf1/S7.pdf>

Subject:) Polity

Topic:) Schedules of the Constitution of India

Subtopic:)

Q.53)

Ans) b

Exp) Option b is the correct answer.

**Pair I is correct: Rakhine State**

**Rakhine State is located in western Myanmar, bordering Bangladesh.**

- The region has been in the news mainly due to the **long-standing crisis involving the Rohingya**, a Muslim minority group in Myanmar.
- Violence and military operations by the **Myanmar armed forces**, along with clashes involving **local insurgent ethnic groups** such as the **Arakan Army**, have led to **massive displacement of people**.
- A large number of Rohingya refugees fled to Bangladesh, making the issue a major humanitarian and geopolitical concern in South Asia.

**Pair II is correct: Durand Line**

The Durand Line is the **2,640-km boundary separating Pakistan and Afghanistan**, demarcated in 1893 during British colonial rule.

- **Pakistan recognizes it as the international border, but Afghanistan has historically contested its legitimacy.**
- Because of this disagreement, along with issues related to militant movement and border fencing, the Durand Line frequently becomes a source of **diplomatic tensions and occasional clashes between the two countries**.

**Pair III is incorrect: Golan Heights**

The Golan Heights is a **strategic plateau originally part of Syria**.

- It was **captured by Israel during the Six-Day War (1967)** and later annexed by Israel in 1981, though this annexation is not widely recognized internationally.
- The **dispute over the Golan Heights** is therefore **primarily between Israel and Syria, not between Israel and Lebanon**.
- Lebanon's territorial disputes with Israel mainly involve areas such as the Blue line and **Shebaa Farms**, not the Golan Heights.

Source:) <https://www.thehindu.com/news/international/the-durand-line-fragile-frontier/article70339470.ece>

<https://www.thehindu.com/news/international/residents-of-hunger-racked-myanmar-region-say-junta-air-strike-kills-12/article69979802.ece>

<https://www.thehindu.com/news/international/israeli-army-in-golan-buffer-zone-violates-1974-deal-un/article68966832.ece>

<https://www.thehindu.com/news/international/israel-plans-to-double-population-on-occupied-golan-cites-threats-from-syria/article68989214.ece>

Subject:) International Relations



Topic:) Places in News

Subtopic:)

Q.54)

Ans) c

Exp) Option c is the correct answer.

**Statement I is correct:** The **European Union Artificial Intelligence Act** (EU AI Act) passed by the European Union is regarded as the **world's first comprehensive legal framework for regulating Artificial Intelligence**. It aims to ensure that AI systems used in the EU are safe, transparent, and respect fundamental rights while promoting innovation.

The **Act applies to AI developers, providers, and users whose systems are placed on the EU market or used within EU member states, even if the company is located outside the EU.**

A **key feature of the Act is its risk-based regulatory approach**, under which AI systems are classified into **four categories**:

- **Unacceptable Risk:** AI practices that are **completely prohibited** (e.g., social scoring by governments, manipulative AI systems that exploit vulnerable groups).
- **High Risk:** AI systems used in **sensitive sectors** such as healthcare, education, employment, law enforcement, and critical infrastructure. These are permitted but subject to strict requirements like risk assessment, human oversight, and transparency.
- **Limited Risk:** AI systems such as chatbots that must disclose that users are interacting with AI.
- **Minimal Risk:** Applications such as spam filters or AI in video games that face minimal regulatory obligations.

The Act also prescribes **strict penalties for violations**, which can reach up to €35 million or 7% of a company's global annual turnover, whichever is higher.

**Statement II is incorrect:** The EU AI Act **does not impose a complete ban on "Agentic AI" systems or General Purpose AI (GPAI) models.**

- Instead, the **Act regulates AI systems depending on the level of risk they pose. General-purpose AI models;** such as large language models used in generative AI are **allowed but subject to certain obligations**, including transparency requirements, technical documentation, and safeguards against misuse.
- **Only specific harmful AI practices classified as "unacceptable risk" are prohibited under the Act.** These include manipulative AI systems, exploitative technologies targeting vulnerable groups, and government social-scoring systems.

Knowledge Base: **Uses and Objectives of the EU AI Act**

The Act seeks to establish a **balanced framework for the development and use of Artificial Intelligence** in Europe. Its major objectives include:

- **Protecting fundamental rights** such as privacy, non-discrimination, and human dignity.
- **Ensuring safety and reliability** of AI systems used in critical sectors.
- **Promoting transparency and accountability** in algorithmic decision-making.
- **Creating a harmonised regulatory environment** across EU member states.

By establishing clear legal rules, the EU also aims to **encourage responsible AI innovation while maintaining public trust in digital technologies.**

Source:) <https://www.europarl.europa.eu/topics/en/article/20230601STO93804/eu-ai-act-first-regulation-on-artificial-intelligence>

<https://www.thehindu.com/sci-tech/technology/tech-lobby-group-urges-eu-leaders-to-pause-ai-act/article69738729.ece>

<https://www.thehindu.com/sci-tech/technology/code-of-practice-to-help-companies-with-ai-rules-may-come-end-2025-eu-says/article69771174.ece#:~:text=A%20code%20of%20practice%20designed,European%20Commission%20said%20on%20Thursday>

<https://www.thehindu.com/sci-tech/technology/eu-sticks-with-timeline-for-ai-rules/article69775426.ece>

<https://artificialintelligenceact.eu/high-level-summary/>

[https://www.edpb.europa.eu/system/files/2025-06/spe-training-on-ai-and-data-protection-technical\\_en.pdf](https://www.edpb.europa.eu/system/files/2025-06/spe-training-on-ai-and-data-protection-technical_en.pdf)

Subject:) International Relations

Topic:) Acts/Treaties/Conventions

Subtopic:)

Q.55)

Ans) d

Exp) Option d is the correct answer.

The **Government of India introduced the category of Classical Languages in 2004** to recognize languages that possess high antiquity, a rich and ancient literary tradition, and an independent cultural heritage. As of 2025, a **total of 11 Indian languages** hold the status of Classical Language.



| Classical Languages of India |                   |                 |                 |                  |
|------------------------------|-------------------|-----------------|-----------------|------------------|
| Pali<br>2024                 | Prakrit<br>2024   | Bengali<br>2024 | Marathi<br>2024 | Assamese<br>2024 |
| Odia<br>2014                 | Malayalam<br>2013 | Kannada<br>2008 | Telugu<br>2008  | Sanskrit<br>2005 |
|                              |                   | Tamil<br>2004   |                 |                  |

Among the languages given in the question, the chronological order in which they were granted Classical Language status is as follows:

- 1. Tamil (Option II)** – Tamil was the **first language to be granted Classical Language status in 2004**. It has a continuous literary tradition dating back more than two millennia, including the Sangam literature.
  - 2. Sanskrit (Option III)** – Sanskrit was granted Classical Language status in 2005. It is one of the world's oldest recorded languages and forms the basis of a vast corpus of ancient Indian literature, including the Vedas, Upanishads, epics, and classical philosophical texts.
  - 3. Odia (Option I)** – Odia received Classical Language status in 2014. The language has a long literary history with works such as Sarala Das's Odia Mahabharata and early inscriptions demonstrating its antiquity.
  - 4. Prakrit (Option IV)** – Prakrit was granted Classical Language status in 2024. The Prakrit languages were widely used in ancient India and appear in Jain and Buddhist literature as well as inscriptions such as those of Emperor Ashoka.
- Thus, the **correct chronological order in which the given languages received the status of Classical Language is: Tamil → Sanskrit → Odia → Prakrit**

Hence, the **correct answer is d) II – III – I – IV**.

Knowledge Base: **Criteria for a language to be classified as a Classical Language in India:**

The Government of India, through the Ministry of Culture and in consultation with linguistic and historical experts, has established criteria for designating a language as a Classical Language.

The **criteria for a language to be classified as a Classical Language include the following:**

- **High antiquity of its early texts or recorded history**, spanning a period of **1,500-2,000 years**.
- A **body of ancient literature or texts** that is considered heritage by generations of speakers.
- Knowledge texts, especially prose texts, in addition to poetry, epigraphical and inscriptional evidence.
- The Classical Language and its literature may be **distinct from its current form or show discontinuity with later forms derived from the original**.

Source:) <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2182753&reg=3&lang=2>

Subject:) Art and Culture

Topic:) Classical Languages

Subtopic:)

Q.56)

Ans) a

Exp) Option a is the correct answer.

The **ICC Women's Cricket World Cup 2025** attracted considerable attention as it marked an important milestone for Indian women's cricket and was widely discussed in the sporting landscape. It was played from 30 September to 2 November 2025.

**Statement I is correct.** The **ICC Women's Cricket World Cup 2025 was hosted by India**. The tournament was organised under the authority of the International Cricket Council and marked the return of the women's ODI World Cup to India after a significant gap. Hosting such a prestigious global event reflects the **growing infrastructure, popularity and administrative capability of Indian cricket**. Several major venues across the country were used for the matches, highlighting India's increasing role in promoting **women's international cricket**.

Sri Lanka was chosen as a neutral venue for some specific matches involving Pakistan.

**Statement II is correct.** India achieved a historic milestone by **winning its first-ever ICC Women's Cricket World Cup title in 2025**. Prior to this victory, the Indian women's team had reached the final on a few occasions but had not been able to secure the championship.

**Statement III is incorrect.** India won the **ICC Women's Cricket World Cup 2025 final against South Africa**, not Australia. The match was played at **Dr. D. Y. Patil Stadium**, where India secured a **52-run victory** to clinch the title.

**Statement IV is incorrect.** The ICC Women's Cricket World Cup 2025, featuring eight teams (not 12) in a round-robin format with 28 league matches, was played from 30 September to 2 November 2025.

Source:) <https://www.icc-cricket.com/media-releases/icc-makes-key-announcements-on-icc-women-s-world-cups> ;  
<https://www.icc-cricket.com/media-releases/india-clinch-historic-first-icc-women-s-cricket-world-cup-title> ;  
<https://www.thehindu.com/opinion/editorial/womens-world-on-indias-victory-in-the-world-cup/article70236318.ece>

Subject:) Current Affairs

Topic:) Sports

Subtopic:)

Q.57)

Ans) a

Exp) Option a is the correct answer.

**Pair I is incorrect.** **Natanz Nuclear Facility** is located in **Iran**, not in Turkey. Natanz is one of the most important sites associated with **Iran's nuclear enrichment programme** and has repeatedly been in the global spotlight due to tensions between Iran and Western powers regarding nuclear development. The facility lies in **central Iran, in Isfahan Province**, and has been linked to several cyber-attacks and sabotage incidents over the years.

### Incident at Natanz uranium enrichment facility



**Pair II is correct.** **Darfur** is a large region located in **western Sudan**. It has been the site of a prolonged humanitarian and ethnic conflict since the early 2000s, often referred to as the **Darfur crisis**. The conflict emerged due to tensions between government-backed militias and rebel groups representing non-Arab communities. The region continues to be mentioned in global news due to **ongoing instability, displacement of civilians, and international humanitarian concerns**.



**Pair III is incorrect.** **Novorossiysk** is actually located in **Russia**, not Ukraine. It is a **major Russian port city on the Black Sea** and hosts an important **naval base of the Russian Black Sea Fleet**. Recently, Russia declared a state of emergency at the port after Ukrainian drone attack had damaged residential buildings and injured at least five people, including two children.



Source: <https://www.thehindu.com/news/international/no-sign-irans-nuclear-sites-were-hit-iaea-says-but-iran-alleges-one-was/article70695020.ece> ; <https://www.thehindu.com/news/international/fate-of-thousands-feared-in-darfur-after-reported-rampage-by-sudanese-paramilitary-force/article70224079.ece> ; <https://www.thehindu.com/news/international/russia-declares-state-of-emergency-at-port-after-ukrainian-drone-attack-on-novorossiysk/article69533920.ece>

Subject: International Relations



Topic:) Places in News

Subtopic:)

**Q.58)**

**Ans) d**

**Exp) Option d is the correct answer.**

Timor-Leste, also known as East Timor, has recently been admitted as the **11th full member of the Association of Southeast Asian Nations (ASEAN) during the 47th ASEAN Summit in Malaysia's Kuala Lumpur in October 2025**. . This marks the **first expansion of ASEAN since the 1990s**, when countries such as Vietnam, Laos, Myanmar and Cambodia joined the grouping.

ASEAN is a **regional organisation of Southeast Asian countries formed in 1967** to promote **economic growth, political cooperation, and regional stability**. Membership in ASEAN allows countries to participate in **regional economic integration mechanisms**, including **free trade arrangements, investment flows, and wider market access within the region**.

By becoming a full member, **Timor-Leste gains access to ASEAN's trade frameworks, regional supply chains, and development cooperation programmes**. This is expected to support the country's **economic development, infrastructure growth, and deeper integration with Southeast Asian economies**.

Source:)

<https://asean.org/about-us/> ; <https://www.thehindu.com/news/international/east-timor-formally-admitted-to-asean-in-the-groups-first-expansion-since-the-1990s/article70204087.ece>

Subject:) International Relations

Topic:) International Organisations

Subtopic:)

**Q.59)**

**Ans) a**

**Exp) Option a is the correct answer.**

Firecrackers have long been associated with festivals and celebrations in India. However, conventional firecrackers release **large amounts of particulate matter and toxic gases** into the atmosphere. These emissions worsen **air pollution**, especially during winter months when atmospheric conditions already favour pollution accumulation.

To address this problem, scientists in India developed **Green Firecrackers**. These crackers are designed to **reduce harmful emissions by about 30%** compared to traditional firecrackers. They achieve this by **modifying the chemical composition** of fireworks. Certain toxic substances used in conventional firecrackers are either **removed or reduced**, while safer alternatives are introduced.

**Option I is correct.** Green firecrackers use **Potassium Nitrate as an oxidising agent**. Oxidising agents supply oxygen required for the combustion reaction. In green firecrackers, potassium nitrate is preferred because it **helps control the intensity of the reaction while reducing the emission of particulate matter**.

**On the other hand, Nitrates, Chlorates, and perchlorates** are incorporated in traditional firecrackers as the oxidizing agents.

**Also, the Aluminium powder** is commonly used in fireworks to produce **bright flashes and sparkling effects**. However, aluminium particles contribute to **fine particulate matter (PM)** in the atmosphere. In green firecrackers, aluminium is **not completely removed**, but its **quantity is significantly reduced**. This controlled use ensures that the visual effect of fireworks is retained while **limiting pollution levels**.

**Option II is incorrect.** In conventional firecrackers, copper and **barium salts** are widely used to produce the **green colour/green flames in fireworks**. However, barium compounds are **highly toxic** and contribute to environmental pollution. Green firecrackers are specifically designed **without the use of copper or barium compounds**. Their removal is



one of the key changes that make green crackers **less harmful to both human health and the environment compared to traditional firecrackers**.

**Option III is incorrect.** In traditional firecrackers, **sulphur acts as a reducing agent** and helps sustain the combustion reaction. It is usually combined with charcoal and oxidising agents to produce the explosive effect. However, sulphur-based formulations lead to **release of sulphur dioxide and other pollutants**. Green firecrackers therefore **avoid such traditional reducing agents**, replacing them with safer chemical combinations to reduce pollution.

Source:) [https://www.wbpcb.gov.in/files/Th-08-2021-08-08-58GreenCrackers\\_FAQ\\_28-10-2021.pdf](https://www.wbpcb.gov.in/files/Th-08-2021-08-08-58GreenCrackers_FAQ_28-10-2021.pdf) - Question 3 - Page 1; <https://www.thehindu.com/news/cities/Delhi/what-are-green-firecrackers-that-delhi-ncr-residents-can-burst-this-deepavali/article70166274.ece> ; <https://www.thehindu.com/sci-tech/energy-and-environment/have-green-crackers-brought-down-pollution-explained/article70202688.ece>

Subject:) Environment

Topic:) Pollution

Subtopic:)

**Q.60)**

**Ans) d**

**Exp) Option d is the correct answer.**

The **United Nations Peacekeeping Operations** represent one of the most practical tools available to the international community for maintaining peace in conflict-prone regions. These missions are usually deployed in places where fragile ceasefires exist or where societies are emerging from violent conflict.

**Statement I is correct.** United Nations peacekeeping missions are established through **mandates authorised by the UN Security Council**. These mandates define the objectives of each mission. A central responsibility of peacekeepers today is the **protection of civilians**, particularly in areas affected by armed conflict. In addition, peacekeepers help **monitor ceasefires, reduce violence, assist in implementing peace agreements and support the rebuilding of institutions** that promote long-term stability.

**Statement II is correct.** Peacekeeping operations are financed through **assessed contributions from UN member states**. The share that each country contributes is determined through a formula approved by the **UN General Assembly**. Among all contributors, **the United States is the largest financial contributor**, while **China is the second largest contributor**. Their combined financial support constitutes **a substantial share of the overall peacekeeping budget**, approaching nearly half of the total contributions.

**Statement III is correct.** India has maintained a **long-standing and respected presence in UN peacekeeping missions** since the early years of the organisation. Thousands of Indian soldiers, police personnel and military observers have served in missions across **Africa, Asia and the Middle East**. Since the **1950s**, India has sent over **290,000 peacekeepers** to more than **50 missions worldwide**, making it the largest contributor to UN peacekeeping efforts. Today, over **5,000 Indian troops** serve in **nine of the eleven** active missions, often in dangerous and hostile regions, dedicated to upholding global peace and security.

Indian peacekeepers are widely known for their **professional conduct, operational capability and humanitarian engagement with local communities**. Because of this sustained participation, India continues to be recognised as **one of the major troop-contributing countries to UN peacekeeping operations**

Source:) <https://peacekeeping.un.org/en/about-us> ; <https://www.thehindu.com/news/international/united-nations-to-cut-25-of-its-global-peacekeeping-force-in-response-to-us-funding-strains/article70145224.ece> ; <https://www.thehindu.com/news/national/indian-peacekeepers-in-most-of-our-peace-keeping-missions-serve-with-commitment-un-official/article69267067.ece>

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2109587&reg=3&lang=2>

Subject:) International Relations

Topic:) UN Peacekeeping

Subtopic:)

Q.61)

Ans) b

Exp) Option b is the correct answer.

In the decades preceding the formation of the Indian National Congress, several **regional political associations emerged in different presidencies of British India**. These organisations played a crucial role in **articulating public grievances, promoting political awareness and preparing the ground for organised nationalist politics**.

They were often led by **educated Indian elites such as lawyers, journalists and social reformers**, who sought greater representation of Indians in administration and policy-making. Many of these bodies later became **important feeders to the early nationalist movement**.

The **Madras Mahajana Sabha** was established in **1884 in Madras**. It was founded by **M. Veeraraghavachariar, G. Subramania Iyer, and P. Ananda Charlu**. The organisation aimed to **promote political awareness, discuss public issues and represent Indian grievances before the colonial government**. It soon became one of the most influential political bodies in South India and later played an important role in the **early sessions of the Indian National Congress**.

Knowledge Base:

- The **Madras Native Association** was one of the earliest political organisations in the Madras Presidency. It was founded in **1852 by Gazulu Lakshminarasu Chetty**. Its objective was to represent the interests of Indians before the British authorities and to raise issues concerning governance and administration.
- The **Justice Party** was founded later in **1916 in Madras**. It was led by **Dr. T. M. Nair, P. Tyagaraja Chetty, and C. N. Mudaliar**. The party represented the interests of **non-Brahmin and intermediate castes** and opposed the perceived dominance of Brahmins in administration, education and politics. It was also linked with the formation of the **South Indian Liberal Federation (SILF)**.
- The **Bombay Presidency Association** was founded in **1885 by Pherozeshah Mehta, Badruddin Tyabji, and K. T. Telang**. It functioned as an important political organisation in the Bombay Presidency and worked towards **constitutional reforms and increased Indian participation in governance**.

Source:) <https://www.thehindu.com/news/national/tamil-nadu/madras-mahajana-sabha-and-social-political-renaissance-in-south-india/article68677593.ece>

Subject:) History

Topic:) Modern History

Subtopic:)

Q.62)

Ans) a

Exp) Option a is the correct answer.

Inscriptions constitute one of the **most reliable primary sources for reconstructing ancient Indian history**. Unlike literary texts, which may contain exaggerations or later interpolations, inscriptions are **contemporary records engraved on stone, pillars or copper plates**. They provide valuable information about **rulers, dynasties, administrative practices, religious patronage and social conditions** of their time.

**Pair I is incorrect**. The **Naneghat inscription**, located in present-day Maharashtra, is associated with the **Satavahana dynasty**. It was issued by **Queen Naganika in 1st Century BCE (not by Gautamiputra Satakarni 106 to 130 CE)**, the wife of **Satakarni I**, one of the prominent early Satavahana rulers. Queen Naganika refers to him with the title **“Dakshinapathapati” (Lord of the Deccan)**. This title indicates that Satakarni I had established considerable authority over the **Dakshinapatha or the Deccan region**.

The inscription also records **royal genealogies and Vedic sacrifices performed by the Satavahana rulers**, such as the Ashvamedha. It is an important source for understanding the **early political history of the Satavahanas** in the Deccan region.

**Pair II is correct.** The **Aihole inscription**, also known as the **Aihole Prashasti**, dates to around **634 CE**. It was composed by the court poet **Ravikirti** during the reign of **Pulakeshin II of the Chalukya dynasty**. The inscription glorifies Pulakeshin II's achievements and notably records **his victory over Harshavardhana on the banks of the Narmada**.

**Pair III is incorrect.** The **Uttarmerur inscriptions**, found in Tamil Nadu, belong to the **Chola period**, but they are mainly associated with the reign of **Parantaka I (10th century CE)** rather than Rajaraja I. These inscriptions are historically significant because they provide **detailed information about the functioning of local self-government in Chola villages**, particularly the **Kudavolai system of electing members to village assemblies**.

Source:) <https://www.egyankosh.ac.in/bitstream/123456789/67704/1/Block-1.pdf> - Emergence of Regional Powers - Page 40;

<https://egyankosh.ac.in/bitstream/123456789/21986/1/Unit-7.pdf> - UNIT 7 Early Medieval Polities In Peninsular India 8TH TO 12TH Centuries A.D. - Page 8; <https://egyankosh.ac.in/bitstream/123456789/84170/3/Unit-5.pdf> - Unit 5 - Inscriptions and Prashastis - Types Of Inscriptions - Page 75

Subject:) History

Topic:) Ancient History

Subtopic:)

**Q.63)**

**Ans) a**

**Exp) Option a is the correct answer.**

During the nineteenth century, newspapers gradually emerged as an important instrument of **political discussion and public criticism of colonial policies**. While English newspapers were largely read by Europeans and a small educated elite, **vernacular newspapers reached a much wider Indian readership**. As a result, the British administration increasingly perceived the Indian-language press as a **potential threat to colonial authority**. It was in this political climate that the colonial government introduced the **Vernacular Press Act of 1878**, one of the most controversial press laws in British India.

**Statement I is correct.** The **Vernacular Press Act, 1878 was enacted during the viceroyalty of Lord Lytton**. The Act empowered colonial authorities, particularly District Magistrates, to take action against newspapers published in **Indian vernacular languages** if their writings were considered to be seditious or hostile to the government. Notably, the law applied **only to vernacular newspapers**, while English-language newspapers were not subjected to the same restrictions.

**Statement II is correct.** By the 1870s, several vernacular newspapers had begun to **criticise British administrative policies and raise political awareness among Indians**. The colonial government feared that such writings could influence public opinion and encourage opposition to British rule.

**Statement II explains Statement I.** The growing criticism of colonial policies in **vernacular newspapers** was precisely the reason why the British government enacted the **Vernacular Press Act, 1878** to curb dissent expressed through the Indian-language press. During this period, vernacular newspapers (like *Amrita Bazar Patrika* and *Soma Prakash*) were highly critical of the British government's handling of the **Great Famine of 1876-78** and the expensive Second Anglo-Afghan War. Also, in 1877 the Delhi Durbar, was organised by Lytton to proclaim Queen Victoria as Empress of India, as a lavish, 17-day event held amid the devastating Great Famine of 1876-1878. The vernacular newspapers were critical of such policies of the British.

Source:) [https://digital.nios.ac.in/content/315en/315\\_History\\_Eng\\_Lesson18.pdf](https://digital.nios.ac.in/content/315en/315_History_Eng_Lesson18.pdf) - Chapter 18 - Social Changes in Modern India - Growth of Press in India - Page 38

<https://www.egyankosh.ac.in/bitstream/123456789/78620/1/Unit-15.pdf>

Subject:) History

Topic:) Modern History

Subtopic:)

Q.64)

Ans) c

Exp) Option c is the correct answer.

The **Santhal Uprising of 1855–56**, also known as the **Santhal Hul**, was one of the most significant tribal revolts against colonial rule in nineteenth-century India. It occurred in the region known as **Daman-i-koh**, located in present-day Jharkhand and adjoining areas of Bihar and West Bengal.

**Statement I is correct.** The Santhal uprising was largely a reaction against **economic exploitation and administrative oppression**. The Santhals had originally cleared forest land and began cultivation in this region. Over time, however, the entry of **zamindars, moneylenders, traders and government officials** into the area deeply disrupted their traditional way of life. These outsiders, referred to by the Santhals as “**dikus**”, imposed heavy rents, manipulated market practices and charged extremely high interest on loans. Such exploitative practices gradually created widespread resentment among the Santhal community. When peaceful protests failed and repression by local authorities intensified, the Santhals resorted to armed rebellion

**Statement II is correct.** The rebellion found strong leadership in **two brothers, Sidhu and Kanu**, who emerged as charismatic figures among the Santhals. Chand-Bhairav, and Phulo-Jhano—along with countless other brave tribals were associated with the Santhal Hul. They mobilised thousands of tribal followers and called upon them to resist the oppression of zamindars, moneylenders and colonial authorities. The rebels even envisioned **restoring control over their land and establishing their own system of governance**.

Although the Santhals displayed remarkable courage and mobilised large numbers of participants, the rebellion **did not succeed in overthrowing British authority**. The colonial government eventually suppressed the uprising through **superior military power and modern weapons**. Despite its defeat, the rebellion remains historically important as a powerful expression of **tribal resistance against colonial exploitation**. While the uprising was crushed, it forced the British to create the separate Santhal Parganas district and enact the Santhal Parganas Tenancy Act of 1876 to protect tribal lands.

Source:) <https://ncert.nic.in/textbook/pdf/lehs301.pdf>

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2140737&reg=3&lang=2> ;

<https://egyankosh.ac.in/bitstream/123456789/20350/1/Unit-29.pdf> - UNIT 29 - Peasant and Tribal Uprisings - Santhal Rebellion - Page 9

Subject:) History

Topic:) Modern History

Subtopic:)

Q.65)

Ans) c

Exp) Option c is the correct answer.

By the mid-1940s, the demand for Indian independence had become irresistible and the British government realised that a constitutional settlement had to be negotiated with Indian political leaders. In this context, the **Cabinet Mission was sent to India in 1946** by the British government. The Mission consisted of **Lord Pethick-Lawrence, Sir Stafford Cripps and A.V. Alexander**, and its objective was to **devise a constitutional framework acceptable to the major political parties in India** and facilitate the transfer of power.

The Cabinet Mission proposed a complex constitutional scheme intended to balance **the demand for national unity with the Muslim League's fear of domination by a Hindu-majority centre**. Its recommendations included the formation of a **Constituent Assembly and a federal structure with limited powers for the Union**.

**Statement I is correct.** The Cabinet Mission recommended the formation of a **Constituent Assembly to draft the Constitution of India**. Since there was not enough time to conduct elections on the basis of **universal adult franchise**, the Mission proposed that certain members of the Constituent Assembly should be **elected indirectly by the already elected provincial legislative assemblies**. This arrangement allowed representation of major political groups without organising fresh nationwide elections.

**Statement II is correct.** The Cabinet Mission Plan proposed a **federal structure in which the Union government would have limited powers**. The central authority was to control only a few subjects of national importance, namely **defence, foreign affairs and communications**, along with the power to raise finances necessary for administering these subjects. Most other powers were to remain with the **provinces and princely states**, reflecting the Mission's attempt to reassure communities that feared an overly powerful central government.

The Cabinet Mission also introduced a **grouping scheme**, under which provinces were placed into three sections (A, B and C). Provinces within each section were expected to frame a **group constitution and share certain subjects in common**. A province could reconsider its membership **only after the first general elections held under the new constitutional arrangements**.

Source:) <https://egyankosh.ac.in/bitstream/123456789/21043/1/Unit-5.pdf> - Unit 5 - The Making of the Indian Constitution - THE CABINET MISSION PLAN - Page 6

<https://sansad.in/ls/about/constituent-assembly>

Subject:) History

Topic:) Modern History

Subtopic:)

Q.66)

Ans) d

Exp) Option d is the correct answer.

**Statement I is incorrect:** While the **United States is the top producer**, it is **not the world's largest exporter of crude oil by value**. That title is held by **Saudi Arabia**. The U.S. is also the world's leading consumer (18.7% of global demand), meaning a significant portion of its production is used domestically.

**Statement II is correct:** According to the Energy Institute's (EI) 2025 Statistical Review of World Energy, the **United States** remains the **world's largest oil producer**, pumping **20,135 thousand barrels daily**. This output accounts for **20.8% of global supply**, nearly equaling the combined production of **Saudi Arabia and Russia**, which **rank second and third respectively**.

**Hence, Statement I is not correct but Statement II is correct.**

Source:) <https://indianexpress.com/article/trending/top-10-listing/top-10-largest-oil-producing-and-consuming-countries-2025-10227247/>

<https://www.thehindu.com/news/international/oil-and-iran-us-war-key-takeaways-in-graphics/article70721149.ece>

Subject:) Economy

Topic:) External Sector

Subtopic:)

Q.67)

Ans) c

Exp) Option c is the correct answer.

The **Open Network for Digital Commerce (ONDC)** was **launched in 2022 to transform and democratize the digital commerce ecosystem in the country**. It aims at promoting open networks for all aspects of exchange of goods and services over digital or electronic networks

**Statement I is incorrect:** The **Open Network for Digital Commerce (ONDC)** is an **initiative of the Department for Promotion of Industry and Internal Trade (DPIIT)** under the **Ministry of Commerce and Industry**, **not the Ministry of Electronics and Information Technology (MeitY)**.

- It was launched to democratize digital commerce by enabling wider participation of sellers and service providers.
- It envisions creating a **level playing field for sellers, buyers, and service providers** across India, particularly **small and medium enterprises (MSMEs)**.



**Statement II is correct:** ONDC seeks to **shift digital commerce from a platform-centric model to an open, interoperable network**. This allows **buyers and sellers to transact across different applications or platforms**, meaning a buyer on one application can purchase from a seller listed on another participating application within the same network.

**Statement III is correct:** ONDC is **built on open-source methodology**. It uses **open specifications and open network protocols** (specifically the Bechn Protocol). These protocols are independent of any specific platform, similar to how HTTP allows different browsers to access the web or UPI allows different bank apps to transfer money. This ensures that no single entity controls the entire ecosystem.

Source:) <https://www.thehindu.com/news/national/ondc-empowering-small-businesses-and-revolutionising-e-commerce-says-pm-modi/article69053357.ece>

<https://www.thehindu.com/news/national/karnataka/indias-growing-e-com-dominance-masks-a-persistent-gender-digital-divide-nasscom-foundation/article70245635.ece>

<https://www.pib.gov.in/PressReleaseIframePage.aspx?PRID=2090097&reg=3&lang=2>

Subject:) Economy

Topic:) Indian Economy

Subtopic:)

Q.68)

Ans) a

Exp) Option a is the correct answer.

**Statement I is correct:** The **International Monetary Fund (IMF)**, in its **Fiscal Monitor**, has projected that **global public debt may exceed 100% of global GDP by 2029**, reaching the **highest level** since the period **following the Second World War**, reflecting a sustained rise in sovereign borrowing across many economies.

**Statement II is correct and explains Statement I:** **Rising fiscal deficits, government borrowing, and increasing interest payments** lead to a continuous **accumulation of public debt**. When governments consistently spend more than their revenues, they finance the gap through borrowing, which increases the overall debt stock and pushes the global debt-to-GDP ratio upward over time. **This structural trend in many major economies is one of the key reasons behind the IMF's projection of global debt exceeding 100% of GDP by 2029.**

Therefore, **Both Statement I and Statement II are correct and Statement II explains Statement I.**

Source:) <https://ddnews.gov.in/en/imf-sounds-alarm-about-high-global-public-debt-urges-countries-to-build-buffers/>

<https://indianexpress.com/article/explained/explained-economics/public-debt-gdp-deficit-meaning-rise-impact-10323853/>

<https://indianexpress.com/article/trending/top-10-listing/top-10-countries-with-the-highest-debt-to-gdp-in-2025-where-does-india-rank-9976330/>

Subject:) Economy

Topic:) External Sector

Subtopic:)

Q.69)

Ans) d

Exp) Option d is the correct answer.

**Statement I is correct:** Nominal GDP is calculated using the GDP Deflator relationship.

$$\text{Nominal GDP} = (\text{Real GDP} \times \text{GDP Deflator}) / 100$$

$$\text{Nominal GDP} = (60,000 \times 125) / 100$$

$$\text{Nominal GDP} = 75,000 \text{ crores}$$

**Statement II is correct:** Using the expenditure method:

$$\text{Nominal GDP} = C + I + G + \text{Net Exports (X-M)} \quad [X = \text{Exports}, M = \text{Imports}]$$

$$75,000 = 50,000 + 15,000 + 12,000 + \text{Net Exports}$$

$75,000 = 77,000 + \text{Net Exports}$

**Net Exports** =  $75,000 - 77,000 = -2,000$  crores

**Statement III is correct:** Net Exports is defined as:

**Net Exports = Exports – Imports (X – M)**

Given Exports (X) = 8,000 crores and Net Exports = -2,000 crores

$-2,000 = 8,000 - M$

**M = 8,000 + 2,000 = 10,000 crores**

Source:) <https://ncert.nic.in/textbook/pdf/leec102.pdf>

Subject:) Economy

Topic:) National Income Accounting

Subtopic:)

**Q.70)**

**Ans) a**

**Exp) Option a is the correct answer.**

The **Insurance Regulatory and Development Authority of India (IRDAI)** has **proposed mandatory adoption of Indian Accounting Standards (Ind AS) for all insurers** starting from **April 1, 2026** to improve transparency and align reporting with global practices.

**Statement I is correct:** **Insurance penetration** is defined as **the ratio of total insurance premiums to GDP**. In India, **insurance penetration** is about **3.7%**, which is **significantly lower than the global average of about 7.3%**, indicating relatively low insurance coverage in the country.

**Statement II is incorrect:** **India is not the third-largest insurance market globally**. According to global insurance market assessments, **India ranks at the 10th position** in terms of total premium volume, with around **1.8% share of global premiums**.

**Statement III is incorrect:** FDI in the insurance sector was earlier capped at 74%, but recent reforms have liberalized the sector, **allowing up to 100% FDI under the automatic route**.

Source:) <https://m.economictimes.com/industry/banking/finance/insure/life-insurance-penetration-remained-unchanged-in-2025-at-about-half-of-global-average-irdai-health-premium/articleshow/126259232.cms>

<https://www.thehindu.com/business/irdai-sets-april-1-deadline-for-insurers-to-adopt-indian-accounting-standards/article70702982.ece>

<https://www.thehindu.com/business/insurers-can-invest-in-select-bonds-of-aifis/article70426973.ece>

<https://www.thehindu.com/news/national/lok-sabha-passes-bill-on-100-fdi-in-insurance/article70404622.ece>

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2216048&reg=3&lang=1>

<https://irdai.gov.in/documents/37343/8885151/Press+Release+on+Ind+AS+Implementation+03.03.2026.pdf/b32bd846-542d-027a-8103-2618531d1246?version=1.5&t=1772539962446>

Subject:) Economy

Topic:) Insurance Sector in India

Subtopic:)

**Q.71)**

**Ans) d**

**Exp) Option d is the correct answer.**

Iran occupies a strategically significant position in **West Asia**, acting as a bridge between **the Middle East, Central Asia, and South Asia**. These borders influence Iran's **security dynamics, regional diplomacy, trade routes, and ethnic linkages**.

**Option I is correct.** Iran shares a **north-western land border with Azerbaijan**. The two countries are connected through the **Caucasus region**, and the boundary runs along the **Aras River** in several stretches.

**Option II is correct.** Iran shares a **short but strategically important border with Armenia** in the north-west. This corridor provides Iran with **direct access to the South Caucasus and Eurasian trade routes**.

**Option III is correct.** Iran shares to the **north-east**, a long land border with **Turkmenistan**, linking it with **Central Asia**. This border region is important for **energy cooperation and trade connectivity**.

**Option IV is correct.** Iran shares a **south-eastern land border with Pakistan**, particularly with Pakistan's **Balochistan province**. The region is sensitive due to **cross-border militancy and security concerns**.

**Option V is correct.** Iran also shares a **long eastern border with Afghanistan**, which is strategically important due to issues related to **migration, security, and water sharing**.

**Option VI is correct.** Iran shares, to the **north-west**, a border with **Turkey**, connecting the **Middle East with Europe and the Caucasus region**. The border has historically been significant for **trade routes and regional geopolitics**.

Knowledge Base:



Source:) <https://indianexpress.com/article/explained/explained-global/iran-neighbours-why-tehran-complicated-relationship-10558320/>

Oxford Student Atlas

Subject:) International Relations

Topic:) Countries in News

Subtopic:)

Q.72)

Ans) b

Exp) Option b is the correct answer.

The **Earth formed about 4.6 billion years ago** as part of the early solar system. In its earliest stage, the planet looked very different from what we see today. It was a **hot, barren and rocky body**, and the atmosphere was extremely primitive. Unlike the present atmosphere rich in oxygen and nitrogen, the early Earth was surrounded by **light gases that were captured during planetary formation**. Over millions of years, geological and atmospheric processes gradually transformed Earth into a planet capable of supporting life.

In the earliest phase of its existence, **Earth had a very thin atmosphere composed mainly of hydrogen and helium**, the lightest and most abundant gases in the early solar nebula. However, because of **high temperatures and weak**

**gravitational retention**, much of this primitive atmosphere eventually **escaped into space**, leading to the formation of a secondary atmosphere through volcanic outgassing.

**Option a is incorrect.** The present atmosphere of Earth is dominated by **nitrogen (about 78%) and oxygen (about 21%)**, but these gases **were not present in significant amounts during the earliest stage of Earth's formation**. Oxygen in particular appeared much later through **photosynthesis by early microorganisms**.

**Option c is incorrect.** **Carbon dioxide and nitrogen became dominant components of the secondary atmosphere**, which developed later due to **volcanic eruptions releasing gases from Earth's interior**.

**Option d is incorrect.** Although **methane and other gases** may have been present in trace amounts during later atmospheric evolution, **argon and methane were not the principal gases in the earliest atmosphere of Earth**.

Source:) <https://ncert.nic.in/textbook/pdf/kegy202.pdf> - Chapter 2 - The Origin and Evolution of The Earth – Evolution of The Earth – Page 15

Subject:) Geography

Topic:) Atmosphere

Subtopic:)

Q.73)

Ans) b

Exp) Option b is the correct answer.

India's monsoon system is not uniform across the country. The **distribution of rainfall depends greatly on the direction of monsoon winds, local relief, and the orientation of the coastline**. A striking example of this variation is seen along the **Tamil Nadu coast**, which behaves differently from most other parts of India during the monsoon season.

**Statement I is correct.** The Tamil Nadu coast receives **comparatively little rainfall during the South-West Monsoon months (June–September) compared to the rest of India**. By the time the moisture-laden winds cross the Western Ghats and move further eastwards, **a large part of their moisture has already been released**, leaving the Coromandel Coast relatively dry compared to the western coast of India.

**Statement II is correct.** The coastal districts of Tamil Nadu receive a **significant proportion of their annual rainfall during October to December**, when the **North-East Monsoon or retreating monsoon winds** move across the **Bay of Bengal and gather moisture before reaching the Coromandel Coast**. This makes the early winter season the **principal rainy period for Tamil Nadu**.

**Statement III is correct.** The **orientation of the Tamil Nadu coastline and the presence of the Western Ghats** play a decisive role in limiting rainfall during the South-West Monsoon. The coastline runs **almost parallel to the Bay of Bengal branch of the monsoon**, reducing the chances of strong uplift and rainfall. At the same time, the region lies on the **leeward side of the Western Ghats with respect to the Arabian Sea branch**, creating a **rain-shadow effect**.

**Statement II does not explain Statement I.** The rainfall received during the **North-East Monsoon** simply indicates **when Tamil Nadu gets most of its rainfall**, but it does **not explain the cause of the dryness during the South-West Monsoon months**.

**Statement III explains statement I:** The **Arabian Sea branch of the monsoon sheds most of its moisture on the windward slopes of the Western Ghats**, leaving the areas to the east, including Tamil Nadu, in a **rain-shadow region**. At the same time, the **Bay of Bengal branch of the South-West Monsoon flows almost parallel to the Tamil Nadu coast**, which limits the upward movement of moist air required for rainfall.

Therefore, the **limited rainfall during the South-West Monsoon is mainly due to the rain-shadow effect and the parallel alignment of the coastline with the monsoon winds**.

Source:) <https://ncert.nic.in/textbook/pdf/kegy104.pdf> - Chapter 4 - Climate – Page 28 and 29 and Rainfall – Page 34 and Monsoon Winds of the Bay of Bengal – Page 37

Subject:) Geography

Topic:) Indian Monsoon

Subtopic:)



Q.74)

Ans) c

Exp) Option c is the correct answer.

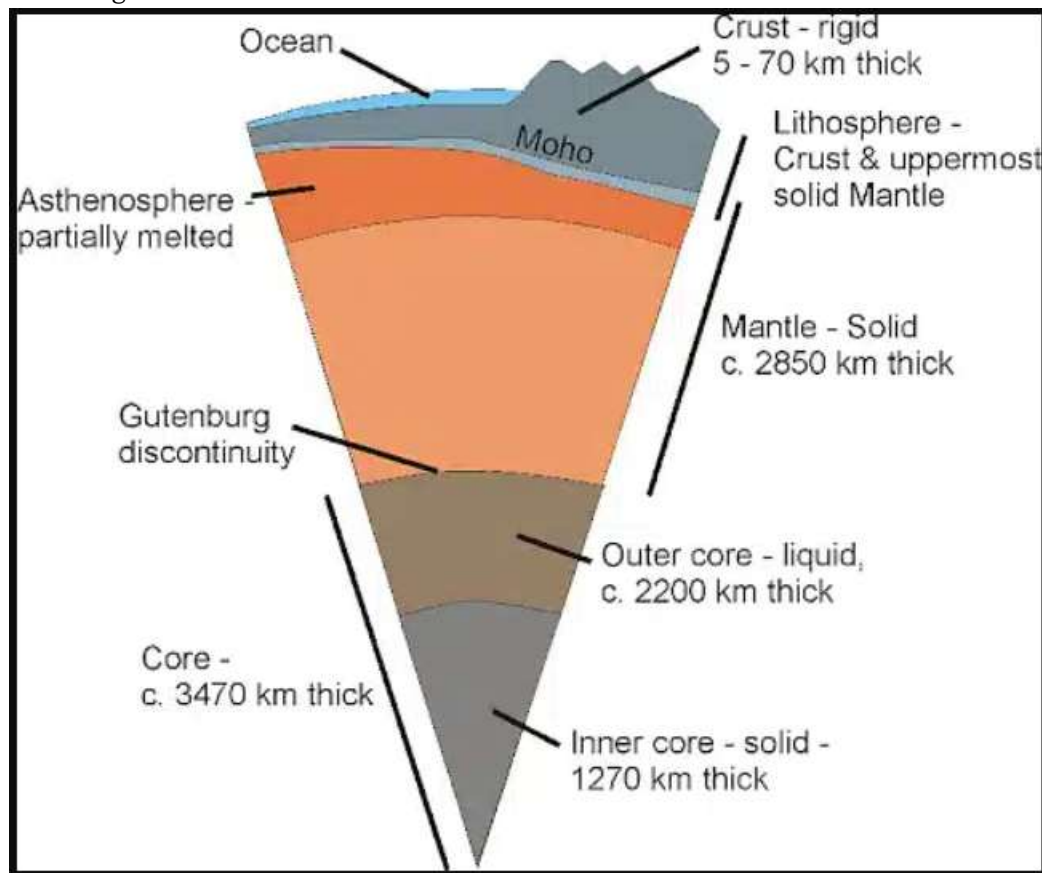
The structure of the Earth's interior plays a crucial role in shaping many geological processes observed on the planet's surface. The interaction between the **rigid outer layers and the relatively weaker zones beneath them** helps explain the movement of large crustal segments and the dynamic nature of the Earth.

**Statement I is correct.** A tectonic plate (also called lithospheric plate) is a massive, irregularly-shaped slab of solid rock, generally composed of both continental and oceanic lithosphere. **Plates move horizontally over the asthenosphere as rigid units.** The lithosphere includes the crust and top mantle with its thickness range varying between 5 and 100 km in oceanic parts and about 200 km in the continental areas. A plate may be referred to as the continental plate or oceanic plate depending on which of the two occupy a larger portion of the plate. The interactions at plate boundaries are responsible for major geological activities such as **earthquakes, volcanic eruptions and crustal deformation.**

**Statement II is incorrect:** The earth was mostly in a volatile state during its primordial stage. Due to gradual increase in density the temperature inside has increased. As a result the material inside started getting separated depending on their densities. This allowed heavier materials (like iron) to sink towards the centre of the earth and the lighter ones to move towards the surface. It is through the process of differentiation that the earth forming material got separated into different layers. **Starting from the surface to the central parts, we have layers like the crust, mantle, outer core and inner core. From the crust to the core, the density of the material increases.**

**What explains statement I.** The **semi-viscous and deformable nature of the asthenosphere** provides a suitable surface over which the rigid lithospheric plates can **float and move gradually.** This weak and ductile layer acts as a **mechanical base that facilitates the horizontal movement of plates**, much like solid rafts drifting over a softer medium. Also the lithospheric plates are less dense than the asthenosphere beneath them, which makes them float over the asthenosphere. Therefore, the **physical properties of the asthenosphere directly enable the mobility of lithospheric plates.**

Knowledge Base:





Source:) <https://ncert.nic.in/textbook/pdf/kegy204.pdf> - Chapter 4 - Distribution of Oceans and Continents - Plate Tectonics - Page 32;

<https://ncert.nic.in/textbook/pdf/kegy202.pdf> Page 15)

<https://egyankosh.ac.in/bitstream/123456789/83715/1/Unit-14.pdf#:~:text=The%20theory%20of%20plate%20tectonics,and%20numerous%20smaller%20plates%2C%20all> - Unit 14 -

Plate Tectonics - Lithosphere And Asthenosphere - Page 90

Subject:) Geography

Topic:) Plate Tectonics

Subtopic:)

Q.75)

Ans) d

Exp) Option d is the correct answer.

Ocean tides represent the **periodic rise and fall of sea levels**, mainly caused by the **gravitational pull of the Moon and the Sun along with the Earth's rotation**. Within a tidal cycle, the terms **ebb** and **flood (or flow)**, which denote the **direction of change in sea level**.

**Statement I is incorrect.** The term **ebb** does not refer to the period when the water level rises from low tide to high tide. In reality, **ebb tide is the phase when the sea level is falling**, that is, the period **between high tide and the subsequent low tide**. During this stage, water gradually **recedes from the shore** as the tide moves outward.

**Statement II is incorrect.** The term **flow or flood tide** does not denote the period when water level falls from high tide to low tide. Instead, **flood tide refers to the phase when the sea level is rising**, which occurs **between low tide and the following high tide**. During this period, the seawater **moves landward and gradually covers more of the coastal area**.

Source:) <https://ncert.nic.in/textbook/pdf/kegy213.pdf> - Chapter 13 - Movements of Ocean Page 110

Subject:) Geography

Topic:) Oceanography

Subtopic:)

Q.76)

Ans) b

Exp) Option b is the correct answer.

### How the Tropical Forest Forever Facility Will Work



Establish a blended finance fund with a target of \$125 billion (\$25 billion from donor countries, \$100 billion from private capital).



Invest in a fixed-income portfolio to earn returns above the capital costs.



Satellite-based monitoring systems calculate canopy cover, deforestation and degradation.



Tropical forest countries are paid per hectare of standing forest discounted for deforestation and fire degradation.



Tropical forest countries allocate at least 20% of payments to Indigenous peoples and local communities; the rest to finance forest conservation and sustainable use.

Source: TFFF Concept Note 3.1.



WORLD RESOURCES INSTITUTE

**Statement I is correct:** The **Tropical Forests Forever Facility (TFFF)** is a global climate-finance initiative **formally launched at the COP30** (30th Conference of the Parties to the United Nations Framework Convention on Climate Change) **held in Belém, Brazil**. The initiative was **spearheaded by Brazil** with the aim of **mobilising long-term international financial resources for the conservation of tropical forests** and reducing deforestation.

**Statement II is incorrect:** **India is not a founding member or primary financial contributor to the TFFF**. Instead, India has participated in the initiative in an **observer** capacity, supporting discussions on global forest conservation but not formally joining as a founding funder.

**Statement III is correct:** The **TFFF operates on a payment-for-performance (or results-based) model**. It aims to **raise \$125 billion** through a "**blended finance**" structure (\$25 billion from sponsors and \$100 billion from private investors). The **returns from these investments are used to pay tropical forest countries an annual reward** (estimated at \$4 per hectare) for **every hectare of forest they preserve**, provided their deforestation rate remains below 0.5%.

A **mandatory requirement** of the TFFF framework is that **participating countries must allocate at least 20% of the received funds to Indigenous Peoples and Local Communities (IPLCs)**. This recognizes their critical role as the traditional guardians of tropical ecosystems.

Source:) <https://www.wri.org/insights/financing-nature-conservation-tropical-forest-forever-facility>

<https://www.thehindu.com/news/international/india-joins-brazil-led-forest-fund-as-observer-calls-for-stronger-ambition-10-years-after-paris-pact/article70255373.ece>

<https://tfff.earth/>

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2187688&reg=3&lang=2>

Subject:) Environment

Topic:) Climate Change

Subtopic:)

Q.77)

Ans) d

Exp) Option d is the correct answer.

The term "**Novel Entities**" is used in the **Planetary Boundaries framework**, which identifies the environmental limits within which humanity can safely operate in order to maintain the stability and resilience of the Earth system.

- Novel Entities refer to **new substances, new forms of existing substances, or modified life forms introduced into the environment through human activities** that were previously absent in nature or existed only in very small quantities. They may have **potentially disruptive effects on ecological and Earth-system processes**.
- These entities include **synthetic chemicals** such as industrial chemicals and pesticides, **plastics** and microplastics, engineered nanomaterials, radioactive materials, and **genetically modified organisms (GMOs)**.
- Many of these substances are **highly persistent, widely dispersed, and capable of accumulating in ecosystems**, while natural environmental processes often **lack the ability to effectively degrade or regulate them**.
- As a result, their large-scale production and release can **disrupt ecological processes and Earth system stability**, potentially causing **long-term or uncertain impacts on ecosystems and human health**.

Source:) <https://www.stapgef.org/resources/advisory-documents/novel-entities>

<https://indianexpress.com/article/explained/explained-climate/humans-breach-planetary-boundaries-8951987/>

<https://www.thehindu.com/sci-tech/energy-and-environment/food-systems-largest-violators-of-planetary-limits-2025-eat-lancet-commission/article70241010.ece>

<https://www.thehindu.com/sci-tech/energy-and-environment/explained-land-restoration-un-global-land-outlook-report/article66813569.ece>

Subject:) Environment

Topic:) Pollution

Subtopic:)

Q.78)

Ans) a

Exp) Option a is the correct answer.

The **Greenhouse Gas Emission Intensity (GEI) Target Rules, 2025** are a set of rules notified by the Government of India under the **Environment (Protection) Act, 1986** to **mandate reductions in greenhouse gas emissions from energy-intensive industries**. They represent India's first legally binding framework that sets sector-specific emission-intensity reduction targets for industrial entities.

**Statement I is correct:** The **Greenhouse Gas Emission Intensity (GEI) Target Rules, 2025** (notified in October 2025) represent **India's first legally binding framework to reduce emissions from the industrial sector**.

Previously, schemes like the Perform, Achieve, and Trade (PAT) scheme focused on energy efficiency, but the GEI rules directly mandate the reduction of carbon emission intensity.

The **rules prescribe sector-specific greenhouse gas emission intensity targets**, which are **measured as emissions per unit of industrial output**. These emissions are **quantified in tonnes of carbon dioxide equivalent (tCO<sub>2</sub>e)** so that different greenhouse gases can be measured under a common standard metric. The Rules set targets on greenhouse gas (GHG) emissions per unit of product output—for example, the gases released in the production of a tonne of cement or aluminium. Industries will earn carbon credits in lieu of meeting emissions targets, while those that fail to do so will have to buy credits or pay environmental compensation.

**Statement II is correct:** These rules serve as the **operational backbone for the Carbon Credit Trading Scheme (CCTS), 2023**.

While the CCTS established the market framework, the GEI Rules define the specific Compliance Mechanism. Industries that over-perform (reduce emissions more than their target) earn Carbon Credit Certificates (CCCs), which they can sell to under-performers in the domestic carbon market.

**Statement III is incorrect:** The **initial phase of the GEI Target Rules applied only to a limited set of energy-intensive industries**, such as:

- **Aluminium**
- **Cement**
- **Chlor-alkali**
- **Pulp and paper**

Later, the notification, issued on 13.01.2026, brought Petroleum Refineries, Petrochemicals, Textiles and Secondary Aluminium under the Greenhouse Gas Emission Intensity (GEI) Target Rules, 2025.

The sectors mentioned; **fertiliser, thermal power, automobile manufacturing, and food processing** are **not covered** under the initial notification of the GEI Target Rules.

Source:) <https://www.thehindu.com/sci-tech/energy-and-environment/india-notifies-first-emission-intensity-targets-for-carbon-intensive-sectors/article70147013.ece>

<https://indianexpress.com/article/india/centre-notifies-first-legally-binding-emission-targets-for-high-polluting-sectors-10299043/>

<https://www.thehindubusinessline.com/news/govt-notifies-ghg-emission-intensity-targets-for-four-more-sectors/article70538659.ece>

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2217239&reg=3&lang=2>

Subject:) Environment

Topic:) Climate Change

Subtopic:)

Q.79)

Ans) a

Exp) Option a is the correct answer.

**GLP-1 receptor agonists** (Glucagon-Like Peptide-1 receptor agonists) are **synthetic drugs** that **mimic the action of the natural hormone GLP-1**, which is released in the intestine in response to food. **They act on multiple pathways:**

1. **Stimulate insulin secretion** – They increase insulin release from the pancreas when blood glucose levels are high.
2. **Suppress glucagon secretion** – This reduces the liver's glucose output.
3. **Slow gastric emptying** – Food moves more slowly through the stomach, helping to control post-meal blood sugar spikes.
4. **Increase satiety** – They reduce appetite, helping with weight management.



Because of these combined effects, GLP-1 receptor agonists are widely used in the management of type-2 diabetes and have recently gained attention as effective weight-loss drugs for obesity management. Drugs like semaglutide, dulaglutide and liraglutide belong to this class and have recently been in the news due to their expanding role in metabolic health.

Source:) <https://www.thehindu.com/sci-tech/health/who-includes-drugs-for-diabetes-and-weight-loss-in-essential-medicines-list/article70058137.ece>

<https://www.thehindu.com/sci-tech/health/how-do-glp-1-drugs-work-this-book-answers-all-your-questions/article69625012.ece>

<https://www.thehindu.com/sci-tech/health/who-backs-weight-loss-drugs-urges-cheap-generics/article70022080.ece>

Subject:) Science and Technology

Topic:) Human Health and Diseases

Subtopic:)

Q.80)

Ans) d

Exp) Option d is the correct answer.

Bio-hacking refers to a do-it-yourself (DIY) approach to modifying or optimising one's own biology through self-experimentation and data-driven lifestyle interventions aimed at improving health, performance, and longevity.

- It involves practices such as dietary changes, intermittent fasting, use of nutritional supplements, sleep optimisation, and monitoring physiological parameters using wearable devices and digital health technologies.
- The concept has gained attention within the broader global longevity movement, which focuses on extending healthy lifespan (healthspan) by making personalised adjustments to diet, metabolism, and daily habits based on biological data.

Source:) <https://www.thehindu.com/sci-tech/science/global-longevity-movement-india-ageing-science/article70448442.ece>

<https://www.thehindubusinessline.com/catalyst/when-bio-hacking-enters-the-supermarket-aisles/article70339605.ece>

<https://www.thehindu.com/sci-tech/health/the-science-and-art-of-aging/article69285518.ece>

<https://indianexpress.com/article/lifestyle/health/explained-biohacking-newest-health-buzzword-tech-billionaire-bryan-johnson-9894781/>

Subject:) Science and Technology

Topic:) Biotechnology

Subtopic:)

**Q.81)**

**Ans) a**

**Exp) Option a is the correct answer.**

**Statement I is incorrect and statement II is correct: Anusandhan National Research Foundation (ANRF) - established through an Act of Parliament: ANRF Act, 2023**, to provide high-level strategic directions for research, innovation, and entrepreneurship in the fields of natural sciences, including mathematical sciences, engineering and technology, environmental and earth sciences, health and agriculture, and scientific and technological interfaces of humanities and social sciences. Anusandhan National Research Foundation (ANRF) has been established to promote research and development and foster a culture of research and innovation throughout India's Universities, Colleges, Research Institutions, and R&D laboratories. **ANRF acts as an apex body to provide high-level strategic direction of scientific research in the country as per recommendations of the National Education Policy.** ANRF forges collaborations among the industry, academia, research institutions and government departments.

The **ANRF has subsumed the Science and Engineering Research Board (SERB)**, which earlier functioned as a **statutory body under the Department of Science and Technology** to fund scientific research. With the creation of ANRF, SERB has been integrated into the new institutional framework.

The **SERB Act, 2008 has been repealed**, and its functions related to funding and promoting scientific research have been integrated into the broader institutional framework of the ANRF.

**Statement III is incorrect:** The **Prime Minister of India serves** as the **ex-officio President of the Governing Board of the ANRF**. The Union Minister of Science and Technology and the Union Minister of Education serve as the Vice-Presidents.

**Statement IV is incorrect:** The **ANRF is not funded exclusively through budgetary allocations from the Union Government**. The Foundation aims to **mobilize ₹50,000 crore during 2023-28** through various funding streams.

Out of this, **₹14,000 crore is to be provided by the Central Government**, while the remaining amount will be mobilized from other sources, including **public sector enterprises**, the **private sector**, philanthropic organizations, foundations, and international bodies.

Hence, its **funding structure is designed to combine government support with contributions from non-government sources**.

Source:) <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2198309&reg=3&lang=1>

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2177845>

<https://serb.gov.in/>

<https://www.thehindu.com/sci-tech/science/anrfs-saral-tool-to-simplify-scientific-research-papers/article70117902.ece>

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2085535>

Subject:) Government Schemes

Topic:) Anusandhan National Research Foundation (ANRF)

Subtopic:)

**Q.82)**

**Ans) a**

**Exp) Option a is the correct answer.**

In government budgeting, receipts are broadly classified into **Revenue Receipts** and **Capital Receipts**.



**Revenue Receipts** are those which neither create a liability nor reduce government assets. These include tax revenue receipts (like GST) and non-tax revenue receipts (like dividends, fees, and interest).

**Capital Receipts**, on the other hand, are either:

1. **Create a liability** for the government (such as borrowings), or
2. **Lead to a reduction in government assets** (such as disinvestment or recovery of loans).

**Option I is correct.** When the government recovers a loan earlier extended, it receives money but simultaneously its **asset (loan outstanding)** reduces. Since it results in a reduction of an asset, it is classified as a capital receipt.

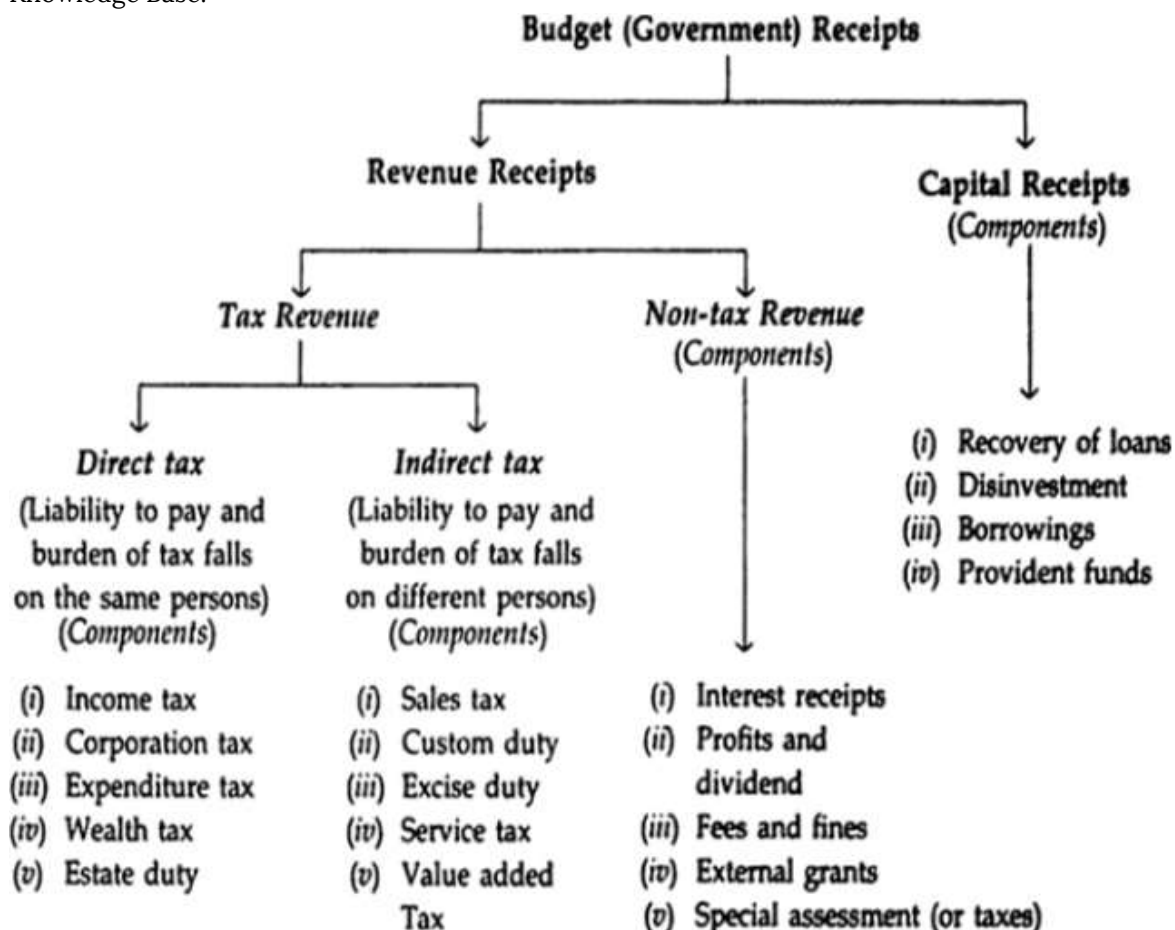
**Option II is correct.** **Disinvestment** involves selling government ownership in Public Sector Undertakings (PSUs). This reduces government assets and brings in funds, thus qualifying as a capital receipt.

**Option III is incorrect.** **Fees and fines** are examples of **Administrative Revenue**, which fall under **Non-Tax Revenue**. They are recurring receipts arising from government services or legal enforcement and do not involve asset sale or borrowing.

**Option IV is incorrect.** Dividends received from Public Sector Undertakings are part of **Non-Tax Revenue**, which falls under **Revenue Receipts**. They represent income earned from government ownership but do not reduce ownership itself.

**Option V is incorrect.** GST is a **Tax Revenue** and therefore a **Revenue Receipt**. It does not create any liability nor does it involve sale of assets.

Knowledge Base:



Source: <https://egyankosh.ac.in/bitstream/123456789/84883/3/Unit-22.pdf> UNIT 22 - Fiscal Policy - Public Revenue - Page 357

Subject:) Economy

Topic:) Budgetary Economics

Subtopic:)

Q.83)

Ans) d

Exp) Option d is the correct answer.

**6G** refers to the **sixth generation of mobile wireless communication technology**, which is expected to **succeed 5G networks in the coming decade** (around 2030). It aims to provide **extremely high data speeds, ultra-low latency**, massive connectivity and intelligent network management, enabling advanced digital applications beyond the capabilities of current mobile networks.

**Statement I is incorrect: 6G (Sixth Generation) wireless communication technology** is expected to **operate in extremely high-frequency spectrum bands**, particularly the **terahertz (THz) band**, generally considered to lie roughly between **100GHz to 10THz**. These frequencies are **higher than the millimetre-wave bands used in 5G**, and they provide extremely large bandwidth, which can support ultra-high data transmission capacities.

**Thus, it will use higher radio frequencies than 5G, letting it handle vast data at once with almost no delay.**

**Statement II is incorrect: The Bharat 6G Alliance (B6GA) is an initiative of Indian industry, Telecom Service providers, academia, national research institutions, and standards organizations.** Supported by the government and industry, it drives indigenous 6G research and development to create a self-reliant, globally competitive ecosystem for advanced communications technologies.

**Thus, it is not a multi-ministerial convergence initiative.**

It was launched by the Honorable Minister of Communications as an initiative aimed at facilitating the realization of the “Bharat 6G Vision“ for fostering the development of 6G technology in the country. It is an industry-led body, facilitated by the government, that provides a collaborative platform to various technology stakeholders comprising public/private companies, academia, research institutions, and Standard Development organizations.

Source:) <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2182603&reg=3&lang=2>

<https://bharat6galliance.com/bharat6G/>

Subject:) Science and Technology

Topic:) IT and Telecommunication

Subtopic:)

Q.84)

Ans) a

Exp) Option a is the correct answer.

**Web3**, represents the **next stage in the evolution of the internet**, characterized by **decentralization, user ownership of data**, and greater use of advanced digital technologies.

- Unlike the earlier **read-only internet (Web 1.0)** and the **interactive but centralized internet (Web 2.0)** dominated by large platforms, Web 3.0 aims to create a more open, transparent, and user-controlled digital ecosystem.
- It is primarily **built on technologies** such as **Blockchain, Smart Contracts, and Artificial Intelligence**, which enable **decentralized applications, secure peer-to-peer transactions**, and intelligent data processing.
- In this model, **users can interact directly through decentralized networks** without relying heavily on centralized intermediaries.

**Statement I is correct: In the Web 2.0 model, data is largely stored and controlled by centralized platforms** such as large technology companies that maintain centralized servers or cloud infrastructures.

In contrast, **Web 3.0 is based on decentralization**, where **data and applications are distributed across a network of computers (nodes) using technologies like blockchain and distributed storage systems**. This allows data to be stored and verified across multiple locations simultaneously, reducing dependence on centralized intermediaries and increasing resilience and transparency.

**Statement II is correct: The Web3 ecosystem includes several emerging technological frameworks built on blockchain networks. These include:**

- **Decentralized Finance (DeFi)**, which provides financial services such as lending, borrowing, and trading without traditional intermediaries like banks.

- **Non-Fungible Tokens (NFTs)**, which represent **unique digital ownership of assets** such as art, music, or virtual items using blockchain verification.
- **Decentralized Autonomous Organizations (DAOs)**, which **enable organizations to be governed collectively by participants** through rules encoded in smart contracts rather than centralized authorities.

These technologies together represent the broader decentralized digital economy emerging within Web3.

**Statement III is correct: Web3 tools** can help authenticate AI models using **zero-knowledge proof (ZKP)**. Blockchain has a cryptographic technique called ZKP that allows one party to prove the validity of a statement to another party without revealing any specific information about that statement itself. Using **this method, credit-scoring firms, for instance, can share information on users with lenders without revealing their personal financial information.** And healthcare-related firms can share information about patients without disclosing specific medical reports. ZKP can assuage security concerns arising from use of data in Large Language Models.

Source: <https://www.deccanherald.com/business/cryptocurrency/evolution-of-a-blockchain-crypto-web3-ecosystem-3823209>

<https://m.economictimes.com/industry/banking/finance/indias-web3-dominance-fueled-by-startup-ecosystem/articleshow/119587637.cms>

<https://www.thehindu.com/sci-tech/technology/how-the-web3-and-ai-worlds-collide/article67247539.ece>

<https://frontline.thehindu.com/science-and-technology/expained-web3/article65396116.ece>

Subject: Science and Technology

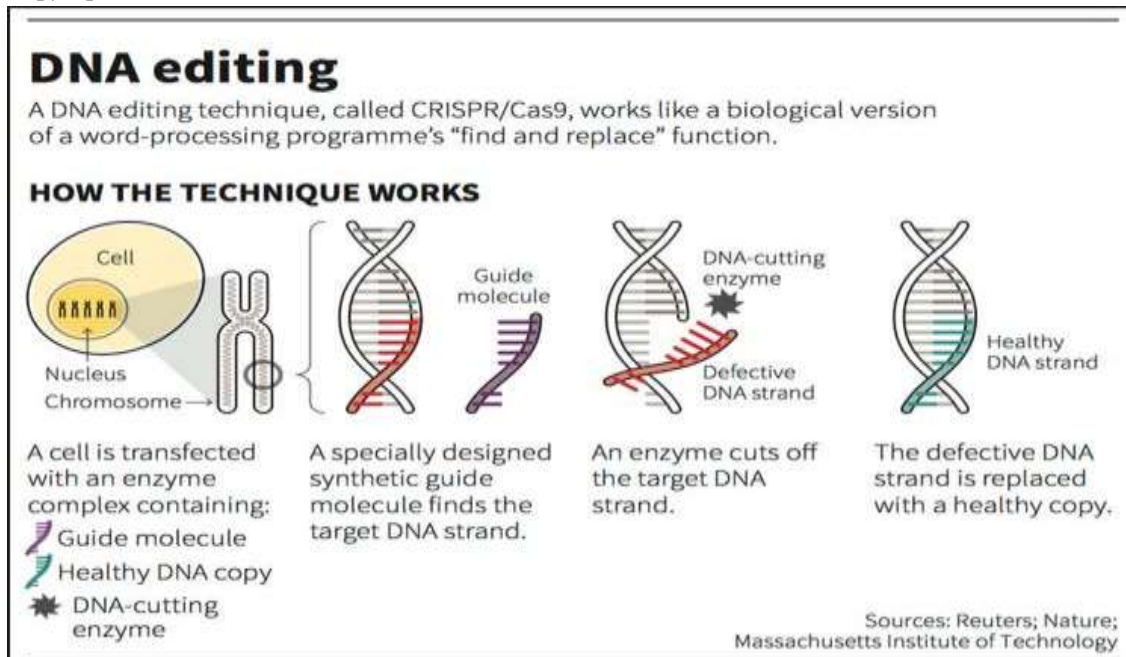
Topic: IT and Telecommunication

Subtopic:

Q.85)

Ans) b

Exp) Option b is the correct answer.



**Statement I is correct: CRISPR-Cas9 technology is derived from the natural adaptive immune system of bacteria and archaea.**

- In this system, microbes capture fragments of viral DNA and store them in clustered DNA sequences known as CRISPR.
- When the virus attacks again, RNA molecules guide Cas proteins to recognize and cut the invading viral DNA.

- Scientists adapted this biological defense mechanism into a powerful genome-editing tool that allows targeted modification of genes in plants, animals and humans.

In **CRISPR-Cas9 gene editing**, a **guide RNA (gRNA)** is designed to match a specific DNA sequence in the genome. This RNA directs the **Cas9 enzyme**, which acts like **molecular scissors**, to that precise location in the DNA. Cas9 then creates a double-strand break in the DNA. During the natural DNA-repair process, scientists can introduce insertions, deletions, or corrections of genes, enabling targeted genetic modification.

**Statement II is incorrect:** **Lyfgenia** is a **gene-addition therapy**, not a **CRISPR-Cas9 gene-editing therapy**. It uses a lentiviral vector to introduce a modified functional hemoglobin gene into a patient's stem cells to treat Sickle Cell Disease. The **first approved therapy that uses CRISPR-Cas9 gene editing for treating sickle cell disease is Casgevy**, which edits a patient's hematopoietic stem cells to increase production of fetal hemoglobin, thereby reducing disease symptoms.

**Statement III is correct:** The **application of CRISPR technology to human embryos** (germline editing) **raises significant ethical concerns**.

- Since changes made in embryos can be heritable, the technology could potentially be used to select or enhance genetic traits such as physical characteristics or intelligence.
- This possibility has led to concerns about the creation of “**designer babies**,” along with issues of safety, unintended mutations, inequality in access, and long-term societal implications.

Source:) <https://www.thehindu.com/sci-tech/health/engineering-medicine-the-future-of-car-t-and-crispr-technologies/article69257064.ece>

<https://www.thehindu.com/sci-tech/science/precision-biotherapeutics-explained/article70288204.ece>

<https://www.nobelprize.org/prizes/chemistry/2020/press-release/>

Subject:) Science and Technology

Topic:) Biotechnology

Subtopic:)

Q.86)

Ans) b

Exp) Option b is the correct answer.

**Pair I is incorrect: TRACERS**

**TRACERS** (Tandem Reconnection and Cusp Electrodynamics Reconnaissance Satellites) is a **NASA mission consisting of two satellites** designed to **study magnetic reconnection** and the interaction between the solar wind and Earth's magnetosphere, particularly in the polar cusp region.

The description given in the pair refers to an **Earth-observation radar mission monitoring land surface deformation, glaciers, and ecosystems**, which actually corresponds to **NASA-ISRO Synthetic Aperture Radar (NISAR)**, a joint mission of NASA and Indian Space Research Organisation that uses **dual-frequency synthetic aperture radar**.

**Pair II is incorrect: Europa Clipper**

**Europa Clipper** is a mission of NASA intended to investigate the **habitability of Europa**, a moon of Jupiter.

The spacecraft will **orbit Jupiter and perform multiple close flybys of Europa**, studying its **ice shell, subsurface ocean, surface composition, and geological activity**.

However, the mission **does not involve landing a rover or drilling through the icy crust**. It is designed as an **orbiter/flyby reconnaissance mission**, not a lander mission.

**Pair III is correct: PLATO**

**PLATO** (PLANetary Transits and Oscillations of stars) is a European Space Agency mission scheduled for launch in late 2026. It is intended to **detect Earth-like exoplanets orbiting bright stars** and **study the internal structure of their host stars**. It will use the transit method and stellar oscillation measurements to determine planetary sizes, ages and characteristics of planetary systems.

**Pair IV is correct: Proba-3**



**Proba-3 is a European Space Agency mission consisting of two satellites flying in precise formation—one acting as an occulter to block the Sun and the other carrying a coronagraph. This configuration creates artificial solar eclipses in space, allowing scientists to study the Sun's corona and solar activity with unprecedented precision.**

Source:) <https://timesofindia.indiatimes.com/etimes/trending/nasa-launches-tracers-twin-satellites-to-discover-the-mysterious-link-between-solar-storms-and-tech-failures/articleshow/122896042.cms>

<https://indianexpress.com/article/technology/science/nasas-europa-clipper-could-soon-cross-the-tail-of-a-comet-why-scientists-are-watching-closely-10323103/>

<https://timesofindia.indiatimes.com/science/new-telescope-cuts-through-space-noise-in-search-for-earth-like-planets-poet-and-plato/articleshow/124324911.cms>

<https://www.ndtv.com/science/europes-sun-observation-satellite-goes-dark-loses-contact-with-ground-control-heres-what-we-know-11184165>

Subject:) Science and Technology

Topic:) IT and Telecommunication

Subtopic:)

**Q.87)**

**Ans) c**

**Exp) Option c is the correct answer.**

**O3b mPOWER, Global Xpress (GX) and EchoStar are satellite-based communication and broadband connectivity systems** that provide high-speed internet and data services across the globe.

**1. O3b mPOWER:**

- It is a **next-generation medium Earth orbit (MEO) satellite constellation** that provides high-throughput, low-latency broadband connectivity to telecom operators, governments, maritime and aviation sectors.

**2. Global Xpress (GX):**

- It is a **Ka-band geostationary satellite communication network** developed by Inmarsat that delivers high-speed global broadband services for maritime, aviation, government and commercial users.

**3. EchoStar:**

- It is a **satellite communications company** that **operates communication satellites and broadband systems** (including those used for HughesNet services) providing satellite internet and broadcasting services worldwide.

**Therefore: All three are commercial satellite communication systems designed to deliver global broadband and connectivity services.**

Source:) <https://www.thehindu.com/sci-tech/technology/how-does-satellite-internet-work-explained/article69924030.ece>

<https://telecom.economictimes.indiatimes.com/news/portal-in-portal/satcom/ses-awaits-security-approval-for-new-meo-satellites-to-enhance-reliance-jios-enterprise-offerings/122877927>

<https://timesofindia.indiatimes.com/technology/tech-news/elon-musks-spacex-has-a-17-billion-plan-for-people-who-like-to-go-trekking/articleshow/124006122.cms>

Subject:) Science and Technology

Topic:) Space

Subtopic:)

**Q.88)**

**Ans) a**

**Exp) Option a is the correct answer.**

**Statement I is correct:** The **S-400 Triumf** is a **long-range surface-to-air missile defence system**. Its longest-range interceptor missile (40N6) can **destroy aerial targets at distances of up to about 400 km**. This makes it one of the most advanced long-range air-defence systems in the world.



The S-400 has multi-target engagement capability. It can track a large number of aerial objects simultaneously and engage multiple threats at the same time, including fighter aircraft, cruise missiles, ballistic missiles, and unmanned aerial vehicles (UAVs).

**Statement II is incorrect:** The system is not jointly developed by India and Russia. It was developed by the Russian defence company Almaz-Antey and exported through Rosoboronexport. India has procured the system from Russia, but it was not developed by the Defence Research and Development Organisation (DRDO).

Source:) <https://www.thehindu.com/news/national/defence-ministry-identifies-domestic-firm-for-s-400-maintenance/article69796670.ece>

<https://www.thehindu.com/news/national/india-to-get-remaining-s-400-missile-systems-by-2026-says-russia/article69650308.ece> "

<https://www.orfonline.org/research/air-defence-mechanisms-a-primer-on-india-and-pakistan>

Subject:) Science and Technology

Topic:) Defence

Subtopic:)

Q.89)

Ans) c

Exp) Option c is the correct answer.

**Statement I is correct:** Launch Vehicle Mark-3 (LVM3), earlier known as GSLV Mk-III, is India's heaviest operational launch vehicle, designed to launch heavy communication satellites into Geosynchronous Transfer Orbit and to undertake complex missions such as lunar and deep-space missions.

LVM3 has a payload capacity of about 4 tonnes to Geosynchronous Transfer Orbit (GTO) and around 8 tonnes to Low Earth Orbit (LEO), making it suitable for launching heavier satellites and complex missions.

**Statement II is incorrect:** LVM3 does not have a two-stage configuration. It is a three-stage launch vehicle consisting of:

- Two solid strap-on boosters (S200)
- A liquid core stage (L110)
- A cryogenic upper stage (C25) powered by the CE-20 cryogenic engine.

**Statement III is correct:** Major lunar missions of India, including Chandrayaan-2 (2019) and Chandrayaan-3 (2023), were launched using the LVM3 launch vehicle.

Source:) [https://www.isro.gov.in/LVM3\\_M6\\_BlueBird\\_Block2\\_Mission.html](https://www.isro.gov.in/LVM3_M6_BlueBird_Block2_Mission.html)

<https://www.thehindu.com/sci-tech/science/success-of-lvm-3-mission-reinforces-indias-growing-role-in-global-commercial-launch-market-pm-modi/article70432669.ece>

<https://www.thehindu.com/sci-tech/science/isros-lvm3-m5-rocket-with-heaviest-communication-satellite-lifts-off/article70232624.ece>

[https://www.isro.gov.in/GSLVmk3\\_CON.html](https://www.isro.gov.in/GSLVmk3_CON.html)

Subject:) Science and Technology

Topic:) Space

Subtopic:)

Q.90)

Ans) a

Exp) Option a is the correct answer.

**Statement I is correct:** Green Methanol is produced by synthesizing green hydrogen (hydrogen generated through electrolysis of water using renewable energy such as solar or wind) with captured carbon dioxide (CO<sub>2</sub>) through Carbon Capture and Utilisation (CCU).

- The captured CO<sub>2</sub> acts as the carbon source and reacts with hydrogen in a catalytic process to form methanol (CH<sub>3</sub>OH). Since renewable energy and recycled CO<sub>2</sub> are used, the fuel is considered low-carbon or carbon-neutral in lifecycle terms.

**Statement II is correct:** Methanol can be used as a fuel in internal combustion (IC) engines. It can either be used directly with certain engine modifications or blended with conventional fuels such as gasoline. Because of its high octane number and clean-burning characteristics, it is also being explored as an alternative fuel in sectors such as road transport and shipping.

**Compared with many conventional fossil fuels** such as diesel or heavy fuel oil, the **combustion of methanol generally produces lower emissions of sulphur oxides (SO<sub>x</sub>), nitrogen oxides (NO<sub>x</sub>) and particulate matter.** This is mainly because methanol contains no sulphur and burns more cleanly, making it a relatively cleaner fuel option.

**Statement III is incorrect:** Although methanol burns relatively cleanly, **its combustion does not produce only water vapour. It also produces carbon dioxide (CO<sub>2</sub>)** and small amounts of other compounds such as formaldehyde.

However, when green methanol is produced using captured CO<sub>2</sub>, the overall carbon cycle can be close to carbon-neutral, but the combustion process itself still releases CO<sub>2</sub>.

Source:) <https://www.thehindubusinessline.com/opinion/the-power-of-green-methanol/article67404914.ece>

<https://www.thehindu.com/business/Industry/how-is-global-shipping-trying-to-decarbonise-explained/article69816234> .

<https://www.thehindu.com/opinion/op-ed/indias-energy-shift-through-the-green-ammonia-route/article70668210.ece>

<https://www.thehindu.com/news/cities/Visakhapatnam/methanol-is-a-potential-renewable-alternative-to-fossil-fuels-says-expert/article67123313.ece>

Subject:) Science and Technology

Topic:) Energy

Subtopic:)

Q.91)

Ans) c

**Exp) Option c is the correct answer.**

**Statement I is correct.** The Union Cabinet has approved the launch of the Urban Challenge Fund (UCF) with a total Central Assistance (CA) of Rs. One lakh crore. **CA will cover 25% of the project cost, subject to raising a minimum 50% of the project cost from the market. A minimum of 50 per cent of project financing has to be mobilised from market sources, including municipal bonds, bank loans and Public-Private Partnerships (PPPs). The remaining share may be contributed by States, Union Territories (UTs), Urban Local Bodies (ULBs) or other sources.** This will lead to a total investment of Rs. Four lakh crore in the urban sector in next five years, marking a paradigm shift in India's urban development approach from grant-based financing to market-linked, reform-driven and outcome-oriented infrastructure creation.

The Urban Challenge Fund will leverage market finance, private participation and citizen-centric reforms for delivery of high-quality urban infrastructure. The Fund aims to build resilient, productive, inclusive and climate-responsive cities, positioning those as key drivers of the country's next phase of economic growth.

The scheme primarily covers three categories of cities:

- 1. All cities with a population of 10 lakh or more** (based on 2025 estimates).
- 2. All State and Union Territory capitals** that are not already included in the first category.
- 3. Major industrial cities with a population of 1 lakh or more.**

**Statement II is correct.** A major challenge faced by many smaller cities in India is their **limited ability to raise funds from financial markets.** Urban Local Bodies in such cities often lack the creditworthiness required to access bank loans or issue municipal bonds.

To address this problem, in order to facilitate first-time access to market finance for all Cities/ULBs in Northeastern & Hilly States/UTs and smaller ULBs (<1,00,000 population) in other States/UTs, **a Credit Repayment Guarantee Scheme**

of ₹5,000 crore has been approved under Urban Challenge Fund. The scheme will provide a Central guarantee of up to ₹7 crore or 70 per cent of the loan amount (whichever is lower) for first-time loans. On successful repayment of the first loan, central guarantee of ₹7 crore or 50 per cent of the loan amount (whichever is lower) will be provided. This will effectively support projects of ₹20 crore for the first time and ₹28 crore for subsequent projects in smaller cities.

Source:) <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2227986&reg=3&lang=1> ;

[https://www.pmindia.gov.in/en/news\\_updates/cabinet-approves-rs-one-lakh-crore-urban-challenge-fund-to-drive-market-led-urban-transformation/](https://www.pmindia.gov.in/en/news_updates/cabinet-approves-rs-one-lakh-crore-urban-challenge-fund-to-drive-market-led-urban-transformation/)

<https://static.pib.gov.in/WriteReadData/specificdocs/documents/2026/feb/doc2026214790101.pdf>

<https://www.thehindu.com/news/national/union-cabinet-clears-urban-challenge-fund-for-cities/article70631961.ece>

Subject:) Economy

Topic:) Infrastructure

Subtopic:)

Q.92)

Ans) c

Exp) Option c is the correct answer

Inflation steadily reduces the **purchasing power of money**. When prices rise, the real return from traditional fixed-income instruments declines because the interest earned may not keep pace with inflation. To address this issue, governments issue **Inflation-Indexed Bonds (IIBs)**.

These bonds are designed to **protect investors against inflation risk** by linking the value of the investment to an inflation index. In India, the Reserve Bank of India introduced Inflation-Indexed Bonds as part of the **government securities programme** to provide investors with an instrument that safeguards the real value of savings.

**Statement I is correct.** Inflation-Indexed Bonds are structured so that the **principal value of the bond is linked to an inflation index**.

This linkage is achieved through an **index ratio**, which adjusts the original face value of the bond according to changes in the inflation index over time. When inflation rises, the **adjusted principal increases**, thereby preserving the purchasing power of the invested amount.

**Statement II is correct:** Inflation-Indexed Bonds also protect investors through their **interest payment mechanism**. These bonds carry a **fixed real coupon rate**, but the interest is calculated on the **inflation-adjusted principal rather than the original face value**. As inflation increases and the principal is revised upward, the **interest payments also increase proportionately**. This feature ensures that investors receive **returns that keep pace with inflation**, thereby maintaining the **real yield on the investment**.

Source:) <https://www.rbi.org.in/commonman/English/scripts/FAQs.aspx?Id=1165> ;

<https://www.thehindu.com/children/government-bonds-in-india-understanding-the-financial-market-component/article69400067.ece>

Subject:) Economy

Topic:) Capital Market

Subtopic:)

Q.93)

Ans) d

Exp) Option d is the correct answer.

Governments frequently rely on **fiscal policy i.e.** changes in public expenditure and taxation, to influence economic activity. When the government increases spending, it aims to stimulate demand and raise national income. However, such policies may sometimes produce an unintended side effect known as the **Crowding Out Effect**, which explains how government borrowing can influence private sector investment decisions.

The **Crowding Out Effect** occurs when **higher government spending leads to increased borrowing from financial markets**, which raises the overall demand for funds in the economy. This refers to a phenomenon where increased

borrowing by the government to meet its spending needs causes a decrease in the quantity of funds that is available to meet the investment needs of the private sector.

As the demand for funds rises, **interest rates tend to increase**. Higher interest rates make borrowing more expensive for businesses. Consequently, firms may postpone or reduce their **private investment in projects such as factories, machinery or infrastructure**. Thus, government borrowing “crowds out” private investment, meaning that part of the economic stimulus created by public spending is offset by a decline in private sector investment.

Source:) <https://egyankosh.ac.in/bitstream/123456789/75063/1/Unit-1.pdf> - Unit 1 - Aggregate Demand - Fiscal Policy and Crowding Out - Page 8; <https://www.thehindu.com/news/international/what-will-be-the-effect-of-rising-military-spending-explained/article69787325.ece>

Subject:) Economy

Topic:) Fiscal Policy

Subtopic:)

Q.94)

Ans) a

Exp) Option a is the correct answer

**The GDP deflator is a broad measure of general price level in an economy. It is defined as the ratio of nominal GDP to real GDP.**

The **GDP deflator** of a sector is the **ratio of its Nominal (current price) GVA to Real (constant price) GVA**, and acts as a price index with the base year value equal to 1. It reflects how prices in that sector change over time.

**The ratio of manufacturing to agriculture sector deflators tells how manufacturing prices have moved relative to agricultural prices. If the ratio is greater than 1, then Manufacturing prices are higher relative to agricultural prices, which implies better terms of trade for manufacturing and vice versa.**

Source:) <https://www.indiabudget.gov.in/economicsurvey/doc/eschapter/echap05.pdf> Page: 216.

Subject:) Economy

Topic:) Economic Survey 2025-26

Subtopic:)

Q.95)

Ans) a

Exp) Option a is the correct answer.

Financing healthcare is a crucial issue for developing economies like India. The structure of health expenditure determines **how much financial burden falls directly on individuals and households**. In many countries with strong public health systems, governments bear a significant share of healthcare costs. However, in countries where public spending on health is relatively low, individuals often have to finance treatment from their own savings.

**Statement I is correct.** India experiences a **relatively high level of Out-of-Pocket Expenditure (OOPE)** on healthcare. OOPE refers to the amount that households directly pay from their own resources for medical services such as consultation, diagnostics, medicines, hospitalisation and follow-up care.

In India, a substantial proportion of total health expenditure comes directly from households. As per National Health Accounts Estimates 2021-22, the Out-of-Pocket Expenditure (OOPE) as a percentage of Total Health Expenditure (THE) is 39.4%. OOPE on health as a percentage of THE in the country for the year 2017-18, 2018-19, 2019-20, 2020-21, and 2021-22 are 48.8%, 48.2%, 47.1%, 44.4%, and 39.4% respectively and therefore there is a declining trend in OOPE as a percentage of THE. This means that **many families must rely on personal savings or borrow money to meet medical expenses**, which can significantly affect their financial stability. Studies and policy reports have repeatedly shown that medical spending can push a large number of people into poverty each year.

**Statement II is correct.** Historically, **government spending on health in India has remained relatively low as a share of GDP**, generally hovering around **1-1.5% of GDP** for a long time.

Compared with many other countries, especially developed economies and several emerging economies, this level of public expenditure is **considered quite modest**. Lower public spending implies fewer publicly funded hospitals, limited free treatment options and reduced coverage of health services through government programmes.

**Statement II explains Statement I. When government spending on healthcare is limited, the public health system cannot fully meet the medical needs of the population.** As a result, individuals often turn to private healthcare providers for treatment. Private healthcare typically requires **direct payment by patients**, which increases the share of household spending in total health expenditure. Therefore, **low public health expenditure contributes significantly to high Out-of-Pocket Expenditure in India.**

Source:) <https://www.thehindu.com/sci-tech/health/how-opaque-healthcare-pricing-burdens-patients-in-india/article70637957.ece>

<https://www.mohfw.gov.in/?q=en/pressrelease-160>

Subject:) Economy

Topic:) Human Development

Subtopic:)

Q.96)

Ans) c

Exp) Option c is the correct answer.

The **structure of legislatures in India has its historical roots in the constitutional reforms introduced during British rule**. In the early twentieth century, the British administration recognised the need to provide Indians with greater participation in governance and law-making. As a result, the **Montagu-Chelmsford Reforms** led to the enactment of the **Government of India Act, 1919**, which introduced **bicameral legislatures at the Centre and in certain provinces**.

Independent India largely adopted this parliamentary model. However, the Constitution under **Article 169** allows states to have either **unicameral or bicameral legislatures**. While Parliament is permanently bicameral, **states have the flexibility to decide whether to establish a Legislative Council**.

**Option I is correct.** Andhra Pradesh has a **bicameral legislature**, consisting of the **Legislative Assembly (Vidhan Sabha)** and the **Legislative Council (Vidhan Parishad)**. The state re-established its Legislative Council in **2007** after it had been abolished earlier.

**Option II is correct.** Telangana, formed in **2014**, also has a **bicameral legislature**. The state inherited the Legislative Council structure from the erstwhile unified Andhra Pradesh and continues to function with both houses of the legislature.

**Option III is correct.** Karnataka is the state with a **bicameral legislature**. It has a **Legislative Council that functions as the upper chamber**, reviewing and revising legislation passed by the Legislative Assembly.

**Option IV is incorrect.** Gujarat does not have a **Legislative Council**. Its legislature consists of **only one house i.e. the Legislative Assembly**, making it a **unicameral state legislature**.

**Option V is correct.** Maharashtra is among the states that follow the **bicameral legislative system**. Its legislature consists of the **Maharashtra Legislative Assembly and the Maharashtra Legislative Council**, making it one of the 6 states in India with two legislative chambers.

Knowledge Base:

- There are Six states with Bicameral Legislative system in India: Andhra Pradesh, Telangana, Karnataka, Maharashtra, Bihar and Uttar Pradesh,

Source:) The Constitution of India – Article 169; Mastering Indian Polity by Forum IAS – 0th Edition – Chapter 16 – Legislative Branch – Unicameral & Bicameral Legislature – Page 336;

<https://www.deccanherald.com/india/karnataka/the-case-for-the-second-house-state-legislative-councils-3028680>

Subject:) Polity

Topic:) Legislature

Subtopic:)



Q.97)

Ans) c

Exp) Option c is the correct answer.

In India, **contempt powers serve to protect the administration of justice**, but they operate within a constitutional framework that also respects **freedom of speech under Article 19(1a)**. The balance between **judicial authority and democratic criticism** is maintained through constitutional provisions and the **Contempt of Courts Act, 1971**.

**Statement I is incorrect.** The **Constitution of India does not define civil contempt or criminal contempt**. Instead, these definitions are provided under **Section 2 of the Contempt of Courts Act, 1971**, which was enacted by Parliament following the recommendations of the **H. N. Sanyal Committee**.

**Statement II is correct.** Under the **Contempt of Courts Act, 1971**, **civil contempt** refers to **wilful disobedience of a court's judgment, decree, direction, order, writ or other process**, or the **wilful breach of an undertaking given to the court**. This form of contempt is primarily concerned with **ensuring compliance with court orders**, thereby safeguarding the effectiveness of the judicial process.

**Statement III is correct.** Not every comment about the judiciary amounts to contempt. The law recognises certain **important safeguards for freedom of expression**. For instance, **innocent publication, fair and accurate reporting of judicial proceedings, and fair criticism of a judicial decision after the case has been decided** are not treated as contempt. These protections ensure that **judicial accountability and public debate remain possible in a democratic society**, while still preserving the authority of the courts.

Knowledge Base: The **Supreme Court's power to punish for contempt originates from constitutional provisions**. Article **129** declares the Supreme Court to be a **Court of Record**, which includes the inherent power to punish for contempt of itself. Additionally, **Article 142(2)** empowers the Supreme Court to **investigate and punish any person for contempt of the Court**. These provisions ensure that the apex court can protect its authority and ensure compliance with its orders.

Source:) The Constitution of India; Mastering Indian Polity by Forum IAS - 0th Edition - Chapter 17 - Judicial Branch - Contempt of Court - Page 470;

<https://www.thehindu.com/news/national/what-constitutes-as-contempt-of-court-in-india-explained/article70245733.ece> ; <https://www.thehindu.com/news/national/no-contempt-if-parliament-legislatures-simply-make-laws-supreme-court/article69653366.ece> ; <https://www.thehindu.com/news/national/contempt-power-not-personal-armour-for-judges-supreme-court/article70380220.ece>

Subject:) Polity

Topic:) Judiciary

Subtopic:)

Q.98)

Ans) b

Exp) Option b is the correct answer.

In recent years, **Haiti** has been in the international spotlight due to **rising gang violence, political instability and a prolonged governance crisis**. Weak law-enforcement institutions and the absence of a stable elected government have allowed organised criminal groups to gain significant influence, particularly in and around the capital city.

**Statement I.** In the capital city of Haiti **Port-au-Prince**, armed gangs are estimated to control **nearly 80-90 per cent of the city**. They dominate major roads and entry points, which allows them to regulate the movement of goods and people. Through extortion, kidnappings and control of transport routes, these gangs have significantly weakened the authority of the state in the capital region.

**Statement II.** Due to its **strategic location in the Caribbean** and its **long coastline with limited surveillance**, Haiti has increasingly become a **transshipment point for drugs** such as cocaine and cannabis. These drugs typically originate in **Latin American production regions** and are transported through the Caribbean towards **North American and European markets**.

**Statement III.** Haiti has **not held national elections for several years**, largely because of persistent insecurity and institutional instability. In response to the political vacuum, a **transitional presidential council** has been formed to restore order, prepare for elections and re-establish democratic governance. However, the process remains challenging due to continued violence and fragile state institutions.

Source:) [https://www.unodc.org/unodc/frontpage/2026/January/explainer\\_-organized-crime-and-gang-violence-in-haiti.html](https://www.unodc.org/unodc/frontpage/2026/January/explainer_-organized-crime-and-gang-violence-in-haiti.html) ;

<https://www.thehindu.com/news/international/un-security-council-priorities-haitis-leaders-to-hold-general-election-as-deadline-nears/article70192264.ece>;

<https://www.thehindu.com/news/international/why-does-instability-and-chaos-persist-in-haiti-explained/article68230016.ece>

Subject:) International Relations

Topic:) Countries in News

Subtopic:)

Q.99)

Ans) d

Exp) Option d is the correct answer.

**Statements I, II and III are correct: The Genetic Engineering Appraisal Committee (GEAC) functions in the Ministry of Environment, Forest and Climate Change (MoEF&CC). As per Rules, 1989, it is responsible for appraisal of activities involving large scale use of hazardous microorganisms and recombinants in research and industrial production from the environmental angle. The committee is also responsible for appraisal of proposals relating to release of genetically engineered (GE) organisms and products into the environment including experimental field trials.**

Source:) <https://www.pib.gov.in/newsite/PrintRelease.aspx?relid=235784&reg=3&lang=2> ;

<http://www.geacindia.gov.in/about-geac-india.aspx> ; <https://www.thehindu.com/news/national/experts-on-gm-crop-committees-to-declare-conflict-of-interest/article69053648.ece> ;

<https://www.thehindubusinessline.com/economy/agri-business/no-new-gm-crops-in-south-asia-this-year-barring-pakistan/article70415458.ece>

Subject:) Environment

Topic:) Genetic Engineering Appraisal Committee (GEAC)

Subtopic:)

Q.100)

Ans) d

Exp) Option d is the correct answer.

The **Comptroller and Auditor-General (CAG) of India** is a key constitutional authority entrusted with ensuring **financial accountability and transparency in the functioning of the government**. Established under **Articles 148 to 151 of the Constitution**, the CAG acts as the **guardian of the public purse** by auditing government receipts and expenditures. Its reports enable Parliament and State Legislatures to exercise effective oversight over the executive's financial administration.

**Statement I is correct.** According to **Article 148 of the Constitution**, There shall be a Comptroller and Auditor-General of India **who shall be appointed by the President by warrant under his hand and seal** and shall only be removed from office in like manner and on the like grounds as a Judge of the Supreme Court.

**Statement II is correct.** The Constitution explicitly states that the Comptroller and Auditor-General shall not be eligible for further office either under the Government of India or under the Government of any State after he has ceased to hold his office.

This restriction is meant to ensure that the CAG functions with **complete neutrality and independence**, without any expectation of post-retirement appointments from the executive.

**Statement III is correct.** The **administrative expenses of the office of the CAG**, including salaries, allowances and pensions of its staff, are **charged upon the Consolidated Fund of India**. Expenditure charged on the Consolidated Fund **does not require annual parliamentary approval**, thereby ensuring the **financial independence of the institution**.

**Statement IV is correct.** Article **149** authorises the CAG to **perform such duties and exercise such powers in relation to the accounts of the Union, States and other authorities or bodies as prescribed by Parliament by law**. In practice, this allows the CAG to conduct various forms of audits, including **financial audits, compliance audits and performance audits** of government programmes.

A recent example is the **CAG performance audit of the Pradhan Mantri Kaushal Vikas Yojana (PMKVY)**, where the auditor examined the implementation of the skill development scheme and highlighted issues related to monitoring, training outcomes and placement tracking. Such audits demonstrate how the CAG's constitutional powers translate into **practical oversight of major public programmes**

Source:) The Constitution of India – Article 148 to 151; <https://indianexpress.com/article/india/1111111111-as-account-number-shut-centres-same-photos-cag-on-gaps-in-pradhan-mantri-kaushal-vikas-yojana-scheme-10429661/>

Subject:) Polity

Topic:) Comptroller and Auditor-General (CAG) of India

Subtopic:)